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CENG - DEPARTMENT OF ELECTRICAL ENGINEERING



Final Project Report

Group B

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Presented to:
Professor Payam Nayeri

Presented By:

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1: Introduction

The purpose of this project is to design, construct, and test a moderately complex microwave system with custom and off-the-shelf building blocks. Specifically, this project will focus on the design of a system that will measure a test antenna at various angles such that an antenna pattern may be derived. This design will involve a variety of electrical and mechanical components such as modulators/demodulators, amplifiers, filters, antennas, mechanical rotators, and other minor components.

2: System Design

The purpose of this system is to measure the antenna pattern of a device-under-test (DUT) antenna. To do this, the system will consist of two sections. The first section will take a known signal, modulate it using a QAM modulator, amplify the signal, filter the signal, and transmit the signal through a horn antenna. The second section will receive this signal at the DUT, filter the signal, amplify the signal, demodulate the signal, and then measure the amplitude of the demodulated signal using a microcontroller. After each measurement, the microcontroller will communicate with a rotator system responsible for rotating the DUT. Then, the measurement process will repeat. Simultaneously, this data will be sent to MATLAB to produce an antenna factor plot. Figure 2.1 presents a detailed block diagram of the system with both the first section (Tx) and the second section (Rx).

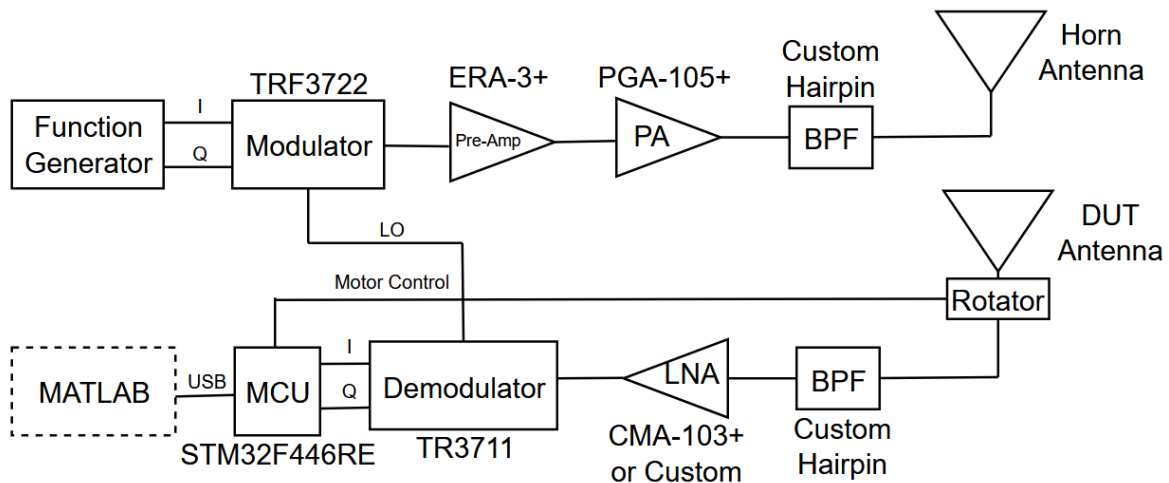


Figure 2.1: System Block Diagram

2.1: System Specification

2.1.1: Frequency

One of the major specifications for any communication system is the operating frequency. For our design, we chose an operating frequency of 2.45 GHz with a bandwidth of 100 MHz. This frequency is used for Wi-Fi meaning that the frequency band will likely be noisy, but there will also be a wide variety of components available.

2.1.2: Antenna Pattern Measurement Resolution

The main purpose of this project is to measure the antenna pattern of a DUT. As such, the resolution of our measurements is critical. Our goal will be to conduct measurements in 5° azimuth increments for the electric-field plane and the magnetic-field plane ($\phi = 0^\circ, 90^\circ$).

2.1.3: Signal-to-Noise Requirements

To appropriately measure the signal strength at the microcontroller, our team is aiming for a signal-to-noise ratio of 20dB at the demodulator.

2.1.4: Physical Specifications

The only major physical specification, besides producing designs that are physically realizable, is the separation between the horn and DUT antennas. Our team decided on 6 meters which is in the far-field of both antennas. To enter the far-field for the AUT will be past 0.5 meters with the standard gain horn past 5.8 meters, thus 6 meters will be in the far-field regions for both antennas.

2.1.5: Extra Specification

Initially, we want to get the system working with a continuous wave being transmitted. In other words, we want the I-Q input of the modulator to be constant; however, if course time permits, we would like to develop a system capable of communicating binary data. Then, the system could measure bit-error rate at various angles and transmission rates. We think a QPSK modulation scheme with a goal transmission rate of 9600 bits per second would be a good goal.

2.2: Block Level Design

Figure 2.1 presented the block diagram of our design; however, it did not include the motivation behind the placement and selection of many of our “blocks”. This section details our justification.

For the modulator and demodulator, we will use the TRF3722 and TRF3711 that former groups that had taken this course had used during previous years. Note that these devices will share a local oscillator (LO) available at the modulator. We will use the lab function generators available to supply the modulator with an I and a Q signal.

After the modulator, we will use a pre-amplifier and a power amplifier to ensure the design has sufficient gain before passing the signal through a bandpass filter. This bandpass filter will help remove unwanted distortion caused by the previous stages. Motivation behind the selection of the specific amplifiers will be elaborated further in the Sub-System Design section. Afterwards, the signal will be passed to a 15 dB standard gain horn antenna available in the lab.

On the receive side, a stand capable of rotating the DUT antenna will receive the microwave signal. This rotator will be controlled via SPI from the microcontroller. After being received by the DUT antenna, the signal will pass through a custom hair-pin bandpass filter to remove noise. A low-noise amplifier (either an off-the-shelf part or a custom designed amplifier) will amplify the signal with minimal noise prior to being demodulated by the TRF3711.

After the demodulator, the ADC available on a common STM32 microcontroller, specifically the Nucleo STM32F446RE, will convert the demodulated I and Q into data that will be sent to MATLAB for further processing. This microcontroller was chosen for its low cost, ease of use, and ADC capabilities.

3: Sub-System Design

3.1: Modulator and Demodulator

For the modulator we will be using the TRF3722 to convert the I-Q signal to a 2.45 GHz microwave signal. On the other side, we will be using the TRF3711 to convert the microwave signal back to an I-Q signal. These two modulators will share a local oscillator; however, due to different system requirements, the TRF3722 will need a 6V power supply while the TRF3711 will need a 5V power supply. Figure 3.1.1 presents the general setup of the modulator and demodulator setup.

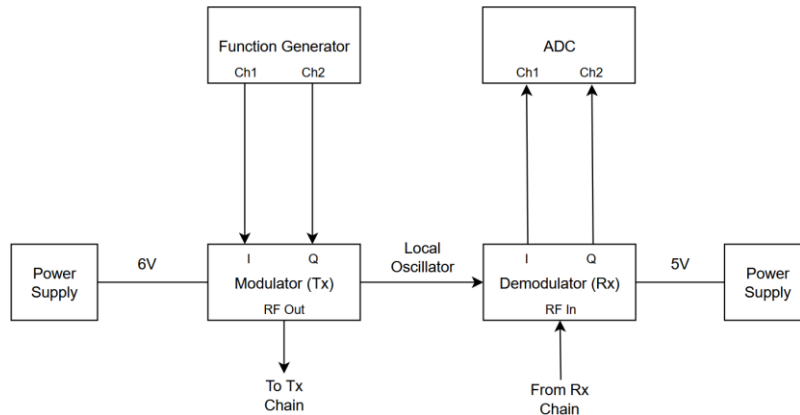


Figure 3.1.1: Modulator and Demodulator Setup

Figure 3.1.2 presents the setup for the physical test of the modulator and demodulator subsystem. For this test setup, the RF out of the modulator is connected directly to the RF in of the demodulator. For this test setup, we set the function generator to provide a 1 MHz signal for the I-input and a 2 MHz signal for the Q-input. Additionally, instead of connecting the demodulator I-Q output to a ADC, they are instead connected directly to the channels of an oscilloscope. The modulator and demodulator were configured for 2.45 GHz.

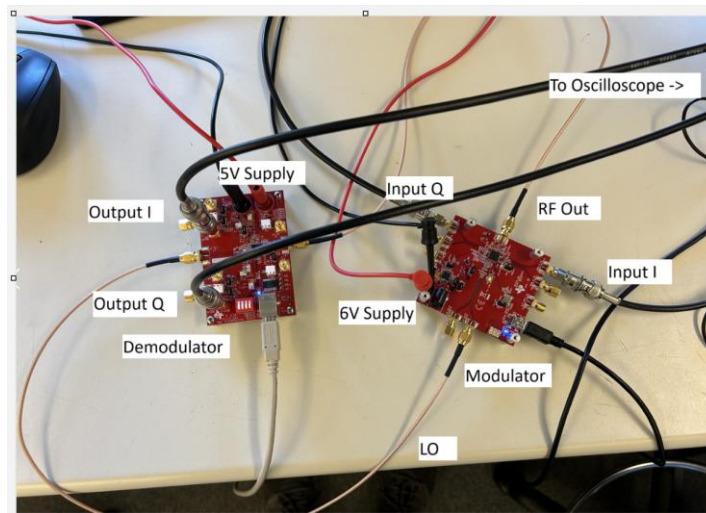


Figure 3.1.2: Modulator and Demodulator Physical Test Setup

Figure 3.1.3 presents the oscilloscope readings for the test setup. Channel 1 (yellow) shows the 1 MHz waveform available at the I-output of the demodulator. Channel 2 (green) shows the 2 MHz waveform available at the Q-output of the demodulator. These results indicate that the modulator and demodulator subsystem will work for our purposes.



Figure 3.1.3: Modulator and Demodulator Physical Test I-Q Data

3.2: Amplifier Selection and Design

3.2.1: Amplifier Selection

Table 3.2.1 presents a selection of amplifiers from Mini Circuits we considered for the LNA alongside the respective goals. For our design, we wanted an LNA that could operate between 2.4 to 2.5 GHz, have at least 10 dB of gain, have a noise figure below 1 dB, have a P1dBm of at least 15 dBm, and have input/output VSWR less than 1.5. After comparing the listed metrics, we decided to go with the CMA-103+ for our low-noise amplifier.

Table 3.2.1: Low Noise Amplifier Selection

Label	Freq. Low (GHz)	Freq. High (GHz)	Gain (dB)	NF (dB)	P1dBm (dBm)	Input VSWR	Output VSWR
Goals	2.4	2.5	10	1	15	1.5	1.5
CMA-103+	0.05	4	11	0.8	23.1	1.38	1.36
PGA-103+	0.05	4	11	0.9	22.5	1.6	1.2
PMA-545+	0.05	6	14.2	0.8	20.3	2.3	1.3
PMA2-123LN5+	0.5	12	15.1	1.2	12.2	1.9	1.3
PSA-545+	0.5	4	14.9	1	20.2	2.1	1.5

Table 3.2.2 presents a selection of power amplifiers from Mini Circuits we considered for the power amplifier alongside the respective goals. For the power amplifier, we were willing to compromise on noise figure (6 dB); however, we wanted a gain greater than 15 dB. After comparing several designs, we decided to go with the PGA-105+ for our power amplifier. Additionally, we decided to add another pre-amplifier stage with the ERA-3+.

Table 3.2.2: Power Amplifier Selection

Label	Freq. Low (GHz)	Freq. High (GHz)	Gain (dB)	NF (dB)	P1dBm (dBm)	Input VSWR	Output VSWR
Goals	2.4	2.5	15	6	20	1.5	1.5
CMA-545+	0.05	6	14.2	0.8	20	2.3	1.5
PGA-105+	0.4	2.6	15.1	1.9	20.5	1.5	1.5
PHA-83W+	0.05	8	15.7	3.3	23.3	1.57	1.28
PSA-0012+	0.05	6	14.2	2.4	22	1.7	1.4
PVGA-123+	0.4	12	15.8	4.3	20.2	1.39	1.34
LEE-049+	0	5	14	5.5	16.4	1.6	1.4
ERA-3+	0	3	18.7	3.5	12.5	1.5	1.4

3.2.2: Custom LNA Design

In addition to utilizing the CMA-103+ from Mini Circuits, our team also wanted to design our own LNA with the same goals. This LNA will be based around a single transistor. Specifically, we will use the SKY65050-372LF, a low cost pHEMT. This transistor was designed for a frequency range from 0.45 to 6.0 GHz with a typical gain of 15.5 dB and a typical noise figure of 0.65 dB. Additionally, the package and resources available on Skyworks' website make this device very friendly to a novice microwave designer.

Figure 3.2.1 presents the recommended application schematic available in the application note. From this schematic, our team derived the bias network, which in addition to the DC blocks and RF chokes, is based largely around resistor R1.

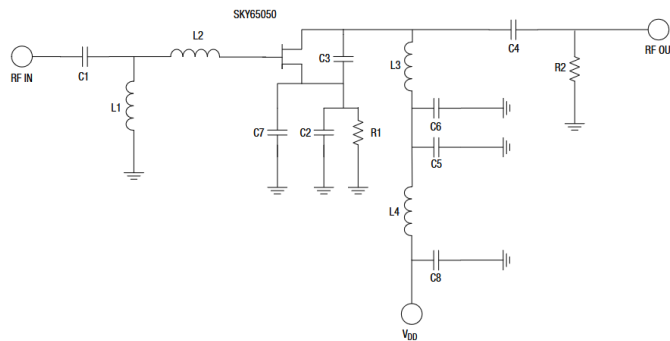


Figure 3.2.1: Recommended Application Schematic

Figure 3.2.2 presents the bias circuit implemented in Keysight ADS along with the building blocks required to plot constant gain, constant noise, and stability circles for the design without a matching network.

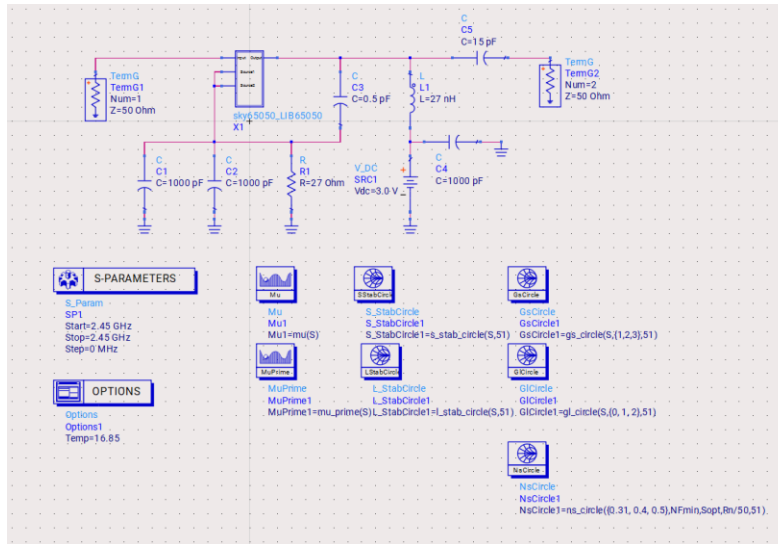


Figure 3.2.2: Bias Circuit in Keysight ADS

Figure 3.2.3 presents the constant gain circles (blue), constant noise figure circles (green), and stability circle (red). Similarly, Figure 3.2.4 presents the output gain and stability circles for the device. Note that for both figures, the stability circles are at least partially within the Smith chart. This means that there are some operating points where the design is not stable. Originally, for our design, we attempted to create matching networks to select operating points on the Smith chart with desired gain and noise figure characteristics. However, after some iterations, we decided to instead design for unconditional stability first.

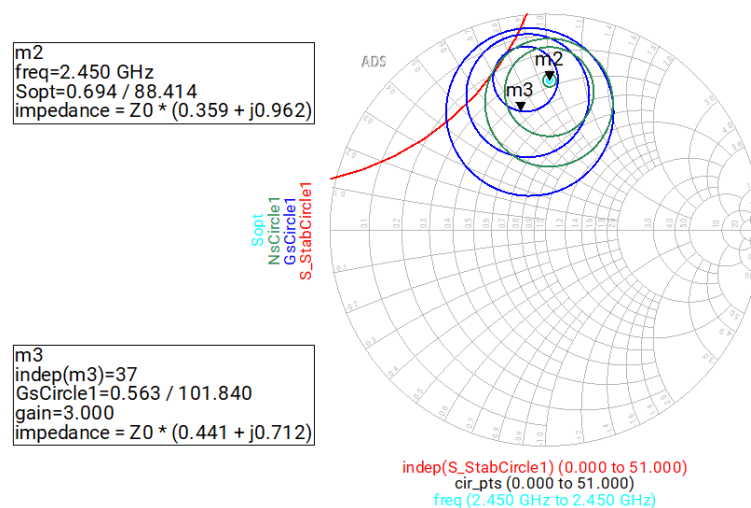


Figure 3.2.3: Input Constant Gain, Constant Noise Figure, and Stability Circles

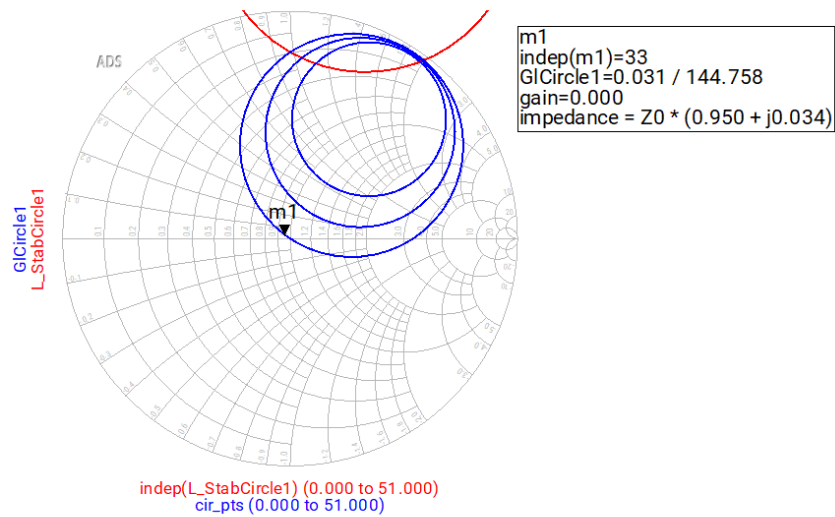
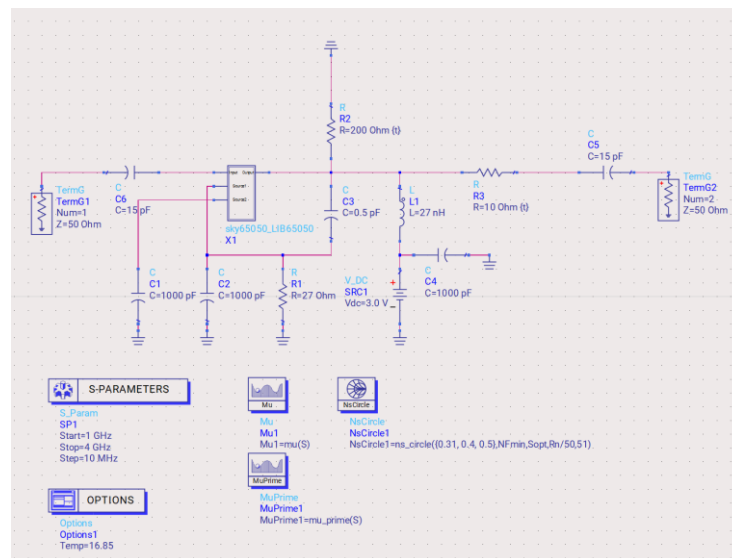


Figure 3.2.4: Output Constant Gain and Stability Circles

Figure 3.2.5 presents the schematic of the unconditionally stable LNA. Note that resistive loading, resistors R2 and R3, contribute to ensuring the stability factor is above 1 for all frequencies within the 2.4 GHz to 2.5 GHz band.



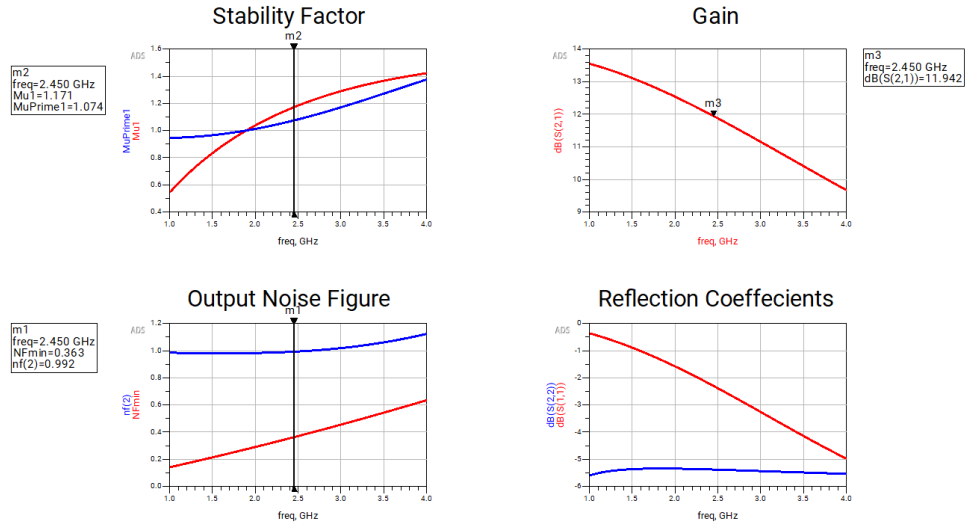


Figure 3.2.6: Unconditional Stability Results

Figure 3.2.7 presents the full LNA schematic with matching networks meant to minimize input and output VSWR, minimize noise figure, and provide decent gain value for the circuit. Additionally, the component values reflect values available for order in 0603 packages from Mouser. Appendix 6.1 elaborates on the LNA bill-of-materials.

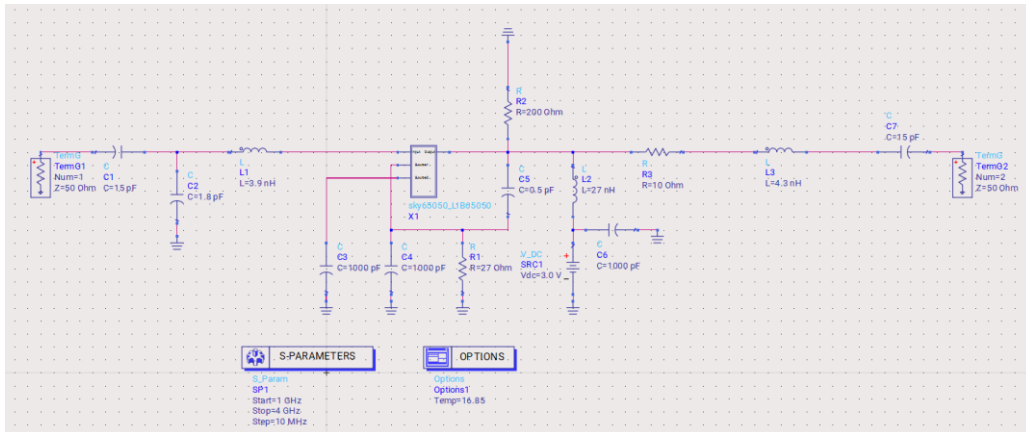


Figure 3.2.7: Full LNA Schematic

Figure 3.2.8 presents the results from the full LNA schematic. At 2.45 GHz, the input and output VSWR values are both below -13 dB. The noise figure is well below our 1.0 dB goal at 0.44 dB. Finally, the gain of the device is 17 dB, which is much greater than our initial goal of 10 dB.

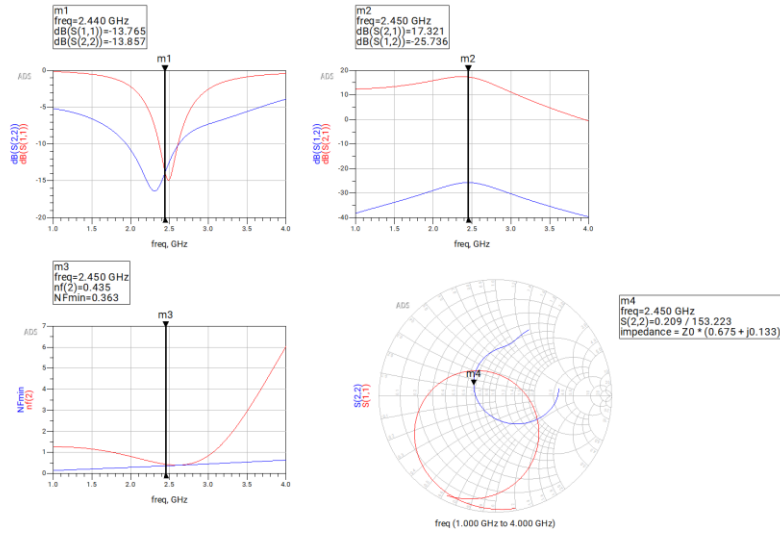


Figure 3.2.8: Full LNA Schematic Results

Figure 3.2.9 presents the full LNA layout. This design uses 0603 packages for passive components. Additionally, this board was designed for the 4-layer Oshpark PCB process.

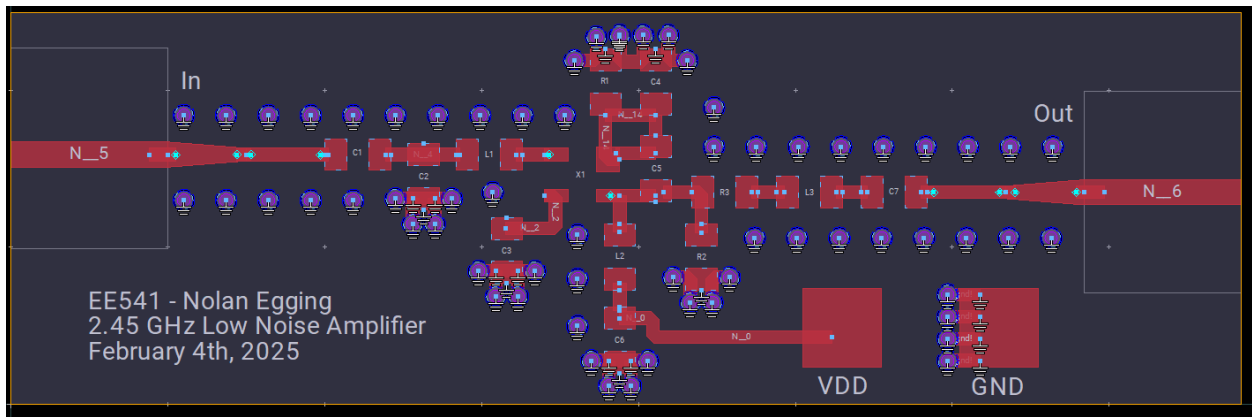


Figure 3.2.9: LNA Layout

Figure 3.2.10 presents the physical LNA constructed during week 9.

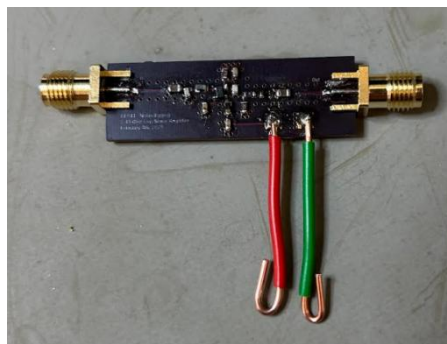


Figure 3.2.10: Physical LNA Construction

3.2.3: Amplifier Characterization

After constructing the custom LNA and receiving the new amplifiers, we characterized each of the amplifiers. Figure 3.2.11 presents a typical measurement setup: the appropriate bias voltage is applied using a power supply, port 1 of the VNA is connected to the input of the amplifier, port 2 of the VNA is connected to a 9dB attenuator which is connected to the output of the amplifier.

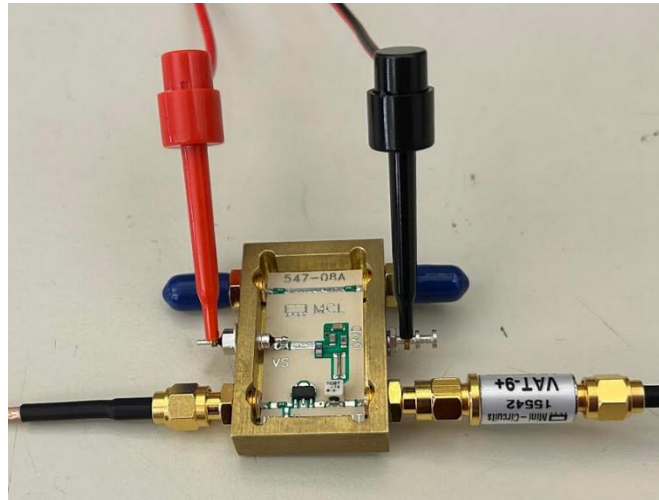


Figure 3.2.11: Typical Amplifier Measurement Physical Setup

Figure 3.2.12, Figure 3.2.13, and Figure 3.2.14 present the S-parameter measurements for the custom LNA. Unfortunately, there is a major discrepancy between simulation and physical results. The overall gain of the amplifier is 4.9dB but design simulations specified as much as 17.3 dB. We hypothesize that this discrepancy is due to the physical distances in the LNA design not being considered which would impact the matching networks and thus the overall amplifier gain. We wanted to experiment with other matching network components; however, our team ran out of time.

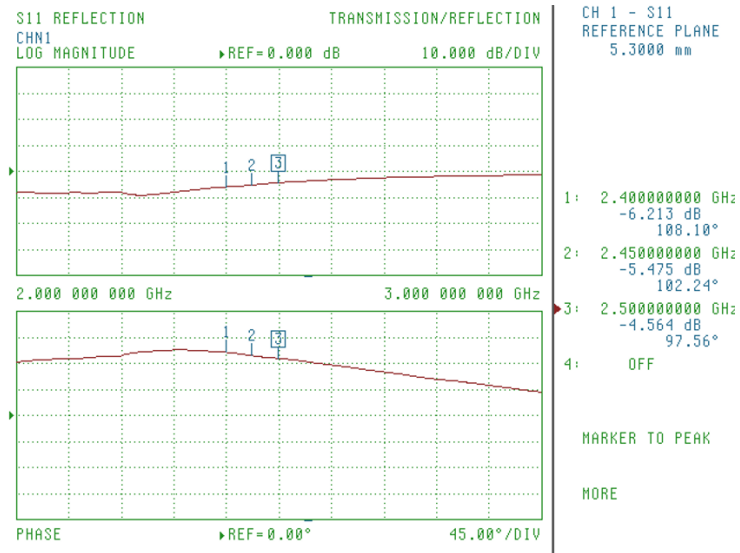


Figure 3.2.12: Custom LNA S_{11}

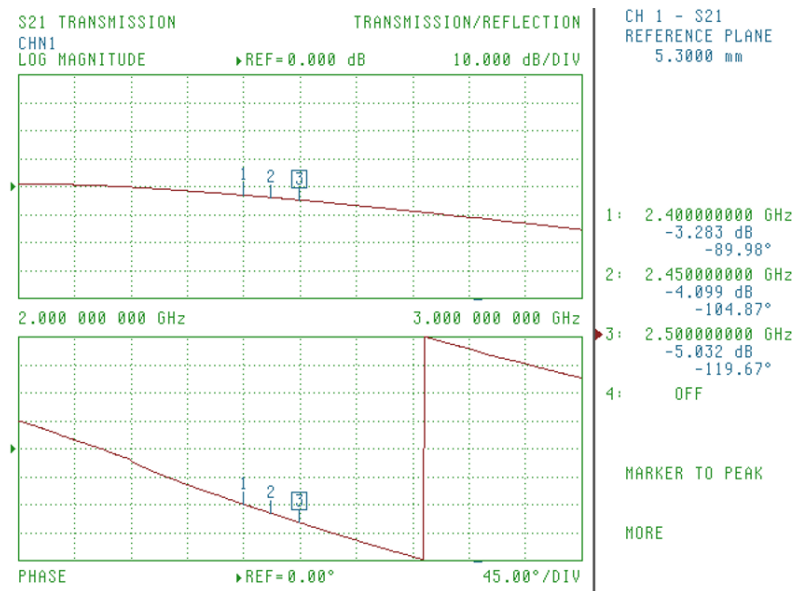


Figure 3.2.13: Custom LNA S_{21}

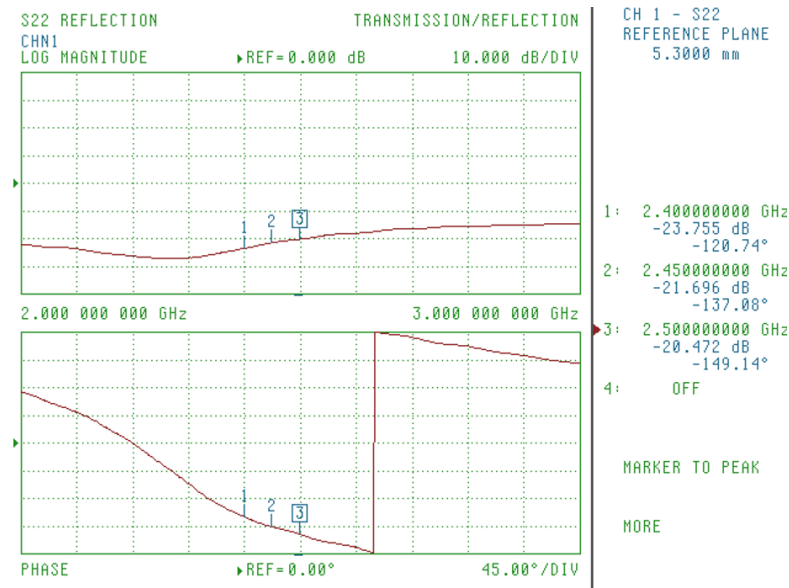


Figure 3.2.14: Custom LNA S_{22}

Table 3.2.1 presents a summary of the characterization of all the amplifiers.

Table 3.2.1: Amplifier Characterizations at 2.45 GHz

Amplifier	Role	VNA Measurements at 2.45 GHz (dB)			Gain (dB)	
		S11	S22	S21	From (S21)	Expected
PGA 105+	PA	-11.5	-27.0	6.1	15.1	15.0
CMA 103+	LNA	-16.2	-25.9	0.2	9.2	10.9
Custom	LNA	-5.5	-21.7	-4.1	4.9	17.3
ERA 3+	Pre-Amp	-14.7	-33.7	3.7	12.7	18.7

3.3: Bandpass Filter Design

3.3.1: Bandpass Filter Design

The requirements for S_{21} are a center frequency of 2.45 GHz, a bandwidth of 100 MHz, and no more than 2dB insertion loss in the passband with a low ripple. Meanwhile, S_{11} requires -10dB or lower in the same bandpass location as S_{21} . In case of design insufficiency or delay order, an order of commercialized filters was purchased [BFCG-252+](#) and [BPF.24.01](#). Keysight ADS was used to create a 6th order hairpin filter to match the stated requirements. An optimizer was used after tuning and multiple iterations of the design.

BPF Optimized Circuit Simulation Response

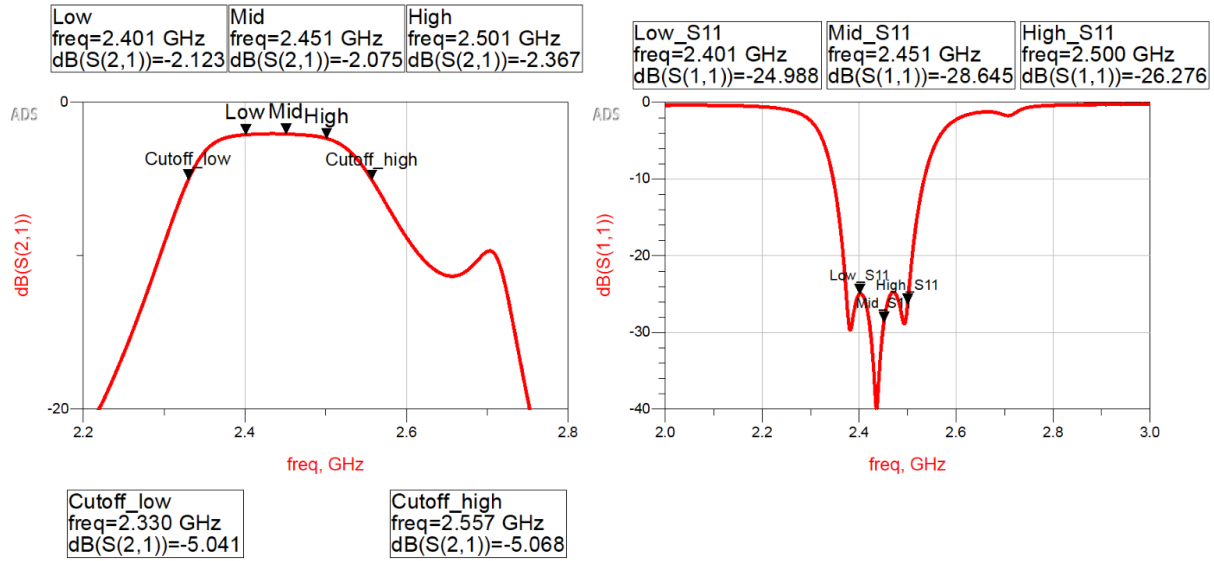


Figure 3.3.1: Results of optimized hairpin bandpass filter

The simulation results are sufficient with the requirements with the highest error being at the passband ripple of S21 with 6.15% error while the S11 are fantastic. The ADS simulation is up to code with the requirements to begin with layout and test with FEM and Momentum.

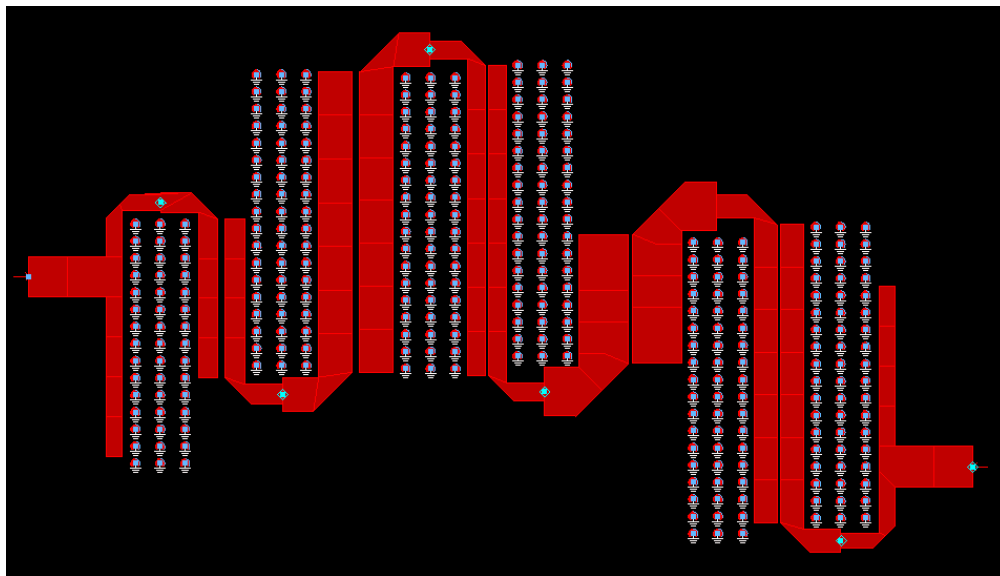


Figure 3.3.4: Layout of Optimized Hairpin Design with vias

BPF Optimized Circuit Simulation Response

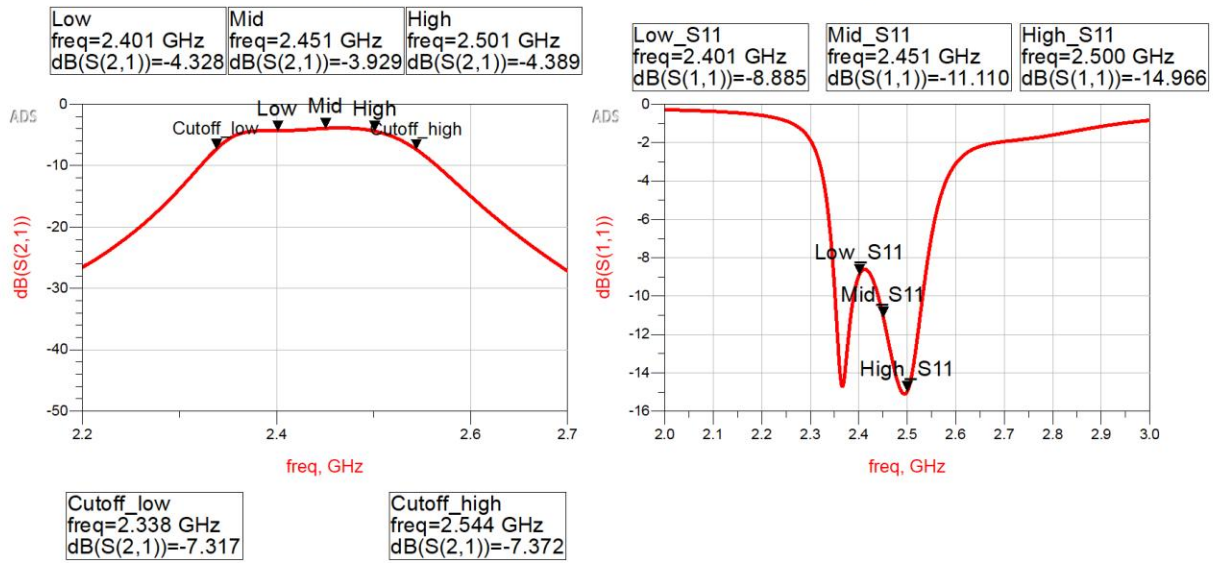


Figure 3.3.5: Optimized Hairpin Design with vias using Momentum

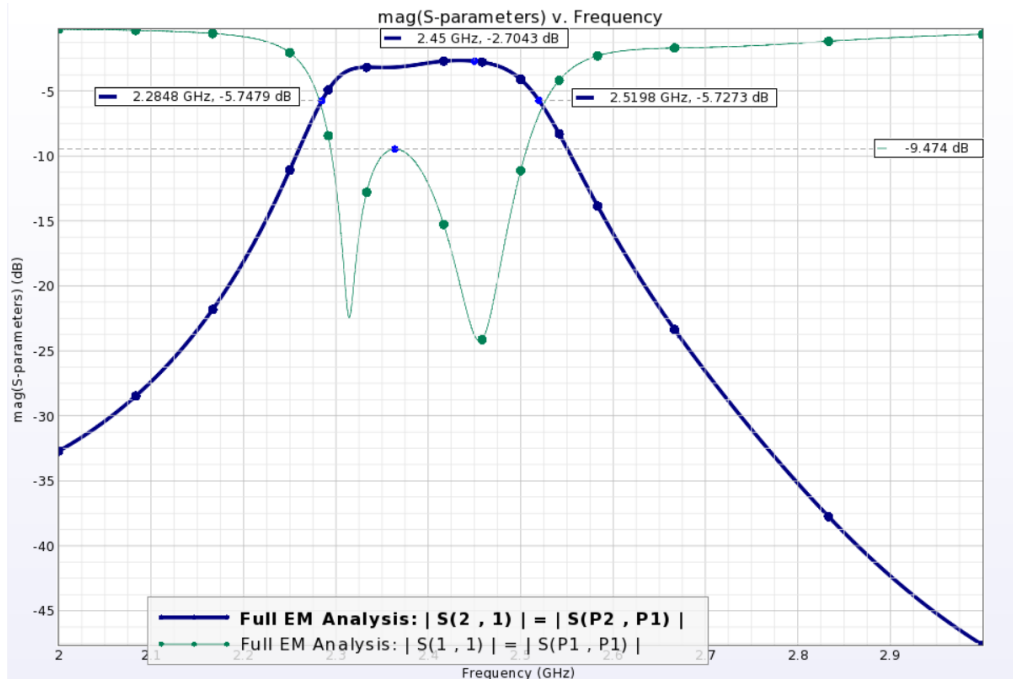


Figure 3.3.6: Optimized Hairpin Design with vias using FEM

By viewing both Momentum and FEM, the simulations are similar in shape but have different frequencies. The bandpass within Momentum is narrower than FEM with 3dB below the passband being located at 2.338GHz and 2.544GHz which is not ideal, but passable with 0.206GHz passband. The FEM is wider and shifted down a few frequencies with 2.2848GHz and 2.5198GHz being the frequencies where the gain is 3dB below the

passband resulting in a 0.235GHz passband. Although the changes in width are large, the gains are also quite different. S_{21} gain is about 1.5dB lower than within the Momentum simulation in comparison to FEM simulation, and the S_{11} gain is higher in Momentum peaking at -8.885dB meanwhile FEM simulation is at -9.474dB . Although both simulations suggest the band widened from the original 0.1GHz by 100%, the experimental results will determine whether this design is sufficient or not.

3.3.2: Bandpass Filter Experimental Results

Based on the design procedures mentioned above, there were two separate harpin filter designs made for this project. This section will focus on the experimental results for both filters, with comparisons to their respective simulations. Both filters were designed to have a center frequency of 2.4 GHz, with a 100 MHz bandwidth.

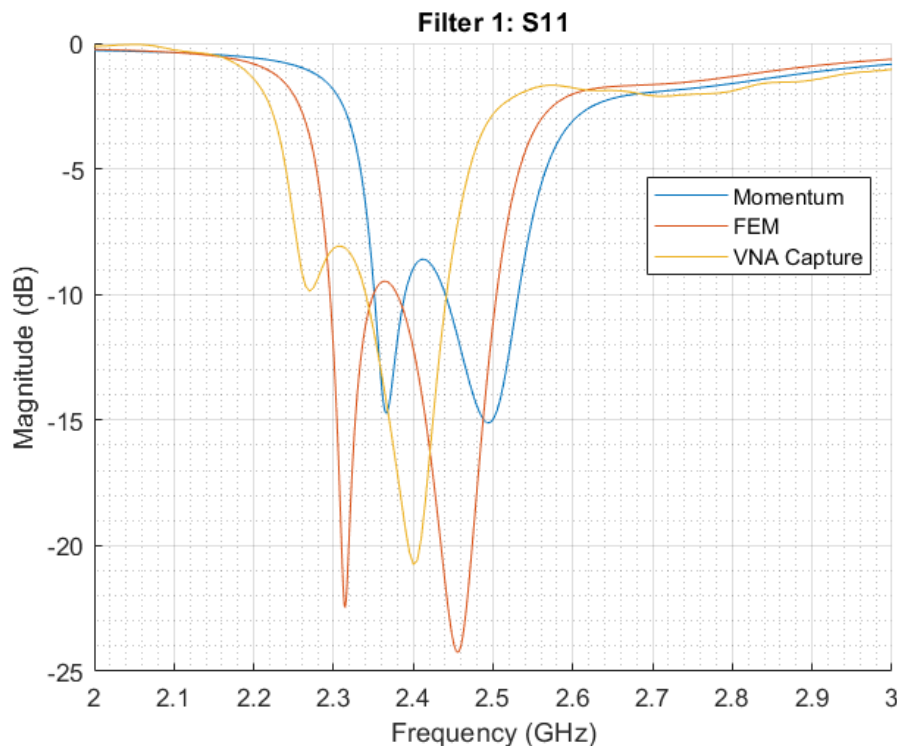


Figure 3.3.7: Filter 1 S_{11} Performance (Experimental vs Simulations)

The center frequency of the experimental model is slightly shifted, and its deepest notch does not occur exactly where the Momentum or FEM simulations predict. Both simulation methods predict the magnitude reaching a minimum below -20 dB (with the FEM curve dipping the most), while the measured notch hovers closer to -18 to -20 dB . As expected, there are return losses due to the conductor and dielectric materials, solder joints, which

prevent achieving the full depth of the ideal simulated notch. For both simulations and the experimental result, the hairpin filters show multiple resonances, and there are at least two distinct dips.

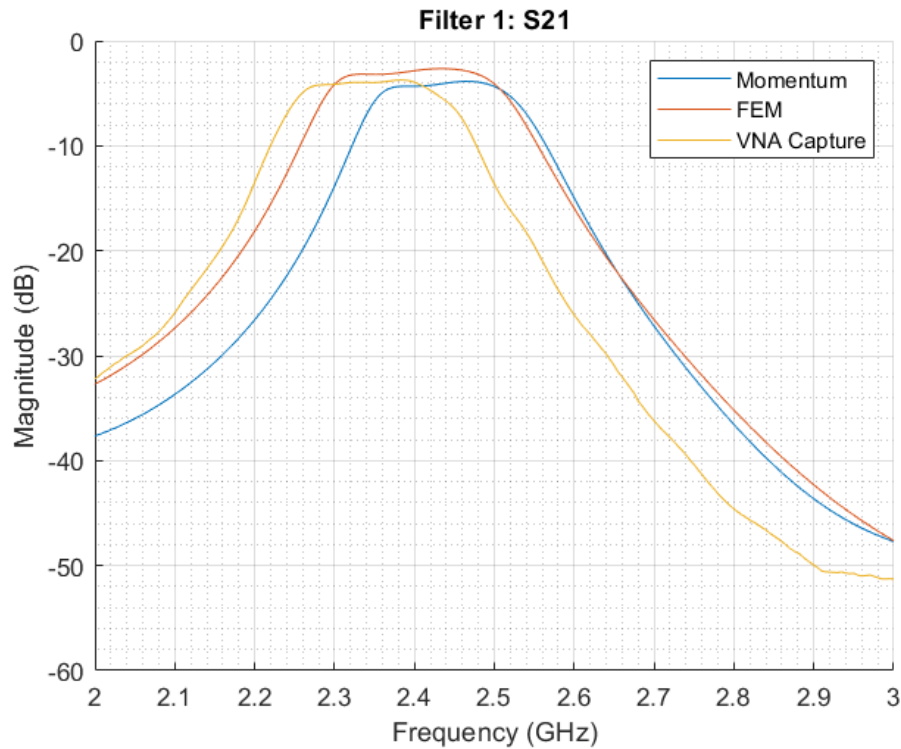


Figure 3.3.8: Filter 1 S_{21} Performance (Experimental vs Simulations)

All three responses peak in the 2.4–2.45 GHz region, as intended. The measured filter's peak is shifted slightly lower in frequency and has a slightly narrower passband compared to both the Momentum and FEM simulations. The measured curve exhibits somewhat higher insertion loss in the passband (around –6 dB to –7 dB) relative to the lower simulated losses (–4 dB to –5 dB). This extra loss typically arises from conductor losses, dielectric losses, connector/solder losses, and other parasitic effects that both simulation models can't fully capture. At frequencies above about 2.7 GHz, the measured filter has a faster roll-off and deeper attenuation than the simulations predict.

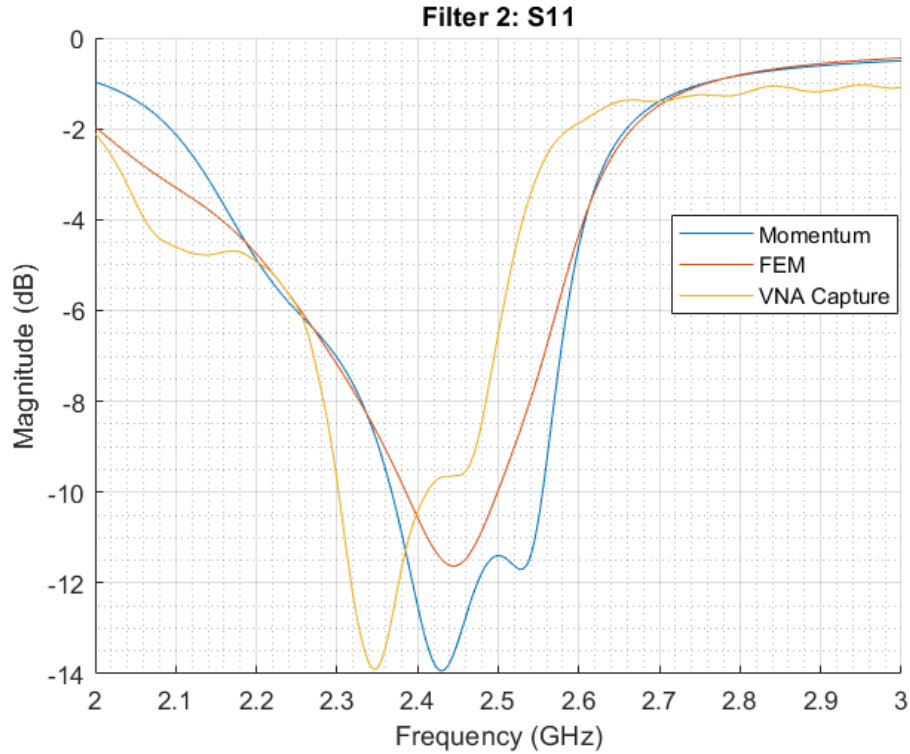


Figure 3.3.9: Filter 2 S₁₁ Performance (Experimental vs Simulations)

The FEM and Momentum models track each other very well between the ranges of 2.2 GHz to 2.35 GHz and 2.6 GHz to 3 GHz. However, for the range in between, the FEM model has a center frequency of 2.45 GHz with a return loss of -11.5dB. The center frequency for the Momentum model shifts downward to about 2.41 GHz with a return loss measured to be -14dB. For the -10dB return loss specification, the passband for the FEM and Momentum models are 3.8 GHz to 2.5 GHz and 3.7 GHz to 2.59 GHz respectively. Although the experimental result and Momentum simulation share a similar shape, the experimental result measured with the VNA is the model with the most deviation. The center frequency significantly shifts downward to around 2.35 GHz. Moreover, the experimental passband is measured to be between 2.3 GHz to 2.41 GHz. Therefore, this filter does act as a bandpass, however, the receiving antenna would need to have a target frequency that is slightly smaller than the 2.45 GHz.

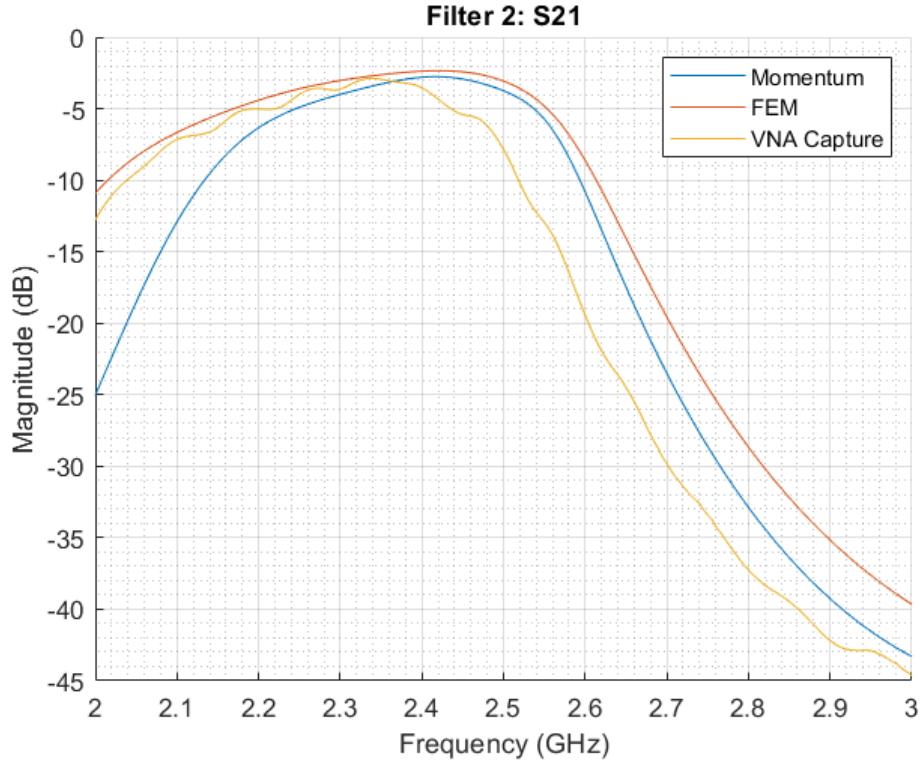


Figure 3.3.10: Filter 2 S_{21} Performance (Experimental vs Simulations)

The simulated insertion loss is less than -5dB for both the FEM and Momentum models at 2.45 GHz. Whereas, for the experimental results, the insertion loss is measured much higher, around -6dB. A higher insertion loss is very common due to conductor and dielectric losses, and imperfect solder connections. Moreover, the roll off slope is noticeably faster, essentially the higher frequencies are more heavily attenuated in the experimental results in comparison to both simulations. Furthermore, the FEM simulation closely tracks the experimental for the lower end of cutoff frequencies. Therefore, these results could indicate inconsistencies in the stack layout of the substrate. Overall, the experimental filter still follows the general bandpass shape, just with a shifted center frequency, and a narrower passband.

3.4: DUT Antenna Design

3.4.1 Single Element Design

The antenna under test needs to be operational at the design frequency, 2.45 GHz, as well as provide sufficient gain in order to transmit the signal across the 6-meter distance. After going through various designs, a patch antenna array was selected with a coaxial feed

being split into 4 using a power splitter. Later renditions may contain a single feed with the signal fed to the four patches using power splitters on the board itself.

The antenna was to be fabricated using OSH Park's 2-layer fabrication, and thus on their FR4 substrate with a dielectric constant of 4.6 and a substrate height of 62 mil. Using Pozar's equations (3.1 – 3.3), the dimensions of the patch were determined for a wavelength at 2.45 GHz or 122.5 mm.

$$W = \frac{\lambda}{2} \sqrt{\frac{2}{\epsilon_r + 1}} \quad (3.1)$$

$$L = \frac{\lambda}{2} - 2\Delta L \quad (3.2)$$

$$\Delta L = 0.412h \frac{(\epsilon_{reff} + 0.3)(W/h + 0.264)}{(\epsilon_{reff} - 0.258)(W/h + 0.8)} \quad (3.3)$$

However, after optimization and tuning, the ideal dimensions for transmission at 2.45 GHz, the patch and substrate dimensions were changed as shown below in Table 3.4.1

Table 3.4.1

Dimensions	Calculated (mm)	Optimized (mm)
W	36.59	49
L	28.93	27.7
W_{sub}	74	100
L_{sub}	58	70

The final single element patch antenna design with dimensions is shown below in Figure 3.4.1.

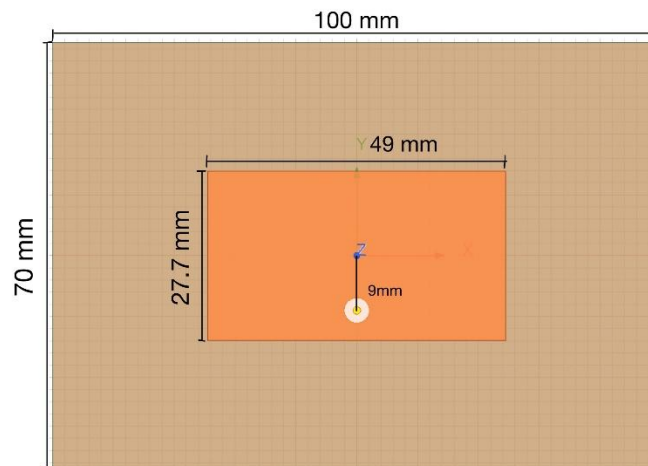


Figure 3.4.1: Single element patch antenna design with dimensions

The main characteristics of the performance of this antenna will be the reflection coefficient and gain across the azimuth angle of the antenna. The results for both measurements for the calculated and optimized dimensions are shown below in Figures 3.4.2, and 3.4.3 respectively.

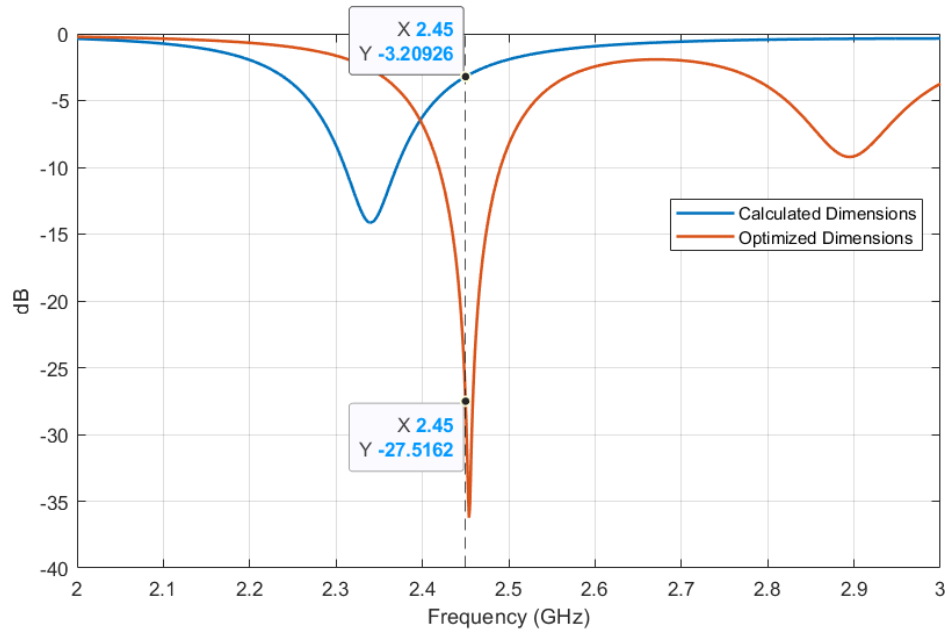


Figure 3.4.2: Reflection coefficient with calculated dimensions and optimized dimensions

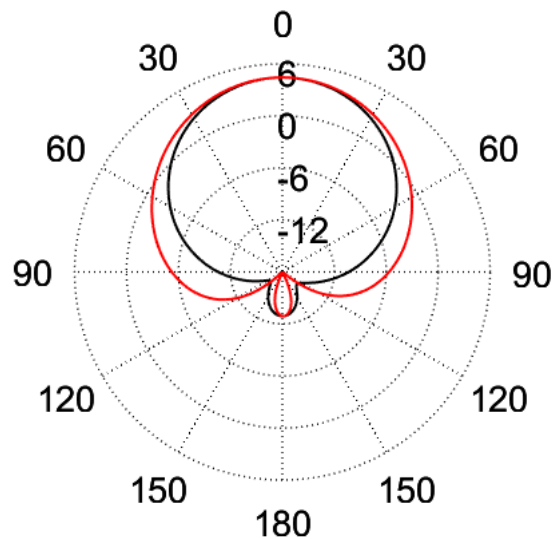


Figure 3.4.3: Gain results with optimized dimensions for $\Phi = 0^\circ$ (black) and $\Phi = 90^\circ$ (red)

Now with the validation of transmission at the design frequency with the optimized dimensions as shown in Figure 3.4.2, any signal fed to the patch at 2.45 GHz will be transmitted and received. Originally the gain with the calculated dimensions was 3.8 dBi, however after optimization the gain has now been increased to a reliable 4.42 dBi. With the results for the single element patch now confirmed operational, the conversion to the array design can begin.

3.4.2 Antenna Array Design

When converting the single element into a patch, several elements need to be considered: the dimensions of the patches may need to be altered along with the x and y dimension spacing between patches and the overall size of the substrate. After several iterations, the size of the patches were found to produce the best transmission with the width of the patch increased to 50 mm from 49 mm. In addition, the spacing between patches lead to the determination of the best coupling for increasing the directivity of the array. With the dimensions shown below, the best results were determined for patch antenna array. Each patch is fed in-phase with its own coaxial cable as shown in Figure 3.4.4.

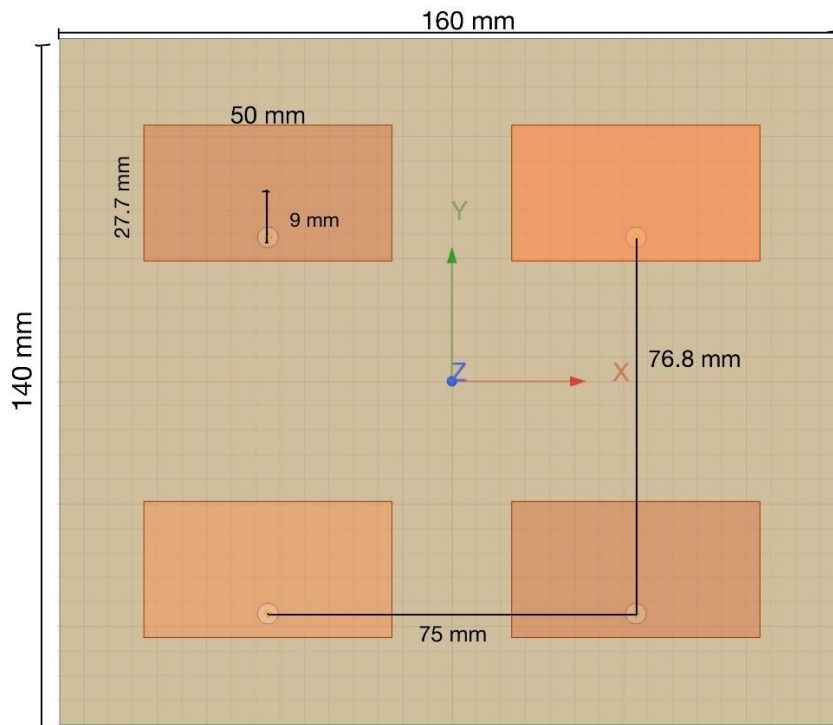


Figure 3.4.4: Microstrip patch antenna array design with dimensions

The results for the patch antenna array are shown in Figures 3.4.5 & 3.4.6.

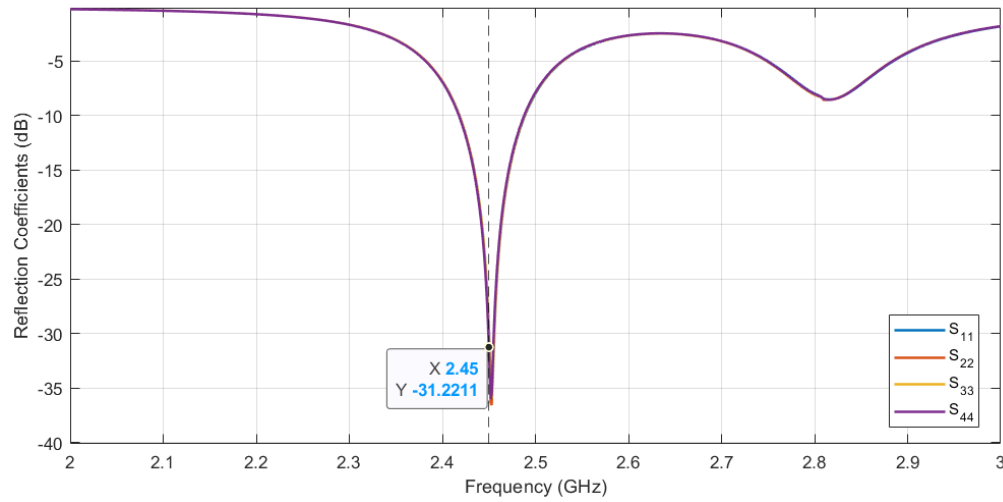


Figure 3.4.5: Reflection coefficient simulation results for each antenna element in the array

As shown with the spike in the results at 2.45 GHz, the patch antenna array works optimally and transmits energy at our design frequency. Each port is matched to 50Ω fairly well and as a result, the gain will be maximized at our design frequency. This is shown in the gain plot versus azimuth angle for both E and H plane measurements as shown below in Figure 3.4.6.

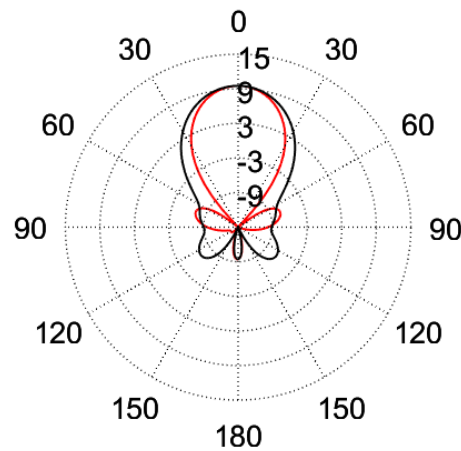


Figure 3.4.6: Simulation array gain versus theta for $\phi = 0^\circ$ (red), 90° (black) cuts

The peak gain for the array was found to be 10.45 dBi. The main lobe of the antenna is directed at boresight as shown with the lobe maximized at $\theta = 0^\circ$. Additionally, the beamwidth of the antenna can be found by determining the sub 3dB points from the peak gain at $\theta = 0^\circ$. The beamwidth determination for both the E and H field cuts is shown in Figure 3.4.7.

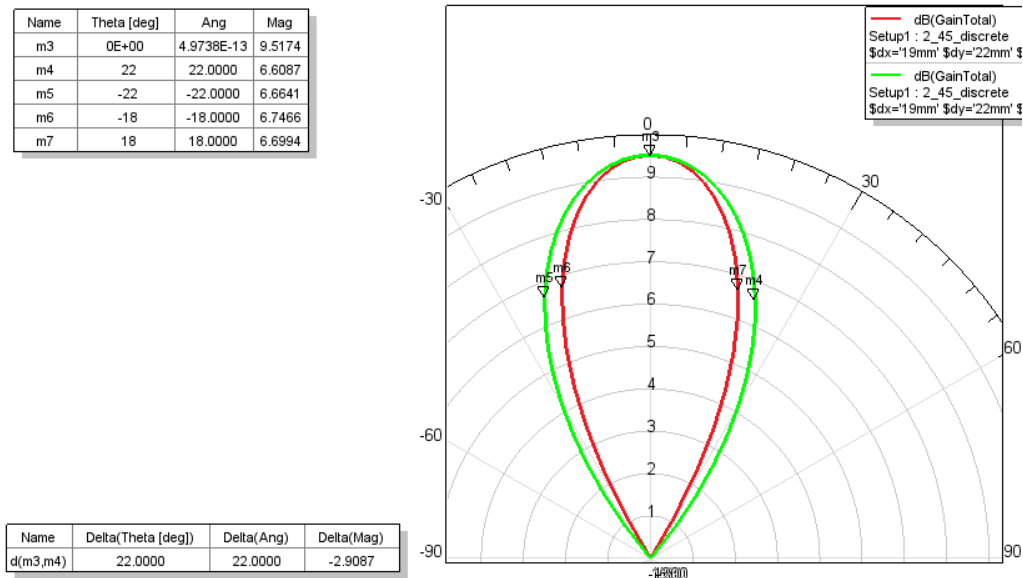


Figure 3.4.7: 3-dB Gain bandwidth of patch antenna array

The single element and array designs were sent to OSH Park for fabrication as shown below in Figure 3.4.8. Once the designs arrive, both the single element antenna and array will have SMA connectors soldered on and measured to verify their performance.

About your board

We detected a 2 layer board of 3.94 x 2.76 inches (100.0 x 70.0mm)
3 boards will cost \$54.25

Board Name*

SingleElementLayout

Email*

Enter your email address

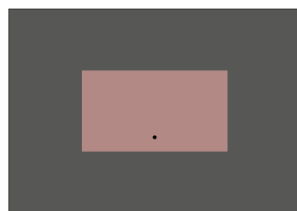
Notes

Processing information

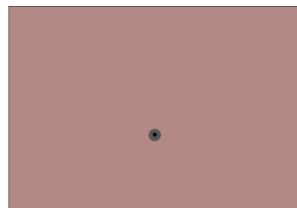
- Processing SingleElementLayout.zip as generic ZIP file.
- 2 layer board of 3.94x2.76 inches.

Warning (non-critical)

- We couldn't find a silkscreen layer. [See help page.](#)



Top



Bottom

About your board

We detected a 2 layer board of 6.30 x 5.51 inches (160.0 x 140.0mm)
3 boards will cost \$173.60

Board Name*

ArrayLayout

Email*

Enter your email address

Notes

Processing information

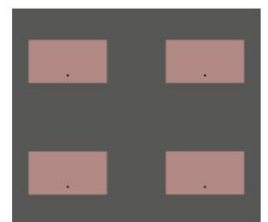
- Processing ArrayLayout.zip as generic ZIP file.
- 2 layer board of 6.30x5.51 inches.
- Correcting drill precision. Please verify drills align correctly. [See help page.](#)

Warning (non-critical)

- We couldn't find a silkscreen layer. [See help page.](#)

Cancel X

Continue



Top



Bottom

Figure 3.4.8 OSH-Park Submissions

3.4.3 DUT Results

With both designs assembled, they were able to be characterized and measured using the VNA and the anechoic chamber located in the microwave lab. First, the reflection coefficient of the single element was measured. This measurement is shown below in Figure 3.4.9.

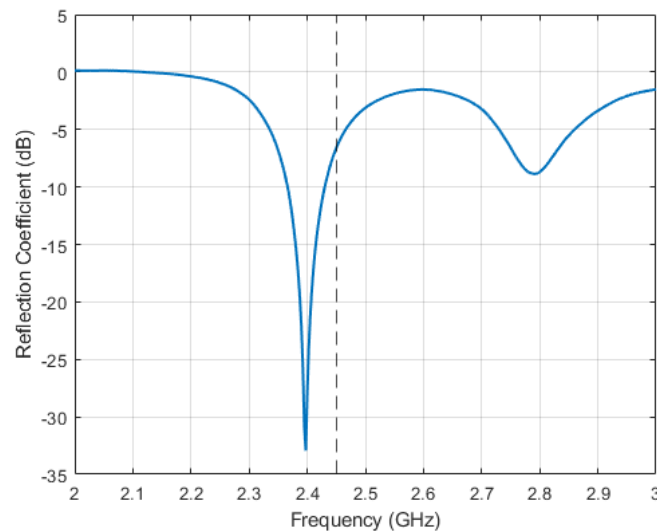


Figure 3.4.9 Measured Reflection Coefficient of Single Patch Element

As shown with the dotted line at 2.45 GHz, the frequency of operation of the antenna has been shifted down towards 2.4 GHz. At our design frequency, there is a reflection coefficient of -6.6 dB, corresponding to a transmission of 78% of the signal. This is not ideal as we would like to see at least -10 dB at our frequency of operation. Upon further analysis of the design and simulation, it was found that the substrate used in the design process of the antenna array was a standard FR4 template with slightly different properties than the substrate which OSH Park uses for fabrication.

With this new information, the substrate used for the antenna design was switched with the parameters used by OSH Park's FR4 substrate with an electrical permittivity of 4.6 and a loss tangent of 0.016. With this new substrate, the previous substrate was compared with the new substrate and the measured antenna as shown below in Figure 3.4.10.

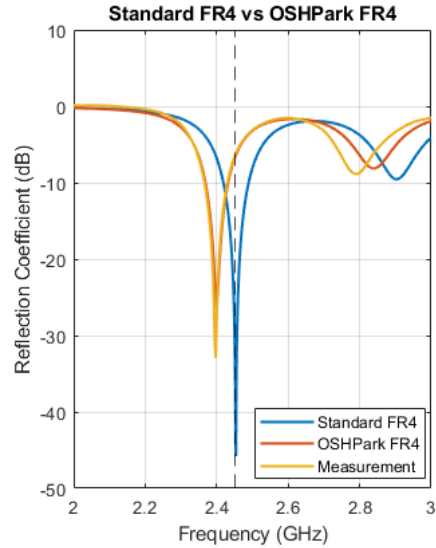


Figure 3.4.10 Comparison of Reflection Coefficient results with different FR4 substrates simulation and measurement

As shown in the plot above, the simulation results with the OSH Park FR4 parameters very closely match the measured patch antenna with a resonance at 2.4 GHz instead of the expected 2.45 GHz. As a result, we should expect this similar type of downshift in results with the array configuration of the antenna. Upon further analysis, with this substrate it was found that to shift the resonance of the patch antenna on this FR4 substate, the length of the patch can be tuned to provide resonance at 2.45 GHz. This analysis is shown below in Figure 15.

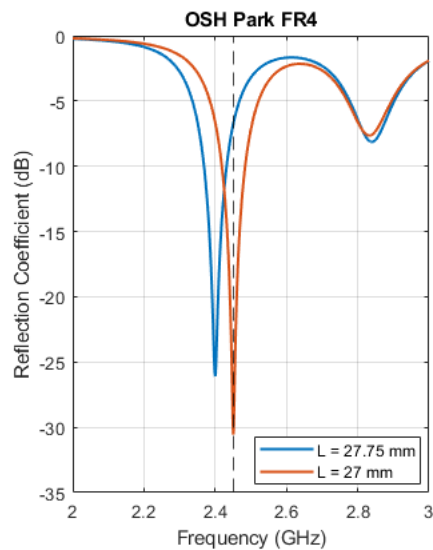


Figure 3.4.11 Comparison of Antenna's Reflection Coefficient on OSH Park's FR4 with different patch lengths

With the previous length of the patch of 27.75 mm, the resonance can be shifted to 2.45 GHz with a slightly smaller length of 27 mm. However, with conjunction of other members of the team, the systems operating frequency was shifted down to 2.4 GHz in order to provide optimal results with each component of the system together.

The next step was to measure and characterize the performance of the array. In order for identical measurement and the beam directed at borefront, each patch needs to be identical and produce transmission equally at 2.4 GHz. As shown below in Figure 3.4.12, the reflection coefficient of each patch individually was measured.

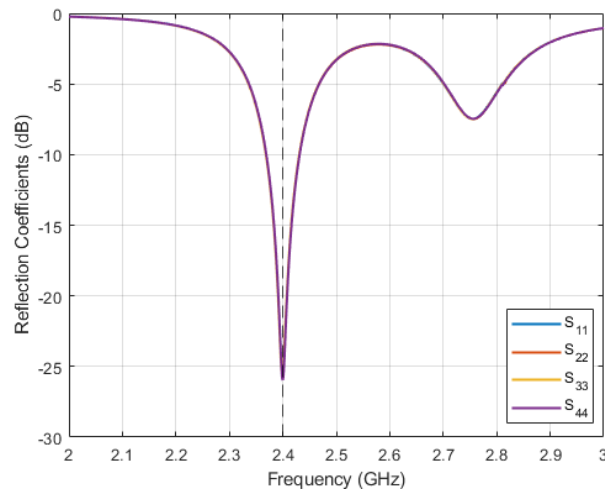


Figure 3.4.12 Reflection coefficient of each individual patch of the array

With each patch verified to be identical in terms of reflection coefficient, the cross talk between patches can be measured to ensure that the radiation pattern integrity is preserved. While beam steering is not a large focus of the application of this array, the cross talk should be minimized to ensure proper impedance matching of each patch and produce the most gain at our design frequency. The cross talk performance of each patch with each was recorded as shown below in Figure 3.4.13.

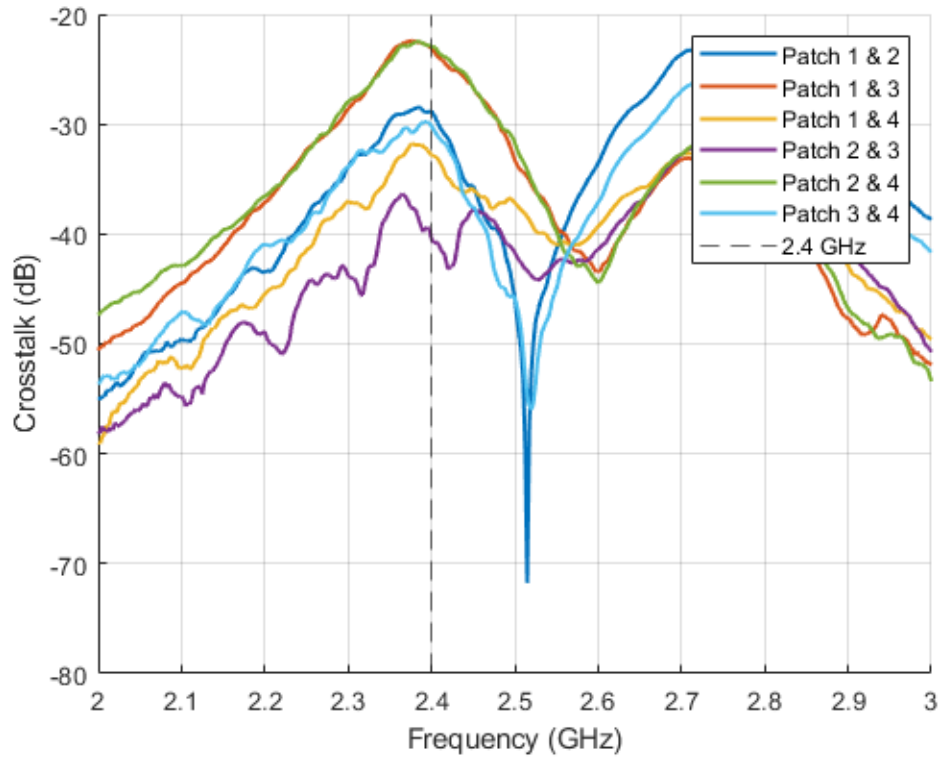


Figure 3.4.13 Cross talk measurement between each patch elements of the array

As shown with the dotted line, the peak cross talk is between the adjacent elements where there is approximately only 25mm between edge to edge of the patches. For these elements, there is approximately -23 dB of cross talk which corresponds to only 0.5% of the input signal being received at the other patch and impacting its performance. This level is within acceptable limits for our design, allowing us to treat the array elements as sufficiently independent for system integration.

With the array verified to be operational at a slightly lower frequency than the intended design of 2.45 GHz, the array can be measured with a 4 to 1 power splitter to take advantage of the array configuration of the patch antenna. The results of the patch array fed with equal length cables, equally excited using a power splitter is shown below in Figure 3.4.13.

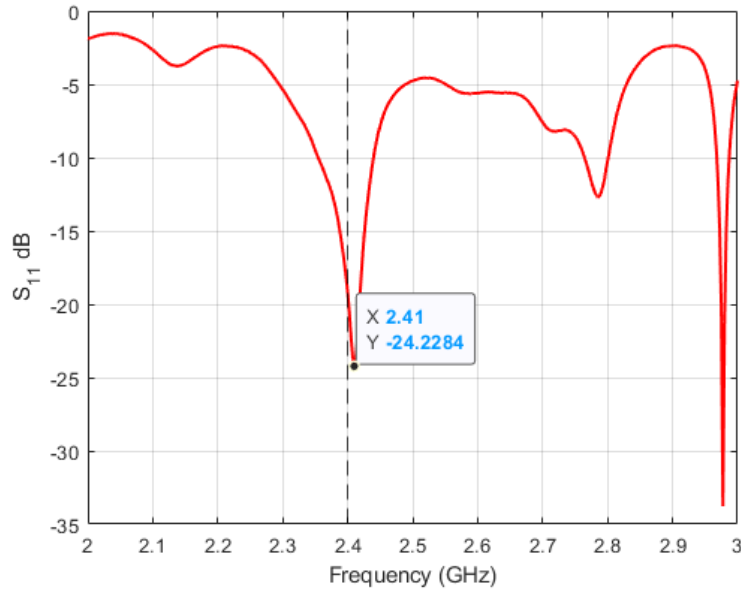


Figure 3.4.13 Measured Reflection Coefficient of Patch Array with power splitter

As shown with the point highlighted, the array transmits the most energy at a slightly up shifted frequency of 2.41 GHz. While expected transmission is at 2.4 GHz, where a reflection coefficient of the antenna system is approximately -18 dB, the system performs optimally at 2.41 GHz. Therefore, the measured together system will be transmitted at 2.41 GHz to ensure precise transmission and reception of the system.

Finally, the radiation pattern performance of the measured antenna with the simulated performance will be compared. As shown below in Figure 3.4.14, the performance of the single patch is compared in simulation (filled line) with the measured result (dotted line) for both $\phi = 0^\circ, 90^\circ$ cuts.

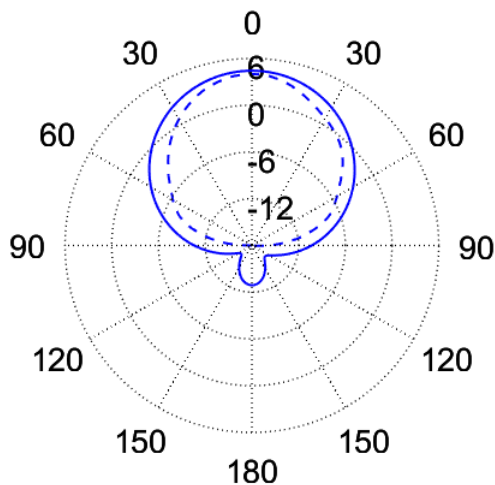


Figure 3.4.14a Gain performance for $\phi = 0^\circ$

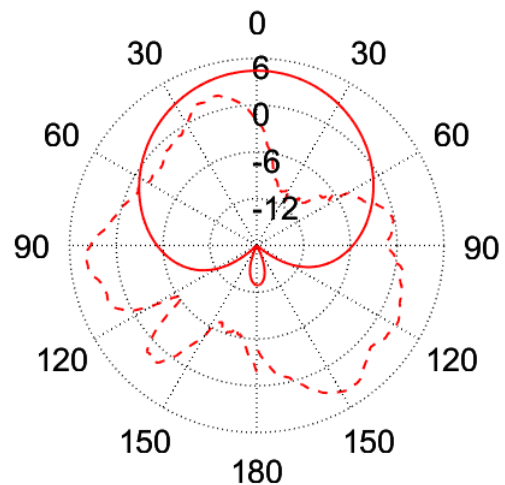


Figure 3.4.14b – Gain performance for $\phi = 90^\circ$

Simulation results (filled line); Measured Results (dotted line)

As shown in the plots, the $\phi = 0^\circ$ performance is quite comparable to the simulation. Note that the measured results are not accurate amplitude wise as the measured gain was not calculated in dBi, rather normalized to 0 dB and added an artificial peak gain of 5 dBi so the radiation patterns are comparable for both plots. The $\phi = 0^\circ$ results match fairly closely for the measurement in the anechoic chamber and with the simulation results, however the H-plane results, $\phi = 90^\circ$ differ quite wildly with the measured result. This is slightly concerning as the performance should not have been degraded this significantly and will have to be explored in future research. Nevertheless, when using the antenna in the correct orientation with $\phi = 0$, the measured patch performance is very similar to simulation and the preferred orientation for the validation of the antenna measurement system.

Finally, the performance of the antenna array will be compared with the single element design. Nevertheless, the use of the results seen in the single patch element will be used in comparison with the experiment. The simulation of the performance of the single element and the array can be compared to compare the beams of both elements and what type of performance is expected when performing the measurement of the entire system. This comparison of simulation results for both the single patch element and array is shown in Figure 3.4.15.

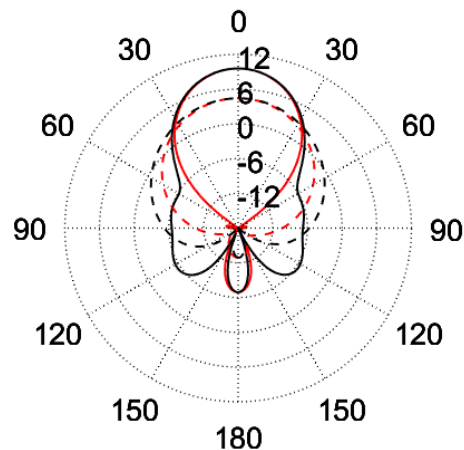


Figure 3.4.15 - Gain simulation performance of single element (dotted) and array (solid) for $\phi = 0^\circ$ (red), and $\phi = 90^\circ$ (black)

As seen in the polar plot, the main lobe is oriented towards bore front for both designs and both Phi cuts, which is to be expected. As shown with the single element results with the dotted lines, there is a very small back lobe which has been increased with the use of the array. The array experiences a significant back lobe as well as some side lobes towards $\pm 120^\circ$. When recording results for the antenna measurement system, these will be

significant enough and need to be investigated to see how the array will receive the transmitted signal when facing away from the transmitting station. The array experiences a peak gain of **9.51** dBi at bore front for both the $\Phi = 0^\circ$, and 90° cuts in simulation. It remains to be seen whether this is the case for the measured result for the fabricated array.

The final comparison will be done with the Array configuration results and comparing the measurement of the array in the anechoic chamber alongside the simulated result. Unfortunately, due to time constraints, the microstrip patch array was not measured in the Anechoic Chamber. This will be done in subsequent weeks following the submission of this report and used to validate the results seen in the antenna measurement system experiment. With the similar results found between the previous design's simulation and measured results, this array should perform as expected and be operational at the transmitting frequency of 2.41 GHz as seen with the reflection results in Figure 3.4.13.

3.4.4 Alternate Antenna Array Design

In addition to the individually fed patch antenna array design, another patch array model was proposed with a single feed splitting out and feeding each patch individually in order to eliminate the need for the power splitter. Both methods of feeding incorporate loss and potential phase mismatches in their designs, however the design of the single feed array will be potentially useful in this application to eliminate the need for the power splitter and to minimize potential issues with needing identical length cables for feeding the previous array.

This design will be compared with the previous design to compare the performance of the two antennas and perhaps make a judgment over the preferred selection of approach when designing patch antenna arrays. In order to compare apples to apples so to say, the dimensions will be kept near the same for both designs to compare the performance at the same cost and size for both designs. With that said, in order to provide resonance at the design frequency of 2.4 GHz, the dimensions of the new patch was forced to be altered slightly. The width remains at 49 mm, however the length of the patch has decreased from 27.75mm to 27.25mm. This difference in length is due to the introduced inset feed which alters the size of the antenna and the way in which the patch radiates energy. The other dimensions of the antenna are shown below on the sketch of the single element design as shown in Figure 3.4.16.

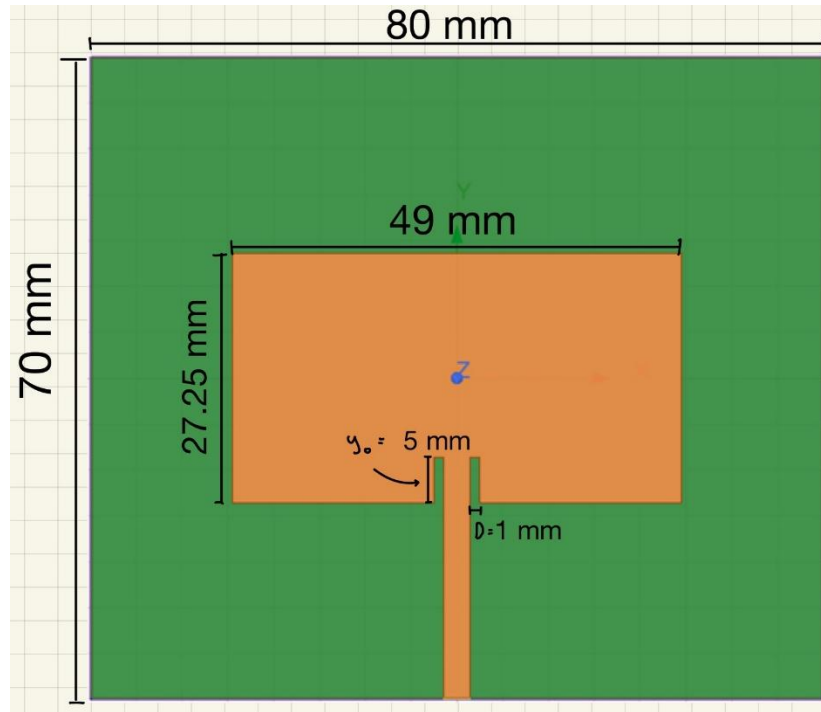


Figure 3.4.16 Single Element Design with inset feed

Note that with the inset feed on the edge of the substrate, the feedline into the antenna must be matched at 50Ω . With the substrate dimensions as provided by OSH Park's fabrication services of $\epsilon_r = 4.6$, $\tan\delta = 0.016$, and 1 oz copper cladding, the width of a 50Ω trace at our intended operating frequency of 2.45 GHz will be approximately 2.89mm. This feed line needs to be fed a certain distance inside the patch where the impedance of the antenna is 50Ω . The distance the trace must feed into the patch can be determined using a few expressions from Balanis' Antenna Theory textbook as shown below. First, the impedance at the outside of the patch must be approximated using the following:

$$R_{in}(y = 0) = 90 \frac{\epsilon_r^2}{\epsilon_r - 1} \left(\frac{W}{L} \right)^2 = 163.6\Omega \quad (3.4)$$

Where W and L are the dimensions of the patch and ϵ_r is 4.6 for the OSH Park FR4 substrate. Using this value, we can input it into another equation from the Balanis book to approximate the length of inset feed using a target input resistance of 50Ω a distance of y_0 into the patch.

$$y_0 = \frac{L}{\pi} \cos^{-1} \left(\sqrt{\frac{R_{in}(y=y_0)}{R_{in}(y=0)}} \right) = 8.5 \text{ mm} \quad (3.5)$$

The separation from the inset feed and the rest of the patch do impact the impedance and performance of the patch due to inductive coupling and fringing fields, however it will be set to a standard 1 mm which is used quite frequently in most other designs. Using this value for the inset length as well as the previous length of patch and the adjusted one, the following reflection coefficient results were found using both the calculated dimensions and optimized as shown in Figure 3.4.17.

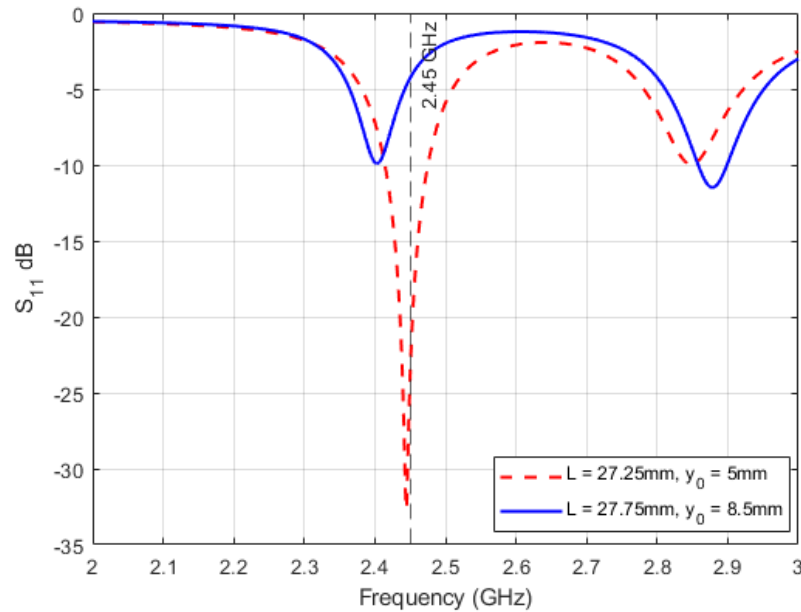


Figure 3.4.17 Inset fed patch reflection coefficient simulation results for calculated vs optimized dimensions

As shown with the new dimensions for the patch with the red dotted trace, there will be transmission at our intended frequency of 2.45 GHz, and thus these will be the dimensions which will be continued further in the design of the array.

The next step in designing the array will be creating a 2x1 array, with both patches needing to be fed equally in phase from a single SMA. In order to do this, a T junction with varying impedances will need to be created to ensure. If the junction feeding each inset feed is just 50Ω , the effective impedance seen looking into the junction would be due to the lines being considered in parallel, causing reflection and poor power transfer. To maximize the efficiency of the antenna, the junction will be designed to transfer half of the input power to each patch and minimize any reflections at the junction. Therefore, the SMA will feed a 50Ω line which will split into smaller 100Ω lines to transfer maximum power.

With these 100Ω lines, they will need to be transformed back into 50Ω to feed the antenna at 50Ω . In order to do this, a quarter wave transformer will be used which will have a

quarter wavelength line at 2.45 GHz and a impedance of the geometric mean of the input and output lines. Therefore, the impedance for this line will be 70.7Ω . The dimensions for each line on the OSH Park substrate are shown below in the table. These calculations were completed using LineCalc and the equations found in Pozar's textbook.

Table 3.1

Impedance	Name	Width
50Ω	Z1	2.89 mm
70.7Ω	Z2	1.5 mm
100Ω	Z3	0.622 mm

The sketch of the 2x1 array with the dimensions displayed in the table is shown below in Figure 3.4.18. Note that in order to produce transmission at 2.45 GHz, the dimension of the patch was optimized further to decrease the length to 27.15 mm and the inset feed length was altered to 2.5 mm. Each lengths of the feed trace were arbitrary to fit onto the board, other than the quarter wave transformer which was a quarter wavelength at our design frequency using effective permittivity of the substrate, and thus $L_{\lambda/4} = 16.95 \text{ mm}$

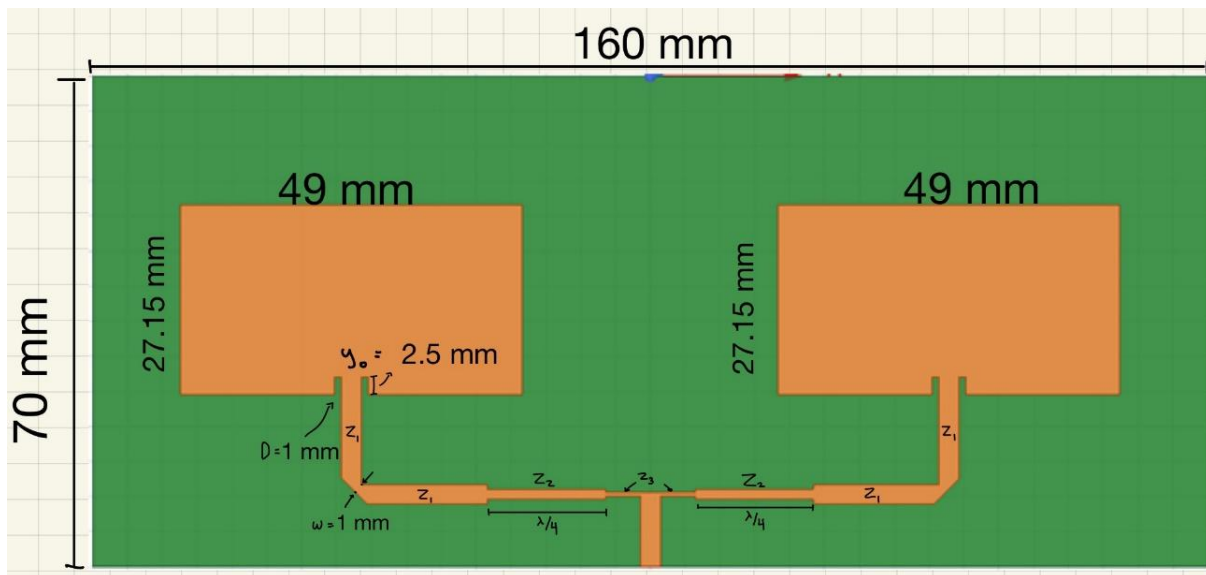


Figure 3.4.18 Sketch of the 2x1 array with dimensions labeled

Another element of the design of this new array is the introduction of the 90° bends in the traces. Ninety-degree bends are not ideal for RF design as they change the characteristic impedance and increase capacitance of the line leading to reflections and increasing return loss at the bends. Therefore, in order to maximize power transfer, the mitered bend will be used. A sketch of this bend is shown below where the triangular section is removed at the corner to improve signal integrity in Figure 3.4.19.

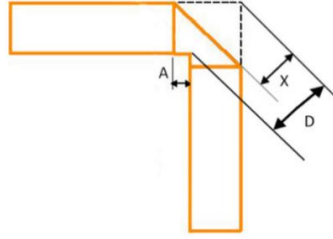


Figure 3.4.19 Mitered bend sketch

To implement, the values of X, A, and D need to be determined. These expressions are shown below for a $Z_0 = 50$ line of width of 2.89mm with a substrate height of 62mils.

$$D = Z_0\sqrt{2} = 4.1 \text{ mm} \quad (3.6)$$

$$X = D * \left(0.52 + 0.65e^{-\frac{1.35*D}{h}}\right) = 2.21 \text{ mm} \quad (3.7)$$

$$A = X\sqrt{2} - Z_0 = 0.23\text{mm} \quad (3.8)$$

Upon implementing this, the power was not found to be maximized with these values, and they were optimized to produce transmission at our design frequency of 2.45 GHz. Upon optimizing this bend and slightly altering the dimensions of the patch, a reflection coefficient for the 2x1 array was realized as shown below in Figure 3.4.20. This new design features a slightly shorter patch with a length of 27.15 mm with similar dimensions for the substrate of the previously designed 2x2 array.

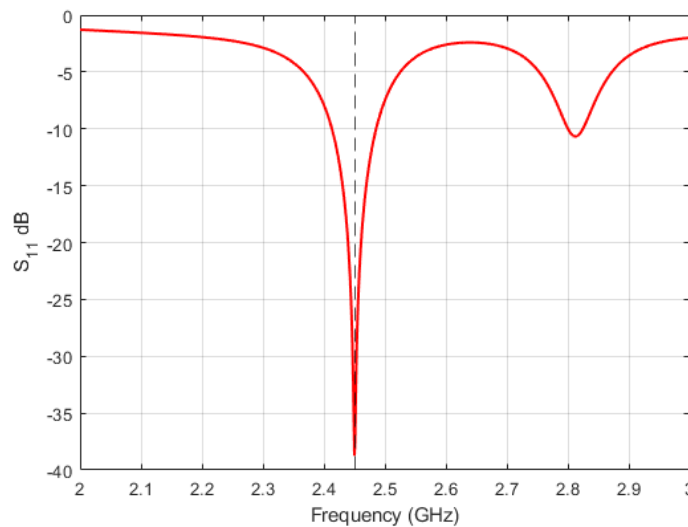


Figure 3.4.20 Reflection coefficient simulation results for 2x1 array

With the 2x1 array verified to be operational at our design frequency, the last step in the process is to design the 2x2 array. While the previous single and double element designs

utilized a single SMA connector at the end of the substrate fed with a 50Ω line, the 2×2 array will need to be altered in order to feed each antenna equally in phase and magnitude. To do this, a back fed SMA connector will be soldered on and fed with several T junctions used to separate the signal and feed each of the 4 patches equally.

A sketch of this is shown below in Figure 3.4.21. The SMA connector is the white circular port seen near the center of the design with equal length feeds feeding each patch,

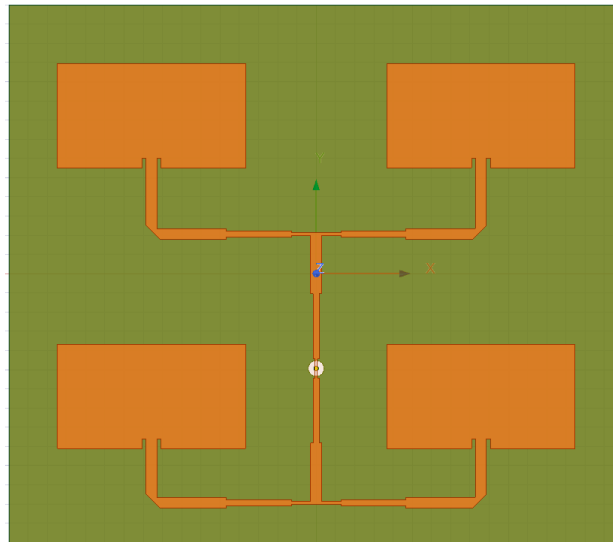


Figure 3.4.21 Sketch of the 2×2 array

To calculate the dimensions of each trace, a similar design process was conducted. First, starting from the SMA connector, a 100Ω line is excited by the SMA which splits in two directions, thus needs to be half of the characteristic impedance. Both traces in both directions are then transformed using quarter wave transformers into 50Ω lines to feed into the previously designed T junctions which require a 50Ω input. At both junctions, the 50Ω traces are then split into two 100Ω traces and are then subsequently transformed using the same quarter wave transformers into the 50Ω feed lines which excite each of the 4 patches.

The widths for the 50Ω , 70.7Ω transformer and 100Ω traces are the same as the 2×1 design and are shown in the previous table. Similarly, the length of the quarter wavelength transformer remains the same at 16.95mm . A sketch with labeled dimensions is shown below in Figure 3.4.22.

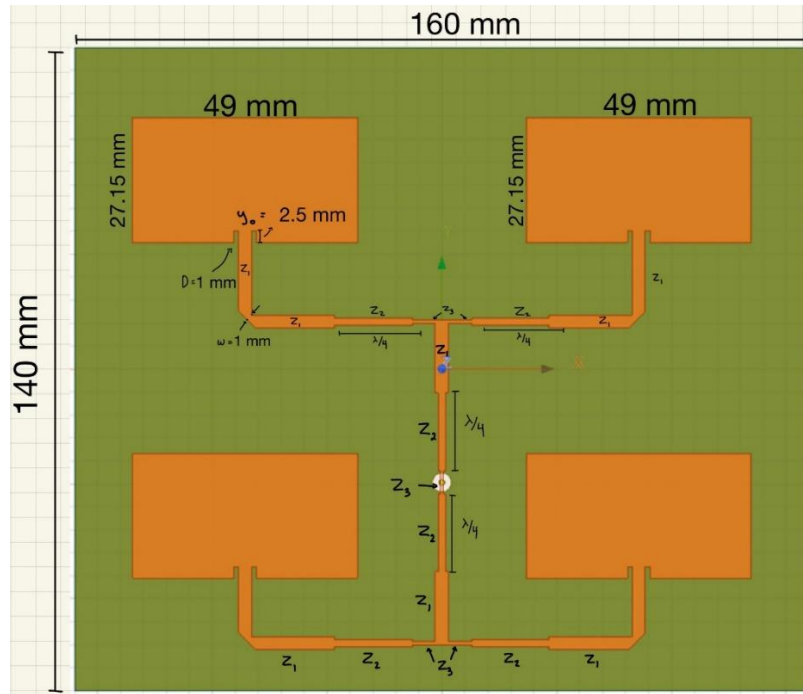


Figure 3.4.22 Sketch of the 2x2 array with dimensions

With this designed 2x2 array for transmission at 2.45 GHz, the reflection coefficient needs to be measured. This result is shown below in Figure 3.4.23 in comparison with the previous single element and 2x1 design.

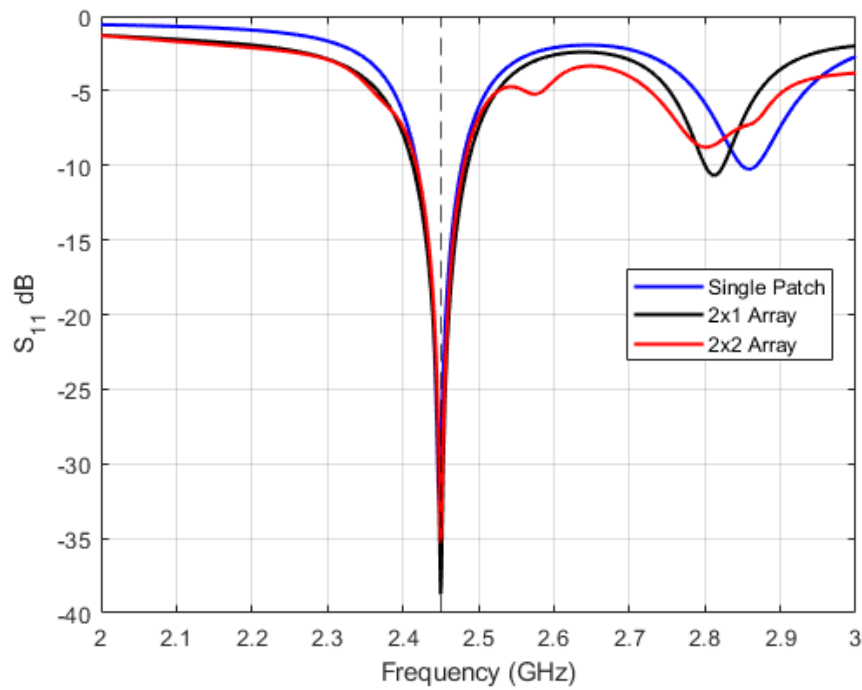


Figure 3.4.23 Reflection coefficient simulation results for each iteration of the alternate antenna design

While there is not noticeable improvement in performance for the 2x2 array with the smaller array and single element, the signal will be transmitted at our design frequency of 2.45 GHz for each variation of this patch antenna. In order to compare the performance of each design, the simulated gain performance of each will be compared at 2.45 GHz as shown below in Figure 3.4.24. Notice the improvement of the transmission of the antenna at borefront for each version.

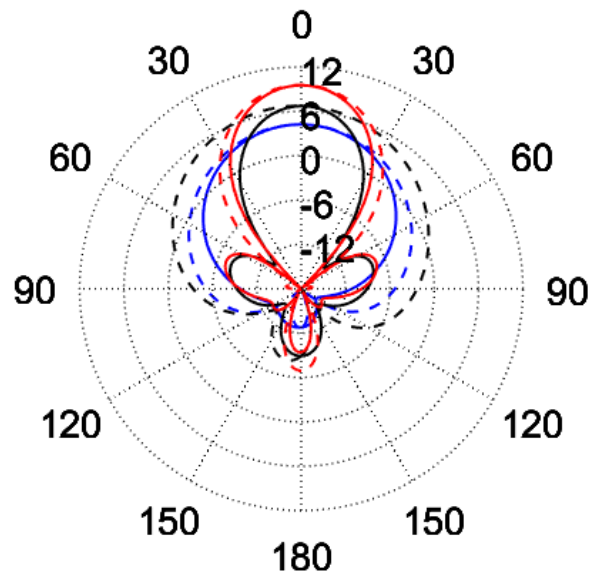
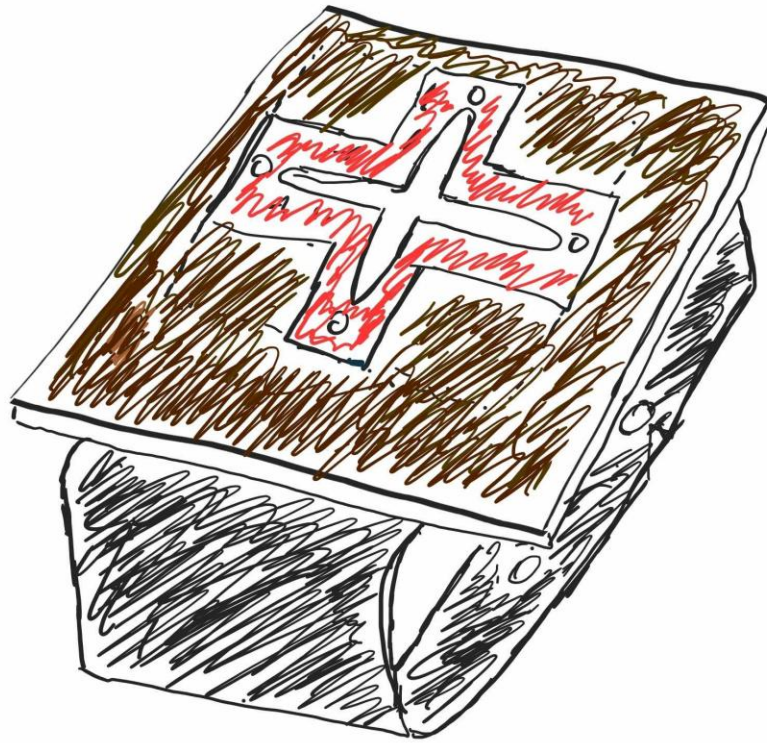


Figure 3.4.24 – Simulated gain performance for both $\Phi = 0^\circ$ and 90° cuts for **Single Patch**, **2x1 Array**, and **2x2 Array**

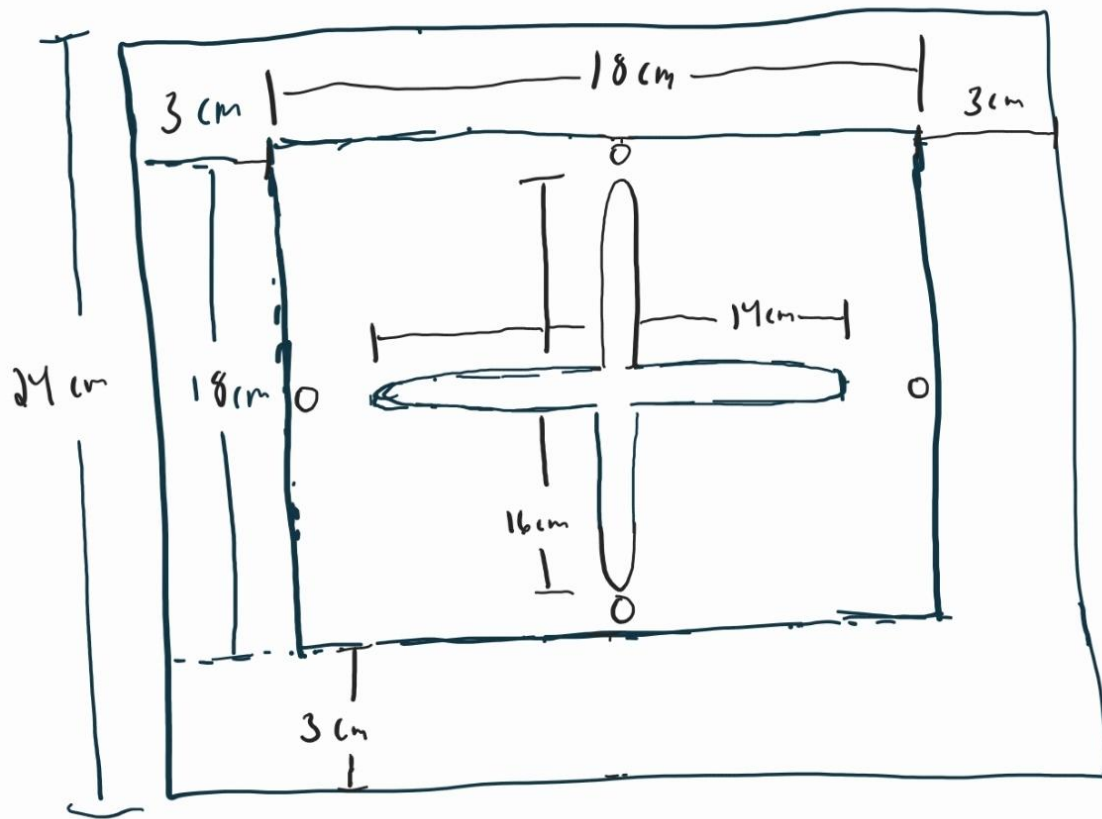
3.5: Antenna Stand Design

3.5.1: Initial Stand Antenna Design



3.5.1 Concept design of the Patch Array Antenna Attachment

Figure 3.5.1 is the original graphic representation of the Antenna attachment for the rotator built from last quarter. The idea was to have a 24x24x3 cm box made of plywood supported by a 3D printed attachment to secure the antenna to the rotator. Moreover, the inside of this plywood box will be hollow to insulate the power splitter from the antenna system.

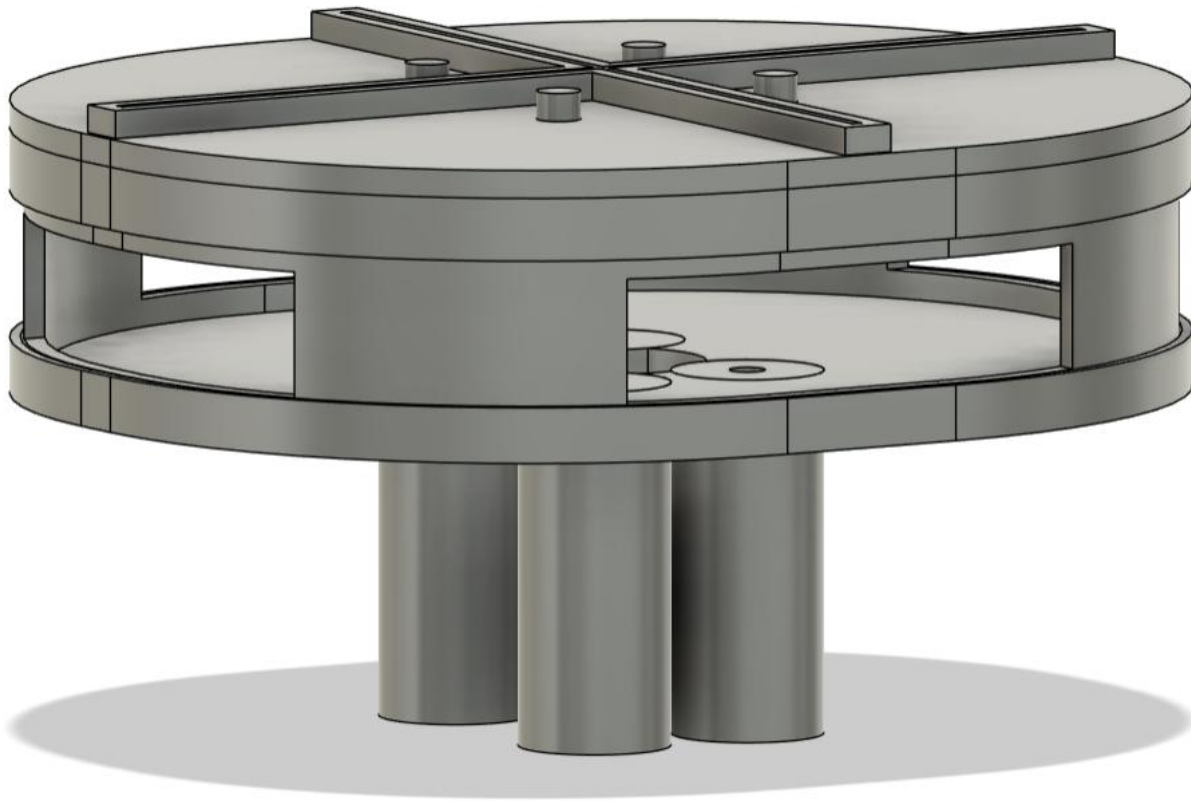


3.5.2 Top overview of the Antenna Stand Attachment

Figure 3.5.2 is the initial rough draft of the design at the top of the attachment. The idea here is to have an 18x18 cm tile with a 14 cm slit horizontally and vertical 16cm slit. The tile itself can be rotated via, unscrewing the sides and manually rotating the tile, which is why the screw holes are equidistant. These slits allow for the patch array to be mounted to the stand in 2 different configurations, one to measure the E field, and the other to measure the H field.

The initial design had multiple critical flaws where if manufactured would have resulted in the design to not function within the system. The first flaw is having only two holes to secure the attachment to the four screws on the rotator. The second flaw is the lack of guard rails preventing the power splitter from sliding out while the rotator spins 360 degrees. Lastly, the cross attachment requiring screws could interfere with EM waves and the location of said screws do not allow for multiple different antenna attachments. All these flaws were addressed with the final design.

3.5.2: Final Stand Antenna Design



3.5.3 Finalized design of the Patch Array Antenna Attachment

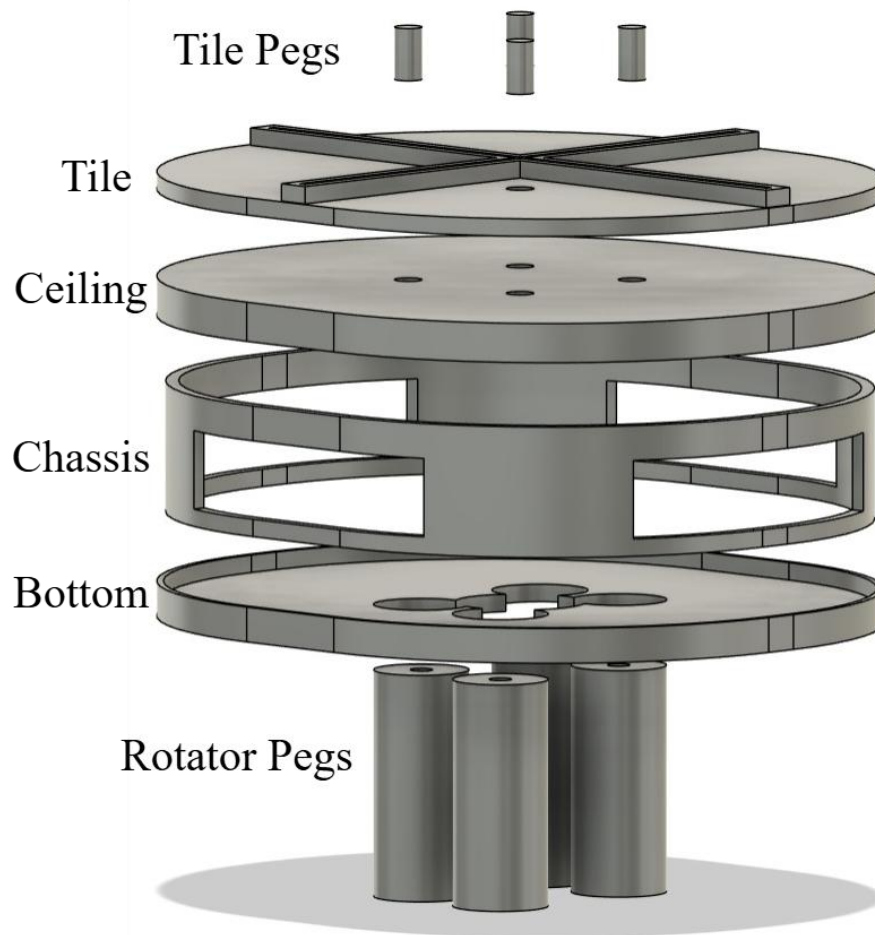
In Figure 3.5.3 is the final design for the entire rotator to antenna attachment. The attachment is held together through friction resulting in no reason to attach hold the attachment with screws reducing the interference and additional tools necessary. The gaping holes within the chassis are meant for the power splitter allowing to be taken out with ease. Additionally, those holes allow for a cable to run through the other side in case the cable is too long.

There are many positives with this final design, where the main one being the modularity of the overall design. If a piece is insufficient for a new design, simply replace it while keeping the rest of the pieces. Modularity is a necessity for this project considering there is a patch array and a singular patch antenna to test thus a different tile needs to be used.

Additionally, if no power splitter is necessary, the whole chassis can be removed and replaced with a thin connector piece between the ceiling and bottom. The second biggest advantage is the chosen material was being PLA plastic thus 3D printing could be used. 3D printing is easier than using plywood and allows for customizability for each piece.

Additionally, this design is quite light considering the infill for each piece was set at 15%

thus resulting in the total weight of the system being less than 500 grams. Lastly, this design addresses all previous flaws as well, such as the need for screws is now removed due to friction-based design. Additionally, there are guardrails and a ceiling in place for the power splitter thus securing it during rotation. Lastly, there are four pegs allowing for a secure attachment to the rotator. Yet this design has some flaws which were only seen in person thus are required to be discussed individually.



3.5.4 Finalized design attachment separated into individual pieces

Figure 3.5.4 shows the naming convention that will be used for this section while showing how each part is connected though either through connecting pegs or by a lip mechanism holding them in place.

There are four main flaws in the final design, which could be easily fixed considering the modular design approached here. The first apparent flaw is the bottom piece did not snugly fit with the chassis due to the bottom lip being too thin resulting in a large gap

between the chassis and the bottom lip. The second flaw is another friction issue with the rotator pegs being much smaller than the holes resulting in them slipping through. Another flaw with the pegs where the holes through them connecting to the screws are smaller than the screws themselves resulting in the pegs not fitting around the screws. Lastly, the tile is designed poorly considering the amount of time provided to rotate the antenna to the other slit.

An additional flaw is how the measurements were chosen, considering they were quite arbitrary. The tile pegs' radius is 3.2mm, the size of compartment between the ceiling and bottom was chosen based on a theoretical model of the divider, and the bottom pegs could have been smaller. This goes against the spirit of a modular design, but all parts followed the bottom shape being an oval.

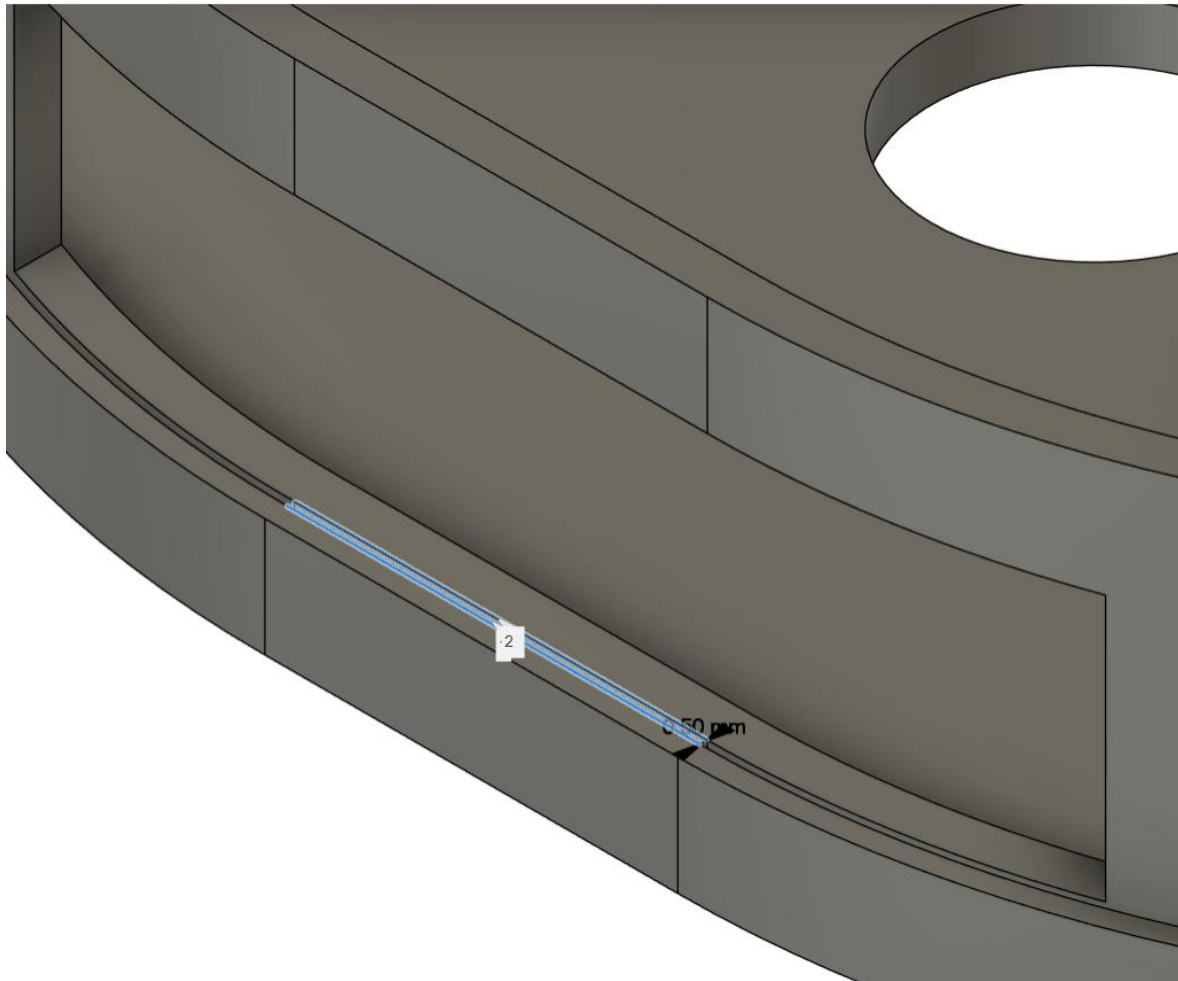


Figure 3.5.5 shows the gap between the chassis and the bottom piece being 0.5mm, for a snug fit 0.1mm gap between the two pieces is required considering that was the same gap used between the ceiling and the chassis.

During this process, we got lucky considering during the 3D print of the chassis, the option for a brim was chosen, thus additional material was attached to the chassis allowing for a tighter fit. For future purposes, simply extruding material inwards resulting in a thicker lip and thus a smaller gap should be sufficient.

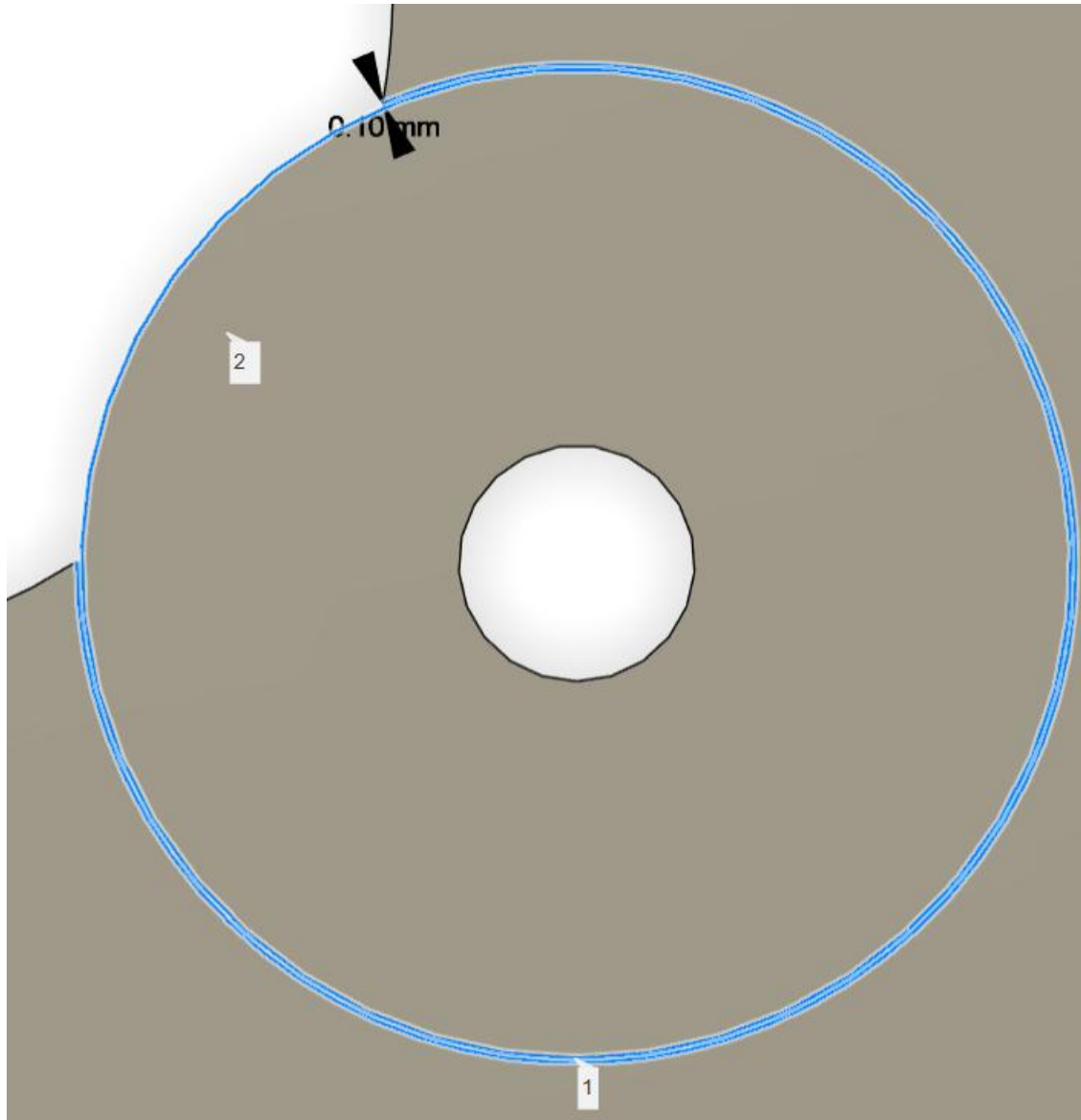


Figure 3.5.5 shows the gap between the bottom piece and the rotator pegs 0.1mm where a snug attachment would only be achieved with 0 tolerance, thus 0mm gap.

Although the gap between the rotator pegs and the bottom piece are 0.1mm gaps which has worked for the ceiling and chassis, here the pegs simply slip through. Within the limited time given, the immediate solution was to add a non-conductive material thus

painter's tape was used. Simply to fix this increase the radius of each peg by 0.1mm in radius and the pegs should match.

The other flaw with the peg is the hole through it connecting to the screws. The whole has a radius of 2.6mm which was not sufficient for the screws, thus the quick solution was to use a drill press with 3/16-inch drill bit and drill new holes. We did not measure the size of the new holes allowing us to produce new rotator pegs, but with some measurements of the screws custom pegs could be created once again.

Lastly, there is a major flaw with the tile considering the time provided to rotate an antenna. The tile slits are exactly the length of the antenna with no tolerance, thus placing the antenna into the slit is quite challenging in the first place. This would not be an issue if Matlab did not crash if the test does not continue within a minute or so. Additionally considering the mechanism does a 360-degree rotation, once the rotation finishes the slit for the H plane measurement is perfectly orthogonal to the horn antenna thus no measurement can be made initially. To fix this, there are two options, modify the code of modify the design of the tile.

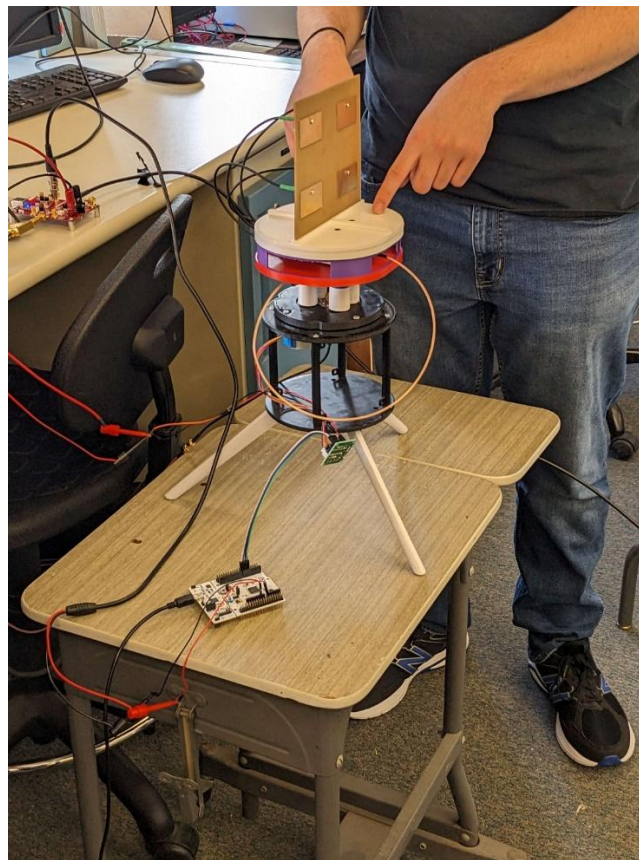


Figure 3.5.6 represents the in person result of the final designed attachment with the rotator and STM-32 connected.

From initially designing the attachment to printing it, the design has improved tremendously over a short period of time. Although the design has its flaws but with the nature of modular design anything can be improved or completely overhauled resulting in a customized design. Yet, even with its flaws it functioned almost flawlessly on test day producing adequate results.

3.6: Firmware and Interface Code

3.6.1: Microcontroller Firmware

For our project, we will be utilizing the Nucleo STM32F446RE microcontroller. This device offers two 12-bit DACs, two general-purpose 32-bit timers, a built-in system clock that can be set to 24 or 48 MHz, and configurations for standard communication interfaces such as SPI, I2C, and UART. Additionally, the Nucleo contains a built-in USB programmer.

For our design, we will be utilizing the two DACs for converting the I and Q values from the demodulator to digital values. Additionally, the microcontroller will need to communicate to the rotate system and to MATLAB via UART. Figure 3.6.1 presents the high-level block diagram of the microcontroller connections.

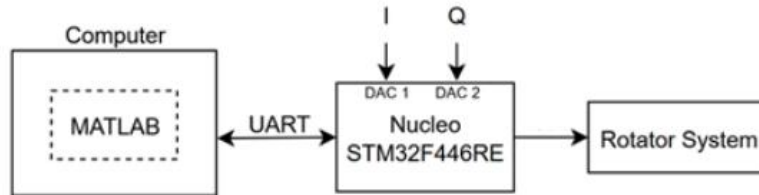


Figure 3.6.1: Microcontroller Hardware Connections

Figure 3.6.2 presents a high-level block diagram for microcontroller hardware. Upon startup, the system will initialize relevant system parameters. Then, the system will take a series of 1024 measurements at a ~4 MHz sample rate. This data will be sent to MATLAB prior to rotating the antenna 5 degrees along the azimuth. If the antenna has completed a full rotation along the azimuth, then the antenna will pause for the antenna to be manually spun 90 degrees. This process will be repeated until the entire E-plane and H-plane have been measured.

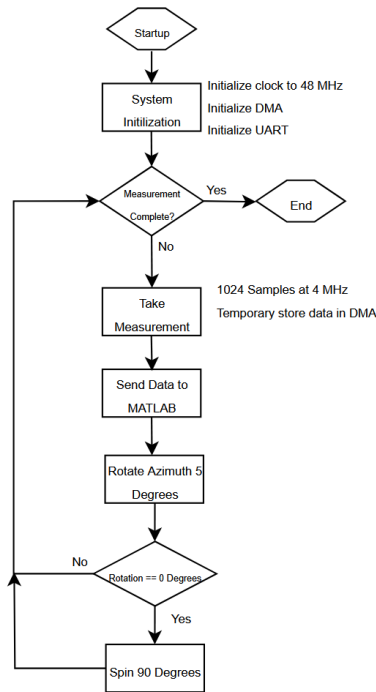


Figure 3.6.2: Microcontroller Firmware Block Diagram

Between these measurements, the data is sent to MATLAB where it will be used to construct graphics for the antenna pattern.

Appendix 6.4.2 contains the completed microcontroller firmware.

3.6.1: MATLAB Interface Code

The MATLAB code works parallel with the microcontroller firmware. Upon startup, the MATLAB interface code establishes communication with the microcontroller and lets it know that it can start collecting samples. After each set of samples is collected, the microcontroller sends the packet of data to MATLAB via UART. The MATLAB interface code then extracts the average value of the I sample data and uses it to find the average absolute value difference of each sample for a given angle. This value is used to update the E-field plot. For simplicity, we only focused on samples from the I channel. Once the appropriate number of samples have been processed for the E-field, the MATLAB interface code pauses to allow for the manual rotation of the antenna system. Afterwards, the process continues for the H-field measurements.

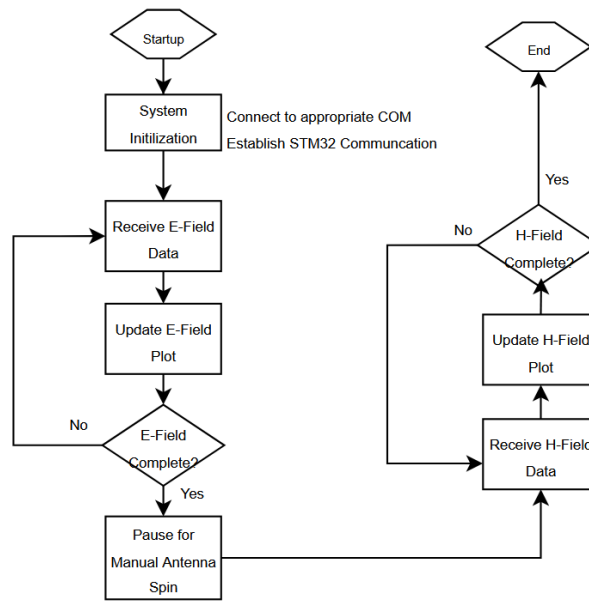


Figure 3.6.3: MATLAB Interface Code Block Diagram

For clarity, Figure 3.6.4 presents a sketch of what MATLAB extracts to plot. The square wave represents the 1 MHz waveform sent on the I channel. The 1.44 DC component of the square wave represents the bias the demodulator adds to the I+ output. The plotted value will be the average magnitude of the value labeled “Plot Value” which will change with the strength of the received signal.

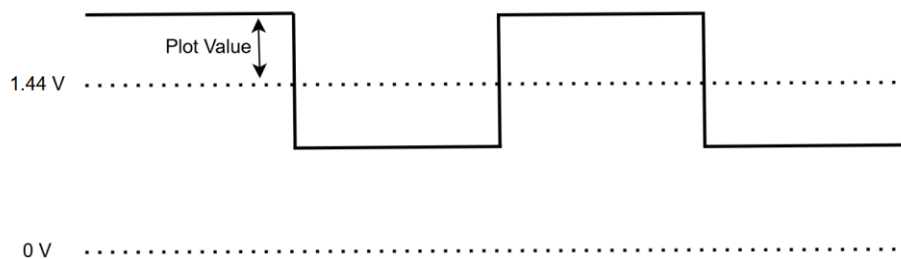


Figure 3.6.4: Sketch of Plot Value

Appendix 6.4.1 contains the completed MATLAB code.

4: System Integration

4.1: Initial Link Budget

After we had selected our initial components, we used the gain and noise specifications of each part to form a link budget for the system.

For the transmit side, we assumed the output power of the modulator would be maximum. According to the datasheet, maximum output power at 2.45 GHz is approximately 1dBm. Similarly, the noise spectral density at this frequency is -156 dBm/Hz. From there, we utilized Equation 4.1 to calculate the noise at each stage to the transmit antenna. Similarly, we utilized Equation 4.2 to calculate the noise figure at each stage. For passive devices, like the bandpass filter, we utilized Equation 4.3 to convert the gain of the device to a noise figure. Finally, after we had calculated the noise figure for the entire transmit side, we could use Equation 4.4 to convert the SNR at the modulator to an SNR at the antenna.

$$G = G_1 G_2 G_3 \cdots \quad (4.1)$$

$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} \cdots \quad (4.2)$$

$$F = \frac{1}{G} \quad (4.3)$$

$$F = \frac{SNR_{in}}{SNR_{out}} \quad (4.4)$$

For the receive side, the process was nearly identical except that we needed to use Friis's Formula, Equation 4.5, to convert the power at the transmit antenna to the power at the receive antenna. Equation 4.6 presents a way to calculate the noise power at the receive antenna given a temperature and an equivalent noise temperature. Unfortunately, with this equation, we were not receiving reliable results, so instead, we used Equation 4.5 with the transmit noise power to calculate the receive antenna noise power.

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2 \quad (4.5)$$

$$P_n = k(T + T_e)B \quad (4.6)$$

Table 1 presents the gain and noise figure of each device derived either during the design process or from the datasheet specification.

Table 4.1: Gain and Noise Figure of Devices

Device	Role	Gain / Insertion Loss (dB)	Noise Figure (dB)
ERA-3+	Pre-Amplifier	23.4	2.8
PGA-105+	Power-Amplifier	15.9	2.1
Hairpin BPF	Band-Pass-Filter	-2	
Horn Antenna	Tx Antenna	15	
Patch Antenna	Rx Antenna	9.5	
CMA-103+	LNA	16.3	0.8
SKY65050	Alt LNA	17.3	0.44
Hairpin BPF	Band-Pass-Filter	-2	

Table 2 presents the initial system link budget for the design. Note that, between the transmit antenna and receive antenna, we are not factoring in external noise, which would be significant at the Wi-Fi frequency band. Additionally, the SNR at the demodulator input is much higher than our system specification of 20 dB which indicates promise that our physical design will meet our desired specification.

Table 4.2: System Link Budget

	Power (dBW)	Noise Figure (dB)	SNR (dB)
Modulator Output	-29.0		77.0
Tx Antenna Input	8.3	2.8	74.2
Rx Antenna Output	-23.0		74.2
Demodulator Input	-8.7	0.8	73.3

4.2: Physical Integration

Figure 4.2.1 presents a system block diagram for integrating the subsystems. How this system operates is described in detail in Section 2, but this diagram is useful for illustrating physical connections.

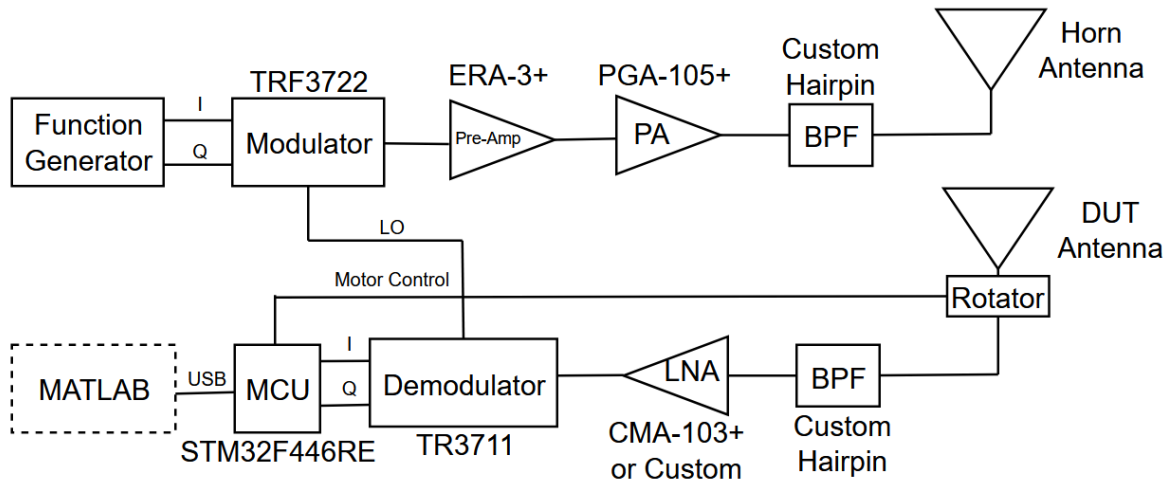


Figure 4.2.1: System Block Diagram

Figure 4.2.2 presents the physical setup for the transmit side from the modulator to the bandpass filter. Note that some of the connections are not present in the image.

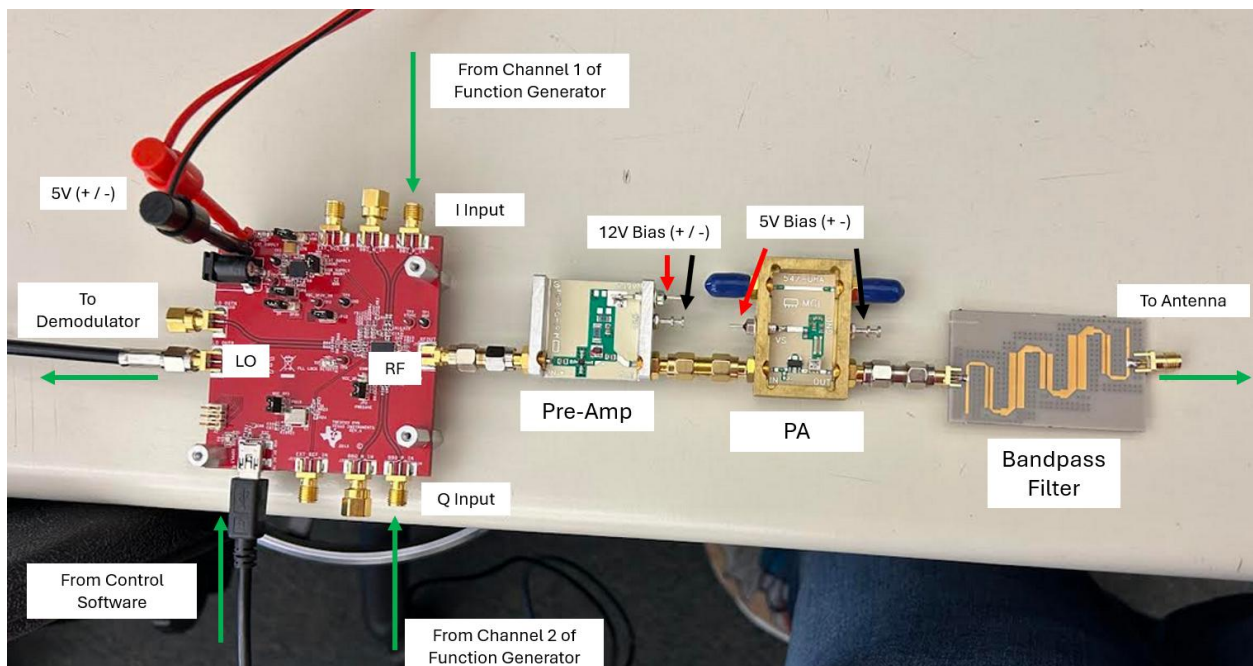


Figure 4.2.2: Transmit Side Physical Setup

Figure 4.2.3 presents the physical setup for the receive side from the bandpass filter to the demodulator.

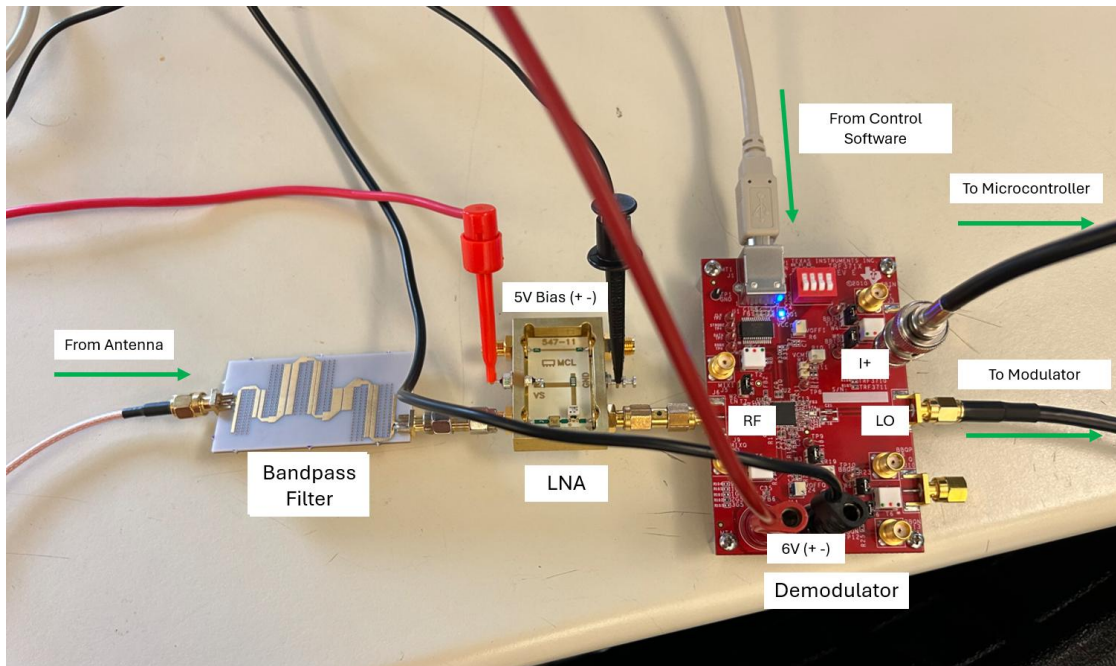


Figure 4.2.3: Receive Side Physical Setup

Figure 4.2.4 presents the antenna stand physical setup.

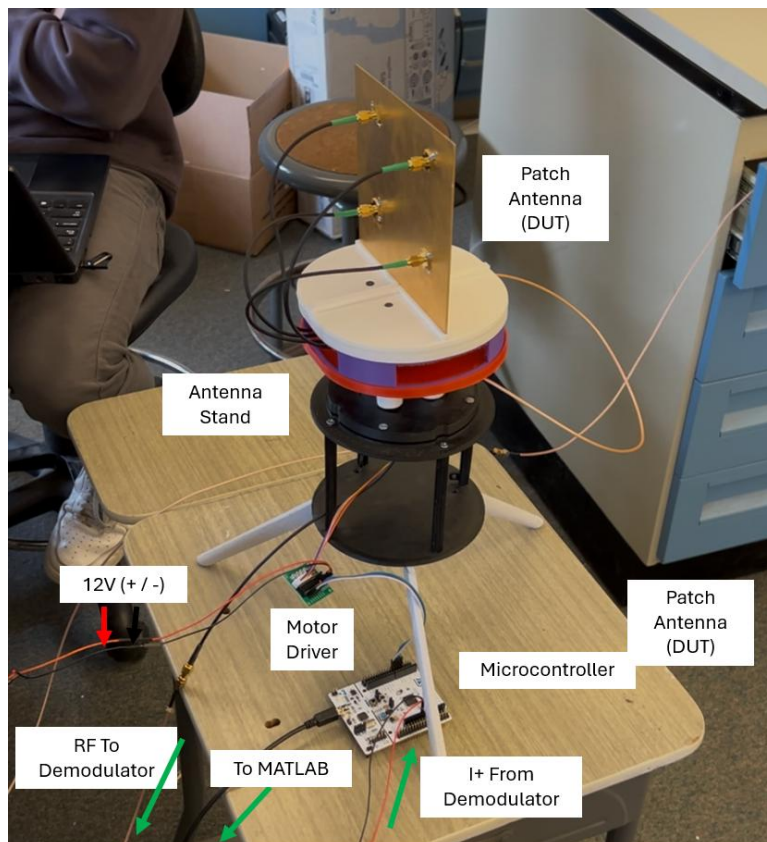


Figure 4.2.4: Antenna Stand Physical Setup

Figure 4.2.5 presents a view of the system from the transmit side. Note that horn antenna is 6 meters apart from the DUT antenna which ensures the measurements are for the far-field.

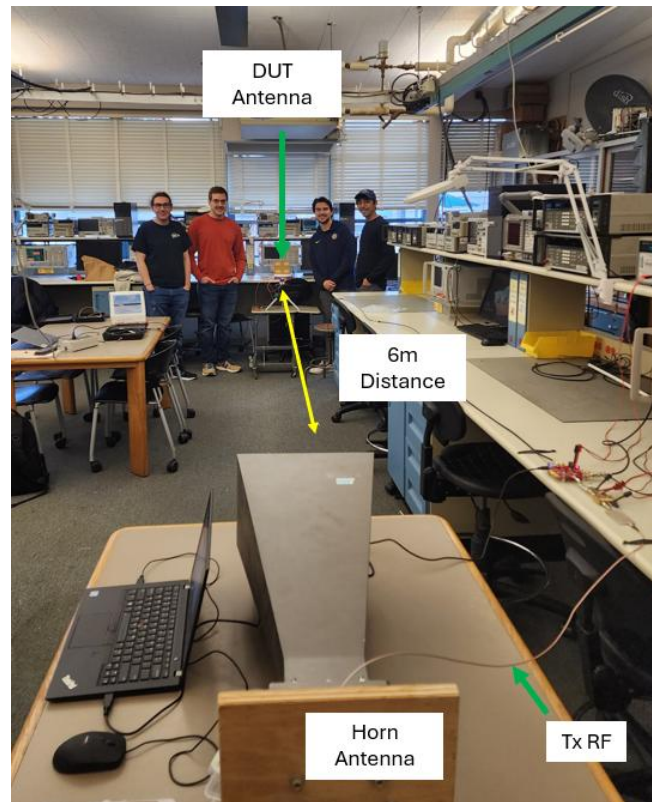


Figure 4.2.5: System Setup

4.3: System Test Procedure

For our system, we completed the following procedure for three separate system tests. Note that the terms “E-field” and “H-field” are used very loosely here to describe the DUT antenna lying along its long-side (E-field) and its short-side (H-field). Additionally, for our last system test, we reversed the “E-field” and “H-field” definitions.

1. Apply the connections illustrated in Section 4.2.
2. Place the DUT antenna in the “E-field” orientation on the antenna stand.
3. Ensure that the horn antenna and the DUT antenna are separated by 6 meters. Additionally, ensure the two antennas are pointing directly at each other.
4. Apply appropriate voltages using power supply to all active components (amplifiers, modulator, demodulator).
5. Use the Texas Instruments software to configure the modulator to send at 2.41 GHz. Recall that the bandpass filters worked best at 2.41 GHz.

6. Use the Texas Instruments software to configure the demodulator to apply a BB gain of 10. Our team found that this was the maximum gain we could apply before saturation occurred.
7. Set the function generator to output a 1 MHz, 1 volt peak-to-peak square wave to the I-input of the modulator. Our team chose to ignore the Q channel for our system tests. Additionally, our team found that a 1-volt peak-to-peak square wave would not saturate the input of the modulator. Additionally, a 1 MHz square wave is slower than the Nyquist frequency of the ADC (2 MHz).
8. Start the MATLAB interface code.
9. Wait for the system to complete the E-field measurements.
10. Manually “spin” the antenna to the H-field orientation.
11. Wait for the system to complete the H-field measurements.

4.4: System Test Results

The results for the first test are shown below in Figure 4.4.1 with the amplitude of the measurements on the polar plot produced in normalized power (relative to both fields).

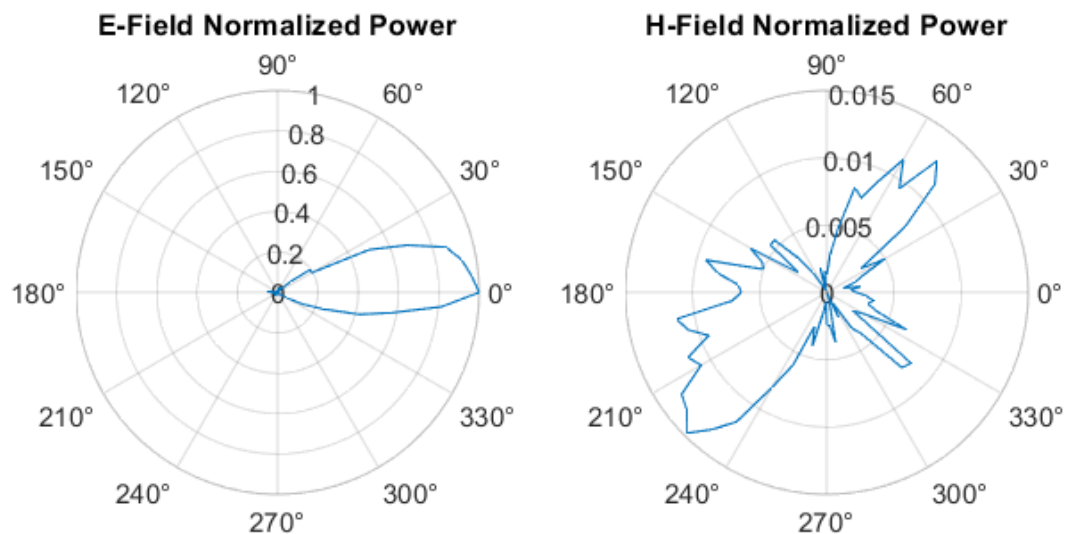


Figure 4.4.1: Test 1 Results

Note the difference between both measurements being quite stark in amplitude and the beam. This is mainly due to the difference between the performance of the microstrip patch array for the E and H field. While comparing the results with the results for the patch array measured in the anechoic chamber, there is some overlap, however additional tests of the setup will be conducted to observe any trends. The results of the second test is shown below in Figure 4.4.2.

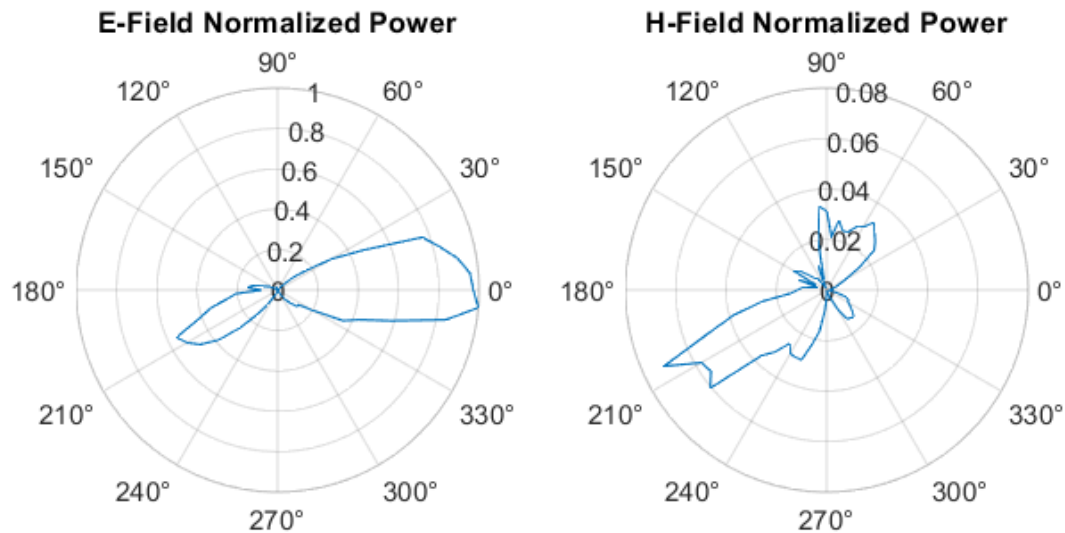


Figure 4.4.2: Test 2 Results

With this second test, we can see the formation of a beam behind the array in the E-plane measurement with significant amplitude, which matches the results seen for the measurement in the Anechoic Chamber shown in Figure 3.4.14. The results for the H-plane recording are also fairly consistent with the anechoic chamber measurements as seen with beams formed in the +80° and -150° directions. Note that in this experiment, the tests were conducted in the Microwave lab with a distance of 6 meters between the antennas, with no other assurances of noise mitigation in the measurement apparatus. In addition, there could be unknown reflections from the transmitting antenna which could've constructively or destructively combined with the received signal of the DUT antenna, resulting in variations seen in the amplitude, phase, and beam distribution of the received signal.

Finally, an additional third test was conducted with the orientations of the antenna swapped such that the definitions of "E-field" and "H-field" are swapped.

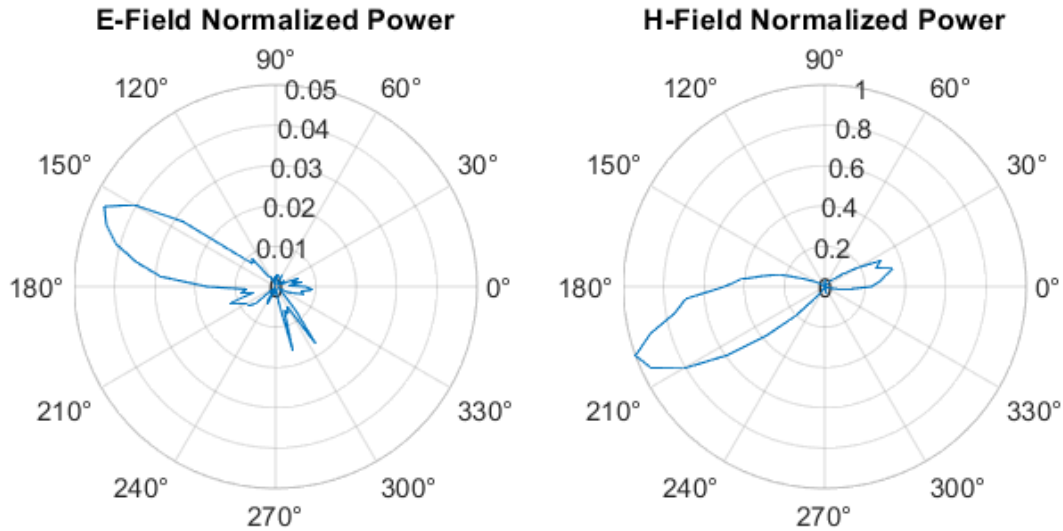


Figure 4.4.3 Test 3 Results

The results of this test match those of the previous test quite closely with beams now swapped. However, in this result, it seems the direction of orientation were swapped such that the main beams in the left measurement are now at +150° and a smaller lobe towards the -80° directions. Note that these are conjugates of the results shown in Figure 4.4. In the new H-field result, the main lobe is in the -165° and the smaller back lobe at +15°, which are both slightly with a 15° bias.

This may be attributed to mechanical issues with the second phase of the experiment relying on a perfect full rotation for the initial measurement of the E-field results. This was found to be slightly off such that the initial measurement is conducted with the antennas not perfectly facing other which should result in a main lobe at 0°. However, due to the mechanical limitations of the setup, the conducted measurement has a 20° bias along with conjugate lobes. All of these trends are shown below in Table 4.1 when comparing the angle for the peak results seen in the Main and side lobes for each of the three measurements and with the Anechoic Chamber measurements.

Table 4.1 – Azimuth Angle for different lobes

Azimuth angle	E-Field Main Lobe	E-Field side lobe	H-Field Main Lobe	H-Field Side Lobe
Experiment 1	0°	180°	-135°	50°
Experiment 2	0°	-150°	-155°	55
Experiment 3	15°	-165°	155°	-55°
Anechoic Chamber Result	0°	180°	148°	-94°

5: Conclusion

This project report outlines the complete antenna measurement system and each subsystem that contributed to the overall functionality of the project. By integrating modulators and demodulators for signal processing, amplifiers to strengthen transmitted and received signals, bandpass filters for frequency selection, and antennas for signal transmission and reception, an operational microwave measurement system was developed. The combination of custom design and off-the-shelf solutions allowed for flexibility and reliability in implementation.

When integrating each component of the system together, a complete measurement was conducted with the DUT antenna in the E-Field and H-Field over the course of a single measurement. To display the repeatability of the system, several tests were conducted with the system to ensure accuracy of the measurement system. By comparing the results of the DUT antenna measured in the microwave lab's anechoic chamber with the results of the antenna measurement system, we can see that the radiation pattern of the DUT fairly matches the results seen in a controlled environment with slight discrepancies. This proves that this antenna measurement system works as intended and can be used with other DUT antennas or applications.

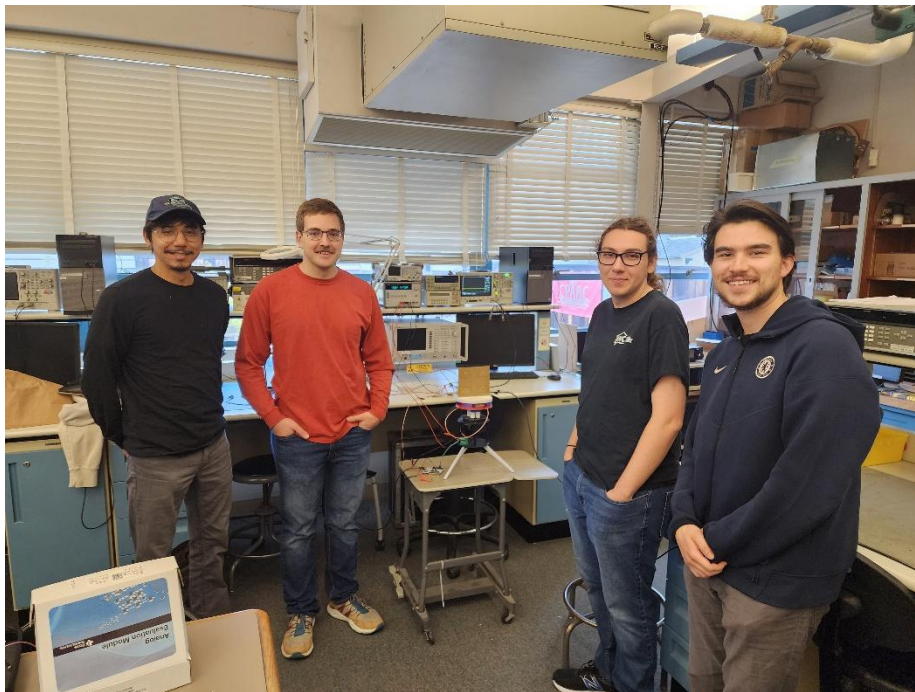


Figure 5.1.1 All group members involved in project, Left to Right: Daniel Guerrero, Nolan Egging, Shiron Bendrihem, and Max Spalek

6: Appendix

6.1: LNA BOM

Component	Quantity (for 3 boards)	Size	Value	Manufacturer	Product	Link
C1, C7	6	US-0603	15pF	Murata	GCG1885G1H150GA01D	https://www.mouser.com/ProductDetail/Murata-Electronics/GCG1885G1H150GA01D?qs=QzBrWTOodeVpr3i7P0eUvw%3D%3D
C2	3	US-0603	1.8pF	Murata	GCM1885C1H188BA16D	https://www.mouser.com/ProductDetail/Murata-Electronics/GCM1885C1H188BA16D?qs=mi1Y6Y2fmr7DWWmlLwu7sQ%3D%3D
C3, C4, C6	9	US-0603	1000pF	Murata	GCM188R71H102KA37J	https://www.mouser.com/ProductDetail/Murata-Electronics/GCM188R71H102KA37J?qs=bGEzvQhTDnbYGrYajR6ZTw%3D%3D
C5	3	US-0603	0.5pF	Murata	GQM1875G2ER50WB12D	https://www.mouser.com/ProductDetail/Murata-Electronics/GQM1875G2ER50WB12D?qs=rrS6PyfT74ddkFPUFZWcEw%3D%3D
L1	3	US-0603	3.9nH	Coilcraft	0603DC-3N9XJRW	https://www.mouser.com/ProductDetail/Coilcraft/0603DC-3N9XJRW?qs=KUoLVG%2F9lIzqC70ihVMag%3D%3D
L2	3	US-0603	27nH	Coilcraft	0603DC-30NXJRW	https://www.mouser.com/ProductDetail/Coilcraft/0603DC-30NXJRW?qs=KUoLVG%2F9lIzqC70ihVMag%3D%3D
L3	3	US-0603	4.3nH	Coilcraft	0603DC-4N3XJRW	https://www.mouser.com/ProductDetail/Coilcraft/0603DC-4N3XJRW?qs=KUoLVG%2F9lIzqC70ihVMag%3D%3D
R1	3	US-0603	27R	Panasonic	ERJ-UP3J270V	https://www.mouser.com/ProductDetail/Panasonic/ERJ-UP3J270V?qs=GedFDfLaBXHovdU9RmRV/Q%3D%3D
R2	3	US-0603	200R	Panasonic	ERJ-UP3J201V	https://www.mouser.com/ProductDetail/Panasonic/ERJ-UP3J201V?qs=GedFDfLaBXHovdU9RmRV/Q%3D%3D
R3	3	US-0603	10R	Panasonic	ERJ-UP3F10R0V	https://www.mouser.com/ProductDetail/Panasonic/ERJ-UP3F10R0V?qs=GedFDfLaBXHovdU9RmRV/Q%3D%3D

6.2: Link Budget Code

```
% EE541 - Link Budget
% 11 Feburary 2025

close all; clear; clc;

% constants
c = 3e8;
k = 1.380649e-23; % boltzman constant
T = 293; % temp in Kelvin

% system specifications
f = 2.45e9; % center frequency
lambda = c / f;
B = 100e6; % bandwidth
d = 6; % distance between antennas in meters
Te = 150;

% transmit side specification
modulator_po_dBm = 1;
modulator_nsd_dBmHz = -156;
preamp_g_dB = 23.4;
preamp_nf_dB = 2.8;
pa_g_dB = 15.9;
pa_nf_dB = 2.1;
bpf_g_dB = -2;
tx_antenna_dB = 15;

% receive side specification
rx_antenna_dB = 9.5;
lna_g_dB = 16.3;
lna_nf_dB = 0.8;
alt_lna_g_dB = 17.3;
alt_lna_nf_dB = 0.44;

% modulator SNR
modulator_ps_W = 10 ^ ((modulator_po_dBm-30)/10);
modulator_nsd_WHz = 10 ^ ((modulator_nsd_dBmHz-30)/10);
modulator_pn_W = modulator_nsd_WHz * B;
modulator_SNR = modulator_ps_W/modulator_pn_W;
modulator_SNR_dB = 10*log10(modulator_SNR);
```

```

disp("Power at Modulator (dBW)" + 10*log10(modulator_ps_W))
disp("Noise at Modulator (dBW)" + 10*log10(modulator_pn_W))
disp("SNR at Modulator (dB): " + modulator_SNR_dB);
disp(" ");

% tx chain
preamp_g = 10^(preamp_g_dB/10);
preamp_f = 10^(preamp_nf_dB/10);
pa_g = 10^(pa_g_dB/10);
pa_f = 10^(pa_nf_dB/10);
bpf_g = 10^(bpf_g_dB/10);
bpf_f = 1/bpf_g;
f_at_tx_antenna = preamp_f + ((pa_f-1)/(preamp_g)) + ((bpf_f-1)/(preamp_g*pa_g));
SNR_at_tx_antenna = modulator_SNR / f_at_tx_antenna;
p_at_tx_antenna = modulator_ps_W * preamp_g * pa_g * bpf_g;
pn_at_tx_antenna = p_at_tx_antenna / SNR_at_tx_antenna; % get noise power at
antenna
disp("Power at Tx Antenna (dBW): " + 10*log10(p_at_tx_antenna));
disp("Noise Figure at Tx Antenna (dB): " + 10*log10(f_at_tx_antenna));
disp("SNR at Tx Antenna (dB): " + 10*log10(SNR_at_tx_antenna));
disp(" ");

% at rx antenna
tx_antenna_g = 10^(tx_antenna_dB/10);
rx_antenna_g = 10^(rx_antenna_dB/10);
p_at_rx_antenna = p_at_tx_antenna * tx_antenna_g * rx_antenna_g *
((lambda/(4*pi*d))^2);
%pn_at_rx_antenna = k*(T+Te)*B; % likely inaccurate
pn_at_rx_antenna = pn_at_tx_antenna * tx_antenna_g * rx_antenna_g *
((lambda/(4*pi*d))^2); % use Friis' with noise
SNR_at_rx_antenna = p_at_rx_antenna / pn_at_rx_antenna;
disp("Power at Rx Antenna (dBW): " + 10*log10(p_at_rx_antenna));
disp("Max SNR at Rx Antenna (dB): " + 10*log10(SNR_at_rx_antenna));
disp(" ");

% rx chain
lna_g = 10^(lna_g_dB/10);
lna_f = 10^(lna_nf_dB/10);
f_at_demod = lna_f + ((bpf_f-1)/(lna_g));
demodulator_SNR = SNR_at_rx_antenna / f_at_demod;
p_at_demodulator = p_at_rx_antenna * lna_g * bpf_g;
disp("Power at Demodulator (dBW): " + 10*log10(p_at_demodulator));
disp("Noise Figure at Demodulator (dB): " + 10*log10(f_at_demod));
disp("Max SNR at Demodulator (dB): " + 10*log10(demodulator_SNR));
disp(" ");

```

6.3 Antenna DUT Code

6.3.1 Antenna DUT Calculator

```

%% EE 541 DUT Calculator
close; clear all; clc; format compact
% Patch dimensions

```

```

% Substrate params
Er = 4.6;
h = 62;      % mils
hmm = h / 39.37; %mm

% Frequency params
f = 2.45e9;
c = 3e8;
Vp = c/sqrt(Er);
lambda = c/f
% Patch Dimensions
W = lambda/2 * sqrt(2/(Er+1));
Wmm = W*1000 % mm
Ereff = (Er+1)/2 + ((Er-1)/2)*(1 + 12*hmm/Wmm)^(-1/2);

dL = 0.412 * hmm * ((Ereff + 0.3)*(Wmm/hmm + 0.264))/((Ereff - 0.258)*(Wmm/hmm + 0.8));
Leff = lambda/(2*sqrt(Ereff)); % m
Lmm = Leff*1000 - dL

lambdag = c/(f*sqrt(Ereff));
BW = 3.77*f * (Er-1)/(Er^2) * ((Wmm*hmm)/(Lmm*lambda*39.37))^2

% Inset feed calculations

Lmm = 27.25;
Wmm = 49;

Rin = 90*(Er^2)/(Er-1) * (Lmm/Wmm)^2
yo = Lmm/pi * acos(sqrt(50/Rin))
W0 = 2.87; %mm

D = Lmm/pi * acos((50/Rin)^(1/4))

%% More calcs
% Microstrip bend calcs
Z1 = 2.893; % 50 ohm line
D = Z1 * sqrt(2)
X = D * (0.52 + 0.65*exp(-1.35*D/hmm))

A = X*sqrt(2) - Z1

```

6.3.2 Antenna Subsystem Plots

```

clear; clc; close all;
addpath('C:\Users\Precision\Desktop\EE 541\Results');
addpath('C:\Users\Precision\Desktop\EE 541\Results\BackFed')
% Imported Results
% Initial dimensions of patch from calcs
calcrefl = readmatrix('C:\Users\Precision\Desktop\EE 541\Results\BackFed\Calc_single_S11.csv');
f = calcrefl(:,4);

```



```

calcS11 = calcS11(:,5);

pmax =15; % maximum of dB scale
dynr=30; % dynamic range of dB scale
rticks=5; % number of rings in polar plot

%% Back Fed results
% optimized dimensions of back fed patch
optrefl = readmatrix('OptimizedDimensionsS11.csv');
optS11 = optrefl(:,5);

% calculated dimensions gain of back fed patch
calcs21 = readmatrix('CalculatedDimensionsGain.csv');
theta = calcs21(1:length(calcs21)/2, 3);
calcGainP0 = calcs21(1:length(calcs21)/2, 4);
calcGainP90 = calcs21(length(calcs21)/2+1 : end, 4);

% optimized dimensions gain of patch
opts21 = readmatrix('OptimizedDimensionsGain.csv');
optGainP0 = opts21(1:length(opts21)/2, 4);
optGainP90 = opts21(length(opts21)/2+1 : end, 4);

% reflection coefficient of each patch from array
arrayrefl = readmatrix('All S11s.csv');
ArrayS11 = arrayrefl(:,5);
ArrayS22 = arrayrefl(:,6);
ArrayS33 = arrayrefl(:,7);
ArrayS44 = arrayrefl(:,8);

% back fed patch gain
PatchGainP0 = readmatrix('SinglePatchGainP0.csv');
PatchGainP90 = readmatrix('SinglePatchGainP90.csv');

% back fed patch array gain
% 2.45 GHz results
load('ArrayGainP0.csv')
ArrayGainP90 = readmatrix('ArrayGainP90.csv');
% 2.4 GHz results
ArrayGainP0_2_4GHz = readmatrix('InsetArrayGainP0.csv');
ArrayGainP90_2_4GHz = readmatrix('InsetArrayGainP90.csv');
%% Inset Fed results
SingleS11 = 'C:\Users\Precision\Desktop\EE
541\Results\Measurements\SinglePatchS11.s2p';

Single = readmatrix('C:\Users\Precision\Desktop\EE
541\Results\InsetFed\Single\SinglePatchS11.csv');
SingleS11 = Single(:,2);

Double = readmatrix('C:\Users\Precision\Desktop\EE
541\Results\InsetFed\Double\DoubleS11.csv');
Db1S11 = Double(:,2);

Db1Db1 = readmatrix('C:\Users\Precision\Desktop\EE
541\Results\InsetFed\DoubleDouble\InsetFeedRef1.csv');
Db1Db1S11 = Db1Db1(:,2);

```



```

% Single calc vs opt
CalcOptS11_Single = readmatrix('C:\Users\Precision\Desktop\EE
541\Results\InsetFed\Single\CalcvsOpt_S11_yoL.csv');
SingleS11_l1y1 = CalcOptS11_Single(:,3);
SingleS11_l2y1 = CalcOptS11_Single(:,6);
SingleS11_l1y2 = CalcOptS11_Single(:,9);
SingleS11_l2y2 = CalcOptS11_Single(:,12);

%% Plots

% S11 for single element
figure; hold on;
grid on; box;

% plot(f, calcS11, 'linewidth', 1.5)
plot(f, optS11, 'linewidth', 1.5)
ylabel('S_{11} (dB)')
xlabel('Frequency (GHz)')
xline(2.45, '--k', 'linewidth', 0.5)
% legend('Calculated Dimensions', 'location', 'best')

% Reflect coefficients for 2.4G array
figure; hold on;
grid on; box;

plot(f, ArrayS11, 'linewidth', 1.5)
plot(f, ArrayS22, 'linewidth', 1.5)
plot(f, ArrayS33, 'linewidth', 1.5)
plot(f, ArrayS44, 'linewidth', 1.5)

ylabel('Reflection Coefficients (dB)')
xlabel('Frequency (GHz)')
xline(2.4, '--k', 'linewidth', 0.5)
legend('S_{11}', 'S_{22}', 'S_{33}', 'S_{44}', 'location', 'best')

% Gain vs theta for array
figure;
hold on;
polardB_90(ArrayGainP0(:,1), ArrayGainP0(:,2), 'r-', 1.5, 0, pmax, dynr, rticks)
polardB_90(ArrayGainP90(:,1), ArrayGainP90(:,2), 'k-', 1.5, 0, pmax, dynr, rticks)
% legend('Phi = 0', 'Phi = 90')
ylabel('Gain (dB)')
xlabel('{\theta} (°)')
% legend('Array Gain Phi = 0°', 'Array Gain Phi = 90°', 'Array Gain', '3', '4',
'5', 'location', 'best')

%% 2/25 Simulation Results updates
% Imported Results
stdFR4 = readmatrix('Standard_FR4_S11.csv');
f = stdFR4(:,4);
stdFR4S11 = stdFR4(:,5);

```

```

oshFR4 = readmatrix('OSH_FR4_S11.csv');
oshFR4S11_f = oshFR4(1:length(oshFR4)/2,5);
oshFR4S11 = oshFR4(length(oshFR4)/2+1:end,5);

SinglePatch_s2p = 'C:\Users\Precision\Desktop\EE
541\Results\Measurements\SinglePatchS11.s2p';
S_mes = sparameters(SinglePatch_s2p);
mes_f = S_mes.Frequencies*1e-9;
mesS11 = rfparam(S_mes,1,1);
single_mesS11dB = 20*log10(abs(mesS11));

% Reflection simulation results
figure; hold on; grid on; box on;
plot(mes_f, single_mesS11dB, 'linewidth', 1.5)
ylabel('Reflection Coefficient (dB)')
xlabel('Frequency (GHz)')
xline(2.45, '--k', 'linewidth', 0.5)

%%
figure;
subplot(1,2,1); hold on; grid on; box on;
plot(f, stdFR4S11, 'linewidth', 1.5)
plot(f, oshFR4S11, 'linewidth', 1.5)
plot(mes_f, single_mesS11dB, 'linewidth', 1.5)
ylabel('Reflection Coefficient (dB)')
xlabel('Frequency (GHz)')
xline(2.45, '--k', 'linewidth', 0.5)
legend('Standard FR4', 'OSHPark FR4', 'Measurement', 'location', 'best')
title('Standard FR4 vs OSHPark FR4')

subplot(1,2,2); hold on; grid on; box on;
plot(f, oshFR4S11, 'linewidth', 1.5)
plot(f, oshFR4S11_f, 'linewidth', 1.5)
ylabel('Reflection Coefficient (dB)')
xlabel('Frequency (GHz)')
xline(2.45, '--k', 'linewidth', 0.5)
legend('L = 27.75 mm', 'L = 27 mm', 'location', 'best')
title('OSH Park FR4')

addpath('C:\Users\Precision\Desktop\EE 541\Results\Measurements\PatchArray');

Patch1_2_s2p = 'Patch1_2.s2p';
Patch1_3_s2p = 'Patch1_3.s2p';
Patch1_4_s2p = 'Patch1_4.s2p';
Patch2_3_s2p = 'Patch2_3.s2p';
Patch2_4_s2p = 'Patch2_4.s2p';
Patch3_4_s2p = 'Patch3_4.s2p';

S12_s2p_1_2 = sparameters(Patch1_2_s2p);
S12_s2p_1_3 = sparameters(Patch1_3_s2p);
S12_s2p_1_4 = sparameters(Patch1_4_s2p);
S12_s2p_2_3 = sparameters(Patch2_3_s2p);
S12_s2p_2_4 = sparameters(Patch2_4_s2p);
S12_s2p_3_4 = sparameters(Patch3_4_s2p);
mes_f = S12_s2p_1_2.Frequencies*1e-9;

```

```

% Define a helper function to extract and compute dB values
function [S11dB, S12dB, S21dB, S22dB] = extractSParams(S2pObj)
    % Extract S-parameters from the sparameters object
    S11 = rfparam(S2pObj, 1, 1);
    S12 = rfparam(S2pObj, 1, 2);
    S21 = rfparam(S2pObj, 2, 1);
    S22 = rfparam(S2pObj, 2, 2);

    % Compute dB values
    S11dB = 20*log10(abs(S11));
    S12dB = 20*log10(abs(S12));
    S21dB = 20*log10(abs(S21));
    S22dB = 20*log10(abs(S22));
end

% Extract and compute dB values for all patches
[S11dB_1_2, S12dB_1_2, S21dB_1_2, S22dB_1_2] = extractSParams(S12_s2p_1_2);
[S11dB_1_3, S12dB_1_3, S21dB_1_3, S22dB_1_3] = extractSParams(S12_s2p_1_3);
[S11dB_1_4, S12dB_1_4, S21dB_1_4, S22dB_1_4] = extractSParams(S12_s2p_1_4);
[S11dB_2_3, S12dB_2_3, S21dB_2_3, S22dB_2_3] = extractSParams(S12_s2p_2_3);
[S11dB_2_4, S12dB_2_4, S21dB_2_4, S22dB_2_4] = extractSParams(S12_s2p_2_4);
[S11dB_3_4, S12dB_3_4, S21dB_3_4, S22dB_3_4] = extractSParams(S12_s2p_3_4);

% Measured S11 of patch array with power splitter
MeasPatch = ('C:\Users\Precision\Desktop\EE
541\Results\Measurements\Patch_powersplitter_s11.s2p');
MeasPatchS = sparameters(MeasPatch);
MeasPatchS11 = rfparam(MeasPatchS, 1, 1);

MeasPatchS11dB = 20*log10(abs(MeasPatchS11));

% Measured S21 from anec chamber
MesS21_Phi0_E = readmatrix('C:\Users\Precision\Desktop\EE
541\Results\Measurements\Phi0\Rad_Pattern_S21_2.4_GHz_E-Co.txt');
MesS21_Phi0_Hco = readmatrix('C:\Users\Precision\Desktop\EE
541\Results\Measurements\Phi0\Rad_Pattern_S21_2.4_GHz_H-Co.txt');
MesS21_Phi0_Hco2 = readmatrix('C:\Users\Precision\Desktop\EE
541\Results\Measurements\Phi0\Rad_Pattern_S21_2.4_GHz_H-Co2.txt');

MesS21_Phi90_E = readmatrix('C:\Users\Precision\Desktop\EE
541\Results\Measurements\Phi90\Rad_Pattern_S21_2.4_GHz_E-Co.txt');
MesS21_Phi90_Hco = readmatrix('C:\Users\Precision\Desktop\EE
541\Results\Measurements\Phi90\Rad_Pattern_S21_2.4_GHz_H-Co.txt');

MesAng = MesS21_Phi0_E(:,1);
MesS21_Phi0_Eco_dB = MesS21_Phi0_E(:,2);
MesS21_Phi0_Hco_dB = MesS21_Phi0_Hco(:,2);
MesS21_Phi0_Hco2_dB = MesS21_Phi0_Hco2(:,2);

MesS21_Phi90_Eco_dB = MesS21_Phi90_E(:,2);
MesS21_Phi90_Hco_dB = MesS21_Phi90_Hco(:,2);

```

```

% Plots of Patch measurements
% Create a figure for S11
figure;
hold on;
plot(mes_f, S11dB_1_2, 'DisplayName', 'Patch1', 'LineWidth', 1.5);
plot(mes_f, S11dB_2_3, 'DisplayName', 'Patch2', 'LineWidth', 1.5);
plot(mes_f, S11dB_3_4, 'DisplayName', 'Patch3', 'LineWidth', 1.5);
plot(mes_f, S22dB_3_4, 'DisplayName', 'Patch4', 'LineWidth', 1.5);
hold off;
title('Measured Reflection Coefficient of each patch');
xlabel('Frequency (GHz)');
ylabel('S_{11} (dB)');
legend('show');
grid on;

% % Create a figure for S21
% figure;
% hold on;
% plot(mes_f, S21dB_1_2, 'DisplayName', 'Patch1_2 S21', 'LineWidth', 1.5);
% plot(mes_f, S21dB_1_3, 'DisplayName', 'Patch1_3 S21', 'LineWidth', 1.5);
% plot(mes_f, S21dB_1_4, 'DisplayName', 'Patch1_4 S21', 'LineWidth', 1.5);
% plot(mes_f, S21dB_2_3, 'DisplayName', 'Patch2_3 S21', 'LineWidth', 1.5);
% plot(mes_f, S21dB_2_4, 'DisplayName', 'Patch2_4 S21', 'LineWidth', 1.5);
% plot(mes_f, S21dB_3_4, 'DisplayName', 'Patch3_4 S21', 'LineWidth', 1.5);
%
% plot(mes_f, S12dB_1_2, 'DisplayName', 'Patch1_2 S12', 'LineWidth', 1.5);
% plot(mes_f, S12dB_1_3, 'DisplayName', 'Patch1_3 S12', 'LineWidth', 1.5);
% plot(mes_f, S12dB_1_4, 'DisplayName', 'Patch1_4 S12', 'LineWidth', 1.5);
% plot(mes_f, S12dB_2_3, 'DisplayName', 'Patch2_3 S12', 'LineWidth', 1.5);
% plot(mes_f, S12dB_2_4, 'DisplayName', 'Patch2_4 S12', 'LineWidth', 1.5);
% plot(mes_f, S12dB_3_4, 'DisplayName', 'Patch3_4 S12', 'LineWidth', 1.5);
% hold off;
% title('Cross talk between patches');
% xlabel('Frequency (Hz)');
% ylabel('S21 (dB)');
% legend('show');
% grid on;

%% FINAL PLOTS
% close all

% Single element S11 calc v opt v mes
figure; hold on;
grid on; box;
plot(f, calcS11, '--r', 'linewidth', 1.5) %calc
% plot(f, optS11, 'linewidth', 1.5)
plot(f, oshFR4S11, '-r', 'linewidth', 1.5) %mes
plot(mes_f, single_mesS11dB, '-k', 'linewidth', 1.5)
ylabel('S_{11} (dB)')
xlabel('Frequency (GHz)')
xline(2.4, '--k', 'linewidth', 0.5)

```

```

legend('Simulation Calculated Dimensions', 'Simuation Optimized Dimensions',
'Measured Results', 'location', 'best')

% Gain vs theta for array 2.45 Ghz
figure;
hold on;
polardB_90(ArrayGainP0(:,1), ArrayGainP0(:,2), 'r-', 1.5, 0, pmax, dynr, rticks)
polardB_90(ArrayGainP90(:,1), ArrayGainP90(:,2), 'k-', 1.5, 0, pmax, dynr, rticks)
legend('Phi = 0', 'Phi = 90')
ylabel('Gain (dB)')
xlabel('{\theta} (°)')
legend('Array Gain Phi = 0°', 'Array Gain Phi = 90°', 'location', 'best')
title('Patch Array gain on std FR4');

% Gain vs theta for array 2.4 Ghz
figure;
hold on;
polardB_90(ArrayGainP0_2_4GHz(:,1), ArrayGainP0_2_4GHz(:,2), 'r-', 1.5, 0, pmax,
dynr, rticks)
polardB_90(ArrayGainP90_2_4GHz(:,1), ArrayGainP90_2_4GHz(:,2), 'k-', 1.5, 0, pmax,
dynr, rticks)
legend('Phi = 0', 'Phi = 90')
ylabel('Gain (dB)')
xlabel('{\theta} (°)')
legend('Array Gain Phi = 0°', 'Array Gain Phi = 90°', 'location', 'best')
title('Patch Array on OSHPark FR4')

% S11 comparison of inset fed variations
figure; hold on;
grid on; box;
plot(f, SingleS11, '-b', 'linewidth', 1.5, 'DisplayName', 'Single Patch')
plot(f, Db1S11, '-k', 'linewidth', 1.5, 'DisplayName', '2x1 Array')
plot(f, Db1Db1S11, '-r', 'linewidth', 1.5, 'DisplayName', '2x2 Array')
ylabel('S_{11} dB')
xlabel('Frequency (GHz)')
xline(2.45, '--k', 'linewidth', 0.5)
legend('Single Patch', '2x1 Array', '2x2 Array', 'location', 'best')

%%
% S11 comparison of calc vs opt
figure; hold on;
grid on; box;
plot(f, SingleS11_l1y1, '--r', 'linewidth', 1.5, 'DisplayName', 'L = 27.25mm, y_0 =
5mm')
% plot(f, SingleS11_l2y1, '-b', 'linewidth', 1.5, 'DisplayName', 'L = 27.75mm, y_0
= 5mm')
% plot(f, SingleS11_l1y2, '-k', 'linewidth', 1.5, 'DisplayName', 'L = 27.25mm, y_0
= 8.5mm')
plot(f, SingleS11_l2y2, '-b', 'linewidth', 1.5, 'DisplayName', 'L = 27.75mm, y_0 =
8.5mm')
ylabel('S_{11} dB')
xlabel('Frequency (GHz)')
xline(2.45, '--', '2.45 GHz', 'HandleVisibility', 'off');

```

```

legend('location', 'best')
%%

% Patch Array cross-talk
figure;
hold on;
plot(mes_f, S21dB_1_2, 'DisplayName', 'Patch 1 & 2', 'LineWidth', 1.5);
plot(mes_f, S21dB_1_3, 'DisplayName', 'Patch 1 & 3', 'LineWidth', 1.5);
plot(mes_f, S21dB_1_4, 'DisplayName', 'Patch 1 & 4', 'LineWidth', 1.5);
plot(mes_f, S21dB_2_3, 'DisplayName', 'Patch 2 & 3', 'LineWidth', 1.5);
plot(mes_f, S21dB_2_4, 'DisplayName', 'Patch 2 & 4', 'LineWidth', 1.5);
plot(mes_f, S21dB_3_4, 'DisplayName', 'Patch 3 & 4', 'LineWidth', 1.5);
xline(2.4, '--', 'linewidth', 0.5, 'DisplayName', '2.4 GHz'); % Hidden in legend

% plot(mes_f, S12dB_1_2, 'DisplayName', 'Patch1_2 S12', 'LineWidth', 1.5);
% plot(mes_f, S12dB_1_3, 'DisplayName', 'Patch1_3 S12', 'LineWidth', 1.5);
% plot(mes_f, S12dB_1_4, 'DisplayName', 'Patch1_4 S12', 'LineWidth', 1.5);
% plot(mes_f, S12dB_2_3, 'DisplayName', 'Patch2_3 S12', 'LineWidth', 1.5);
% plot(mes_f, S12dB_2_4, 'DisplayName', 'Patch2_4 S12', 'LineWidth', 1.5);
% plot(mes_f, S12dB_3_4, 'DisplayName', 'Patch3_4 S12', 'LineWidth', 1.5);
hold off;
% title('Cross talk between patches');
xlabel('Frequency (GHz)');
ylabel('Crosstalk (dB)');
legend('show');
grid on;

% S11 of array with power splitter
figure; hold on;
grid on; box;
plot(mes_f, MeasPatchS11dB, '-r', 'linewidth', 1.5, 'DisplayName', 'Array with
power splitter')
ylabel('S_{11} dB')
xlabel('Frequency (GHz)')
xline(2.4, '--k', 'linewidth', 0.5)
% legend('Single Patch', '2x1 Array', '2x2 Array', 'location', 'best')

% Mes Gain vs theta for single patch 2.4 Ghz
% Phi 0
figure;
hold on;
% polardB_90(MesAng, MesS21_Phi0_Eco_dB - max(MesS21_Phi0_Eco_dB) + 10, 'r-', 1.5,
0)
polardB_90(MesAng, MesS21_Phi0_Hco_dB - max(MesS21_Phi0_Hco_dB) + 4, 'b--', 1.5, 0,
6, 24, 4)
polardB_90(PatchGainP0(:,1), PatchGainP0(:,2), 'b-', 1.5, 0, 6, 24, 4)

ylabel('Gain (dB)')
xlabel('{\theta} (°)')

% Phi 90
figure;
hold on;
% polardB_90(MesAng, MesS21_Phi90_Eco_dB - max(MesS21_Phi90_Eco_dB) + 4, 'r-', 1.5,
0)

```

```

polardB_90(MesAng, MesS21_Phi90_Hco_dB - max(MesS21_Phi90_Hco_dB) + 4, 'r--', 1.5,
0, 6, 24, 4)
polardB_90(PatchGainP90(:,1), PatchGainP90(:,2), 'r-', 1.5, 0, 6, 24, 4)

ylabel('Gain (dB)')
xlabel('{\theta} (°)')

% S11 comparison of inset fed variations
figure; hold on;
grid on; box;
plot(f, DbLS11, '-r', 'linewidth', 1.5, 'DisplayName', '2x1 Array')
ylabel('S_{11} dB')
xlabel('Frequency (GHz)')
xline(2.45, '--k', 'linewidth', 0.5)
% legend('Single Patch', '2x1 Array', '2x2 Array', 'location', 'best')

%%
% Mes Gain vs theta for single patch 2.4 Ghz
% Phi 0
figure;
hold on;
polardB_90(PatchGainP0(:,1), PatchGainP0(:,2), 'k-', 1.5, 0, 6, 24, 4)

polardB_90(PatchGainP90(:,1), PatchGainP90(:,2), 'r-', 1.5, 0, 6, 24, 4)

ylabel('Gain (dB)')
xlabel('{\theta} (°)')

```

6.4: Firmware and Interface Code

6.4.1 MATLAB Communication Code

```

% Nolan Egging
% Plot Data - MATLAB CODE

clear; close all; clc;

% constants (should match STM32 code)
SAMPLE_SIZE = 1024;
ADC_DELAY = 100;
ROTATE_DELAY = 100;
AZIMUTH_INCREMENT = 5;

% display ports
disp("Ports: " + serialportlist);
com = input("Input COM: ");
nucleo = serialport("COM" + com, 115200);

% press r when ready
disp("Press Enter When Ready");

```

```

input("");
write(nucleo, 'r', "char");

% create plot
theta = 0:AZIMUTH_INCREMENT:360;
rho_e = zeros(1, length(theta));
rho_h = zeros(1, length(theta));
figure(1);
subplot(121);
polarplot(theta*(pi/180), rho_e);
title("E-Field ADC Reading");
subplot(122);
polarplot(theta*(pi/180), rho_h);
title("H-Field ADC Reading");

% init arrays
I_array = zeros(1, SAMPLE_SIZE);
Q_array = zeros(1, SAMPLE_SIZE);

% take e-field reading
for i = 1:1:(length(theta)-1)

    % receive I_array
    for j = 1:1:SAMPLE_SIZE
        I_array(j) = str2double(read(nucleo, 4, "char"));
    end

    % receive Q_array
    for j = 1:1:SAMPLE_SIZE
        Q_array(j) = str2double(read(nucleo, 4, "char"));
    end

    % calcualte rho value
    avg = mean(I_array);
    rho_e(i) = mean(abs(I_array - avg));

    % updates plot
    subplot(121);
    polarplot(theta*(pi/180), rho_e);
    title("E-Field ADC Reading");

end
rho_e(end) = rho_e(1); % 360 and 0 should be equal

% updates plot
subplot(121);
polarplot(theta*(pi/180), rho_e);
title("E-Field ADC Reading");

% waits for user to spin antenna
disp("Spin Antenna, Then Press Enter");
input("");
write(nucleo, 's', "char");

% take h-field reading

```



```

for i = 1:1:(length(theta)-1)

    % receive I_array
    for j = 1:1:SAMPLE_SIZE
        I_array(j) = str2double(read(nucleo, 4, "char"));
    end

    % receive Q_array
    for j = 1:1:SAMPLE_SIZE
        Q_array(j) = str2double(read(nucleo, 4, "char"));
    end

    % calcualte rho value
    avg = mean(I_array);
    rho_h(i) = mean(abs(I_array - avg));

    % updates plot
    subplot(122);
    polarplot(theta*(pi/180), rho_h);
    title("H-Field ADC Reading");

end
rho_h(end) = rho_h(1); % 360 and 0 should be equal

% updates plot
subplot(122);
polarplot(theta*(pi/180), rho_h);
title("H-Field ADC Reading");

```

6.4.2 Main Microcontroller Firmware

```

/* Includes -----*/
#include "main.h"

/* Private includes -----*/
/* USER CODE BEGIN Includes */
#include <string.h>
#include <stdio.h>
/* USER CODE END Includes */

/* Private typedef -----*/
/* USER CODE BEGIN PTD */
typedef enum {INIT, IDLE, COLLECT, SEND, ROTATE, SPIN} state_t;
/* USER CODE END PTD */

/* Private define -----*/
/* USER CODE BEGIN PD */
#define SAMPLE_SIZE 1024
#define ADC_DELAY 100
#define ROTATE_DELAY 100
#define AZIMUTH_INCREMENT 5
/* USER CODE END PD */

/* Private macro -----*/
/* USER CODE BEGIN PM */

/* USER CODE END PM */

/* Private variables -----*/
ADC_HandleTypeDef hadcl;

```

```

ADC_HandleTypeDef hadc2;
DMA_HandleTypeDef hdma_adc1;
DMA_HandleTypeDef hdma_adc2;

UART_HandleTypeDef huart2;

/* USER CODE BEGIN PV */
uint16_t I_array[SAMPLE_SIZE];
uint16_t Q_array[SAMPLE_SIZE];
uint8_t i_sample_done_f = 0;
uint8_t q_sample_done_f = 0;
uint8_t rx_buf = 'i';

/* USER CODE END PV */

/* Private function prototypes -----*/
void SystemClock_Config(void);
static void MX_GPIO_Init(void);
static void MX_DMA_Init(void);
static void MX_USART2_UART_Init(void);
static void MX_ADC1_Init(void);
static void MX_ADC2_Init(void);
/* USER CODE BEGIN PFP */
void RotateMotor(uint16_t angle_increment);
/* USER CODE END PFP */

/* Private user code -----*/
/* USER CODE BEGIN 0 */

/* USER CODE END 0 */

/**
 * @brief The application entry point.
 * @retval int
 */
int main(void)
{
    /* USER CODE BEGIN 1 */
    char msg[5];
    state_t state = INIT;
    uint16_t azimuth = 0;
    uint8_t spin = 0;

    /* USER CODE END 1 */

    /* MCU Configuration-----*/

    /* Reset of all peripherals, Initializes the Flash interface and the Systick. */
    HAL_Init();

    /* USER CODE BEGIN Init */

    /* USER CODE END Init */

    /* Configure the system clock */
    SystemClock_Config();

    /* USER CODE BEGIN SysInit */

    /* USER CODE END SysInit */

    /* Initialize all configured peripherals */
    MX_GPIO_Init();
    MX_DMA_Init();
    MX_USART2_UART_Init();
    MX_ADC1_Init();
    MX_ADC2_Init();
    /* USER CODE BEGIN 2 */

    /* USER CODE END 2 */

```

```

/* Infinite loop */
/* USER CODE BEGIN WHILE */
while (1)
{
    /* USER CODE END WHILE */

    /* USER CODE BEGIN 3 */

        switch (state) {

            case (INIT):
                HAL_GPIO_WritePin(LD2_GPIO_Port, LD2_Pin, GPIO_PIN_RESET);
                i_sample_done_f = 0;
                q_sample_done_f = 0;
                rx_buf = 'i';
                azimuth = 0;
                spin = 0;
                state = IDLE;
                break;

            case (IDLE):
                HAL_GPIO_WritePin(LD2_GPIO_Port, LD2_Pin, GPIO_PIN_SET);
                while (rx_buf != 'r') HAL_UART_Receive(&huart2, &rx_buf, 1,
HAL_MAX_DELAY);
                state = COLLECT;
                break;

            case (COLLECT):
                HAL_GPIO_WritePin(LD2_GPIO_Port, LD2_Pin, GPIO_PIN_RESET);
                HAL_Delay(ADC_DELAY); // settling time
                i_sample_done_f = 0;
                q_sample_done_f = 0;
                HAL_ADC_Start_DMA(&hadc1, &I_array[0], SAMPLE_SIZE);
                //HAL_ADC_Start_DMA(&hadc2, &Q_array[0], SAMPLE_SIZE);
                //while ((i_sample_done_f && q_sample_done_f) != 1);
                while(!i_sample_done_f);
                state = SEND;
                break;

            case (SEND):
                for (uint16_t i = 0; i < SAMPLE_SIZE; i++){
                    sprintf(msg, "%4hu", I_array[i]);
                    HAL_UART_Transmit(&huart2, (uint8_t*)msg, strlen(msg),
HAL_MAX_DELAY);
                }
                for (uint16_t i = 0; i < SAMPLE_SIZE; i++){
                    sprintf(msg, "%4hu", Q_array[i]);
                    HAL_UART_Transmit(&huart2, (uint8_t*)msg, strlen(msg),
HAL_MAX_DELAY);
                }
                state = ROTATE;
                break;

            case (ROTATE):
                HAL_Delay(ROTATE_DELAY);
                azimuth += AZIMUTH_INCREMENT;
                RotateMotor(AZIMUTH_INCREMENT);
                if (azimuth >= 360) {
                    azimuth = 0;
                    if (spin == 0) state = SPIN;
                    else state = INIT;
                }
                else state = COLLECT;
                break;

            case (SPIN):
                spin = 90;
                HAL_UART_Receive(&huart2, &rx_buf, 1, HAL_MAX_DELAY);
                if (rx_buf == 's') state = COLLECT;
                break;

```

```

        default:
            state = INIT;
            break;
    }

}

/* USER CODE END 3 */
}

/**
 * @brief System Clock Configuration
 * @retval None
 */
void SystemClock_Config(void)
{
    RCC_OscInitTypeDef RCC_OscInitStruct = {0};
    RCC_ClkInitTypeDef RCC_ClkInitStruct = {0};

    /** Configure the main internal regulator output voltage
    */
    __HAL_RCC_PWR_CLK_ENABLE();
    __HAL_PWR_VOLTAGESCALING_CONFIG(PWR_REGULATOR_VOLTAGE_SCALE3);

    /** Initializes the RCC Oscillators according to the specified parameters
    * in the RCC_OscInitTypeDef structure.
    */
    RCC_OscInitStruct.OscillatorType = RCC_OSCILLATORTYPE_HSI;
    RCC_OscInitStruct.HSISState = RCC_HSI_ON;
    RCC_OscInitStruct.HSICalibrationValue = RCC_HSICALIBRATION_DEFAULT;
    RCC_OscInitStruct.PLL.PLLState = RCC_PLL_ON;
    RCC_OscInitStruct.PLL.PLLSource = RCC_PLLSOURCE_HSI;
    RCC_OscInitStruct.PLL.PLLM = 16;
    RCC_OscInitStruct.PLL.PLLN = 336;
    RCC_OscInitStruct.PLL.PLLP = RCC_PLLP_DIV4;
    RCC_OscInitStruct.PLL.PLLQ = 2;
    RCC_OscInitStruct.PLL.PLLR = 2;
    if (HAL_RCC_OscConfig(&RCC_OscInitStruct) != HAL_OK)
    {
        Error_Handler();
    }

    /** Initializes the CPU, AHB and APB buses clocks
    */
    RCC_ClkInitStruct.ClockType = RCC_CLOCKTYPE_HCLK|RCC_CLOCKTYPE_SYSCLK
                                  |RCC_CLOCKTYPE_PCLK1|RCC_CLOCKTYPE_PCLK2;
    RCC_ClkInitStruct.SYSCLKSource = RCC_SYSCLKSOURCE_PLLCLK;
    RCC_ClkInitStruct.AHBCLKDivider = RCC_SYSCLK_DIV1;
    RCC_ClkInitStruct.APB1CLKDivider = RCC_HCLK_DIV2;
    RCC_ClkInitStruct.APB2CLKDivider = RCC_HCLK_DIV1;

    if (HAL_RCC_ClockConfig(&RCC_ClkInitStruct, FLASH_LATENCY_2) != HAL_OK)
    {
        Error_Handler();
    }
}

/**
 * @brief ADC1 Initialization Function
 * @param None
 * @retval None
 */
static void MX_ADC1_Init(void)
{
    /* USER CODE BEGIN ADC1_Init 0 */

    /* USER CODE END ADC1_Init 0 */

    ADC_ChannelConfTypeDef sConfig = {0};

```

```

/* USER CODE BEGIN ADC1_Init 1 */

/* USER CODE END ADC1_Init 1 */

/** Configure the global features of the ADC (Clock, Resolution, Data Alignment and number of
conversion)
*/
hadc1.Instance = ADC1;
hadc1.Init.ClockPrescaler = ADC_CLOCK_SYNC_PCLK_DIV4;
hadc1.Init.Resolution = ADC_RESOLUTION_12B;
hadc1.Init.ScanConvMode = DISABLE;
hadc1.Init.ContinuousConvMode = ENABLE;
hadc1.Init.DiscontinuousConvMode = DISABLE;
hadc1.Init.ExternalTrigConvEdge = ADC_EXTERNALTRIGCONVEDGE_NONE;
hadc1.Init.ExternalTrigConv = ADC_SOFTWARE_START;
hadc1.Init.DataAlign = ADC_DATAALIGN_RIGHT;
hadc1.Init.NbrOfConversion = 1;
hadc1.Init.DMAContinuousRequests = ENABLE;
hadc1.Init.EOCSelection = ADC_EOC_SINGLE_CONV;
if (HAL_ADC_Init(&hadc1) != HAL_OK)
{
    Error_Handler();
}

/** Configure for the selected ADC regular channel its corresponding rank in the sequencer
and its sample time.
*/
sConfig.Channel = ADC_CHANNEL_0;
sConfig.Rank = 1;
sConfig.SamplingTime = ADC_SAMPLETIME_3CYCLES;
if (HAL_ADC_ConfigChannel(&hadc1, &sConfig) != HAL_OK)
{
    Error_Handler();
}
/* USER CODE BEGIN ADC1_Init 2 */

/* USER CODE END ADC1_Init 2 */
}

/**
 * @brief ADC2 Initialization Function
 * @param None
 * @retval None
 */
static void MX_ADC2_Init(void)
{
    /* USER CODE BEGIN ADC2_Init 0 */

    /* USER CODE END ADC2_Init 0 */

    ADC_ChannelConfTypeDef sConfig = {0};

    /* USER CODE BEGIN ADC2_Init 1 */

    /* USER CODE END ADC2_Init 1 */

    /** Configure the global features of the ADC (Clock, Resolution, Data Alignment and number of
conversion)
    */
    hadc2.Instance = ADC2;
    hadc2.Init.ClockPrescaler = ADC_CLOCK_SYNC_PCLK_DIV4;
    hadc2.Init.Resolution = ADC_RESOLUTION_12B;
    hadc2.Init.ScanConvMode = DISABLE;
    hadc2.Init.ContinuousConvMode = ENABLE;
    hadc2.Init.DiscontinuousConvMode = DISABLE;
    hadc2.Init.ExternalTrigConvEdge = ADC_EXTERNALTRIGCONVEDGE_NONE;
    hadc2.Init.ExternalTrigConv = ADC_SOFTWARE_START;
    hadc2.Init.DataAlign = ADC_DATAALIGN_RIGHT;

```

```

hadc2.Init.NbrOfConversion = 1;
hadc2.Init.DMAContinuousRequests = ENABLE;
hadc2.Init.EOCSelection = ADC_EOC_SINGLE_CONV;
if (HAL_ADC_Init(&hadc2) != HAL_OK)
{
    Error_Handler();
}

/** Configure for the selected ADC regular channel its corresponding rank in the sequencer
and its sample time.
*/
sConfig.Channel = ADC_CHANNEL_1;
sConfig.Rank = 1;
sConfig.SamplingTime = ADC_SAMPLETIME_3CYCLES;
if (HAL_ADC_ConfigChannel(&hadc2, &sConfig) != HAL_OK)
{
    Error_Handler();
}
/* USER CODE BEGIN ADC2_Init 2 */

/* USER CODE END ADC2_Init 2 */
}

/**
 * @brief USART2 Initialization Function
 * @param None
 * @retval None
 */
static void MX_USART2_UART_Init(void)
{
    /* USER CODE BEGIN USART2_Init 0 */

    /* USER CODE END USART2_Init 0 */

    /* USER CODE BEGIN USART2_Init 1 */

    /* USER CODE END USART2_Init 1 */
    huart2.Instance = USART2;
    huart2.Init.BaudRate = 115200;
    huart2.Init.WordLength = UART_WORDLENGTH_8B;
    huart2.Init.StopBits = UART_STOPBITS_1;
    huart2.Init.Parity = UART_PARITY_NONE;
    huart2.Init.Mode = UART_MODE_TX_RX;
    huart2.Init.HwFlowCtl = UART_HWCONTROL_NONE;
    huart2.Init.OverSampling = UART_OVERSAMPLING_16;
    if (HAL_UART_Init(&huart2) != HAL_OK)
    {
        Error_Handler();
    }
    /* USER CODE BEGIN USART2_Init 2 */

    /* USER CODE END USART2_Init 2 */
}

/**
 * Enable DMA controller clock
 */
static void MX_DMA_Init(void)
{
    /* DMA controller clock enable */
    __HAL_RCC_DMA2_CLK_ENABLE();

    /* DMA interrupt init */
    /* DMA2_Stream0_IRQn interrupt configuration */
    HAL_NVIC_SetPriority(DMA2_Stream0_IRQn, 0, 0);
    HAL_NVIC_EnableIRQ(DMA2_Stream0_IRQn);
    /* DMA2_Stream2_IRQn interrupt configuration */

```

```

HAL_NVIC_SetPriority(DMA2_Stream2_IRQn, 0, 0);
HAL_NVIC_EnableIRQ(DMA2_Stream2_IRQn);

}

/**
 * @brief GPIO Initialization Function
 * @param None
 * @retval None
 */
static void MX_GPIO_Init(void)
{
    GPIO_InitTypeDef GPIO_InitStruct = {0};
    /* USER CODE BEGIN MX_GPIO_Init_1 */
    /* USER CODE END MX_GPIO_Init_1 */

    /* GPIO Ports Clock Enable */
    __HAL_RCC_GPIOC_CLK_ENABLE();
    __HAL_RCC_GPIOH_CLK_ENABLE();
    __HAL_RCC_GPIOA_CLK_ENABLE();
    __HAL_RCC_GPIOB_CLK_ENABLE();

    /*Configure GPIO pin Output Level */
    HAL_GPIO_WritePin(LD2_GPIO_Port, LD2_Pin, GPIO_PIN_RESET);

    /*Configure GPIO pin Output Level */
    HAL_GPIO_WritePin(GPIOB, SMD1_Pin|SMD2_Pin|SMD3_Pin|SMD4_Pin, GPIO_PIN_RESET);

    /*Configure GPIO pin : B1_Pin */
    GPIO_InitStruct.Pin = B1_Pin;
    GPIO_InitStruct.Mode = GPIO_MODE_IT_FALLING;
    GPIO_InitStruct.Pull = GPIO_NOPULL;
    HAL_GPIO_Init(B1_GPIO_Port, &GPIO_InitStruct);

    /*Configure GPIO pin : LD2_Pin */
    GPIO_InitStruct.Pin = LD2_Pin;
    GPIO_InitStruct.Mode = GPIO_MODE_OUTPUT_PP;
    GPIO_InitStruct.Pull = GPIO_NOPULL;
    GPIO_InitStruct.Speed = GPIO_SPEED_FREQ_LOW;
    HAL_GPIO_Init(LD2_GPIO_Port, &GPIO_InitStruct);

    /*Configure GPIO pins : SMD1_Pin SMD2_Pin SMD3_Pin SMD4_Pin */
    GPIO_InitStruct.Pin = SMD1_Pin|SMD2_Pin|SMD3_Pin|SMD4_Pin;
    GPIO_InitStruct.Mode = GPIO_MODE_OUTPUT_PP;
    GPIO_InitStruct.Pull = GPIO_NOPULL;
    GPIO_InitStruct.Speed = GPIO_SPEED_FREQ_LOW;
    HAL_GPIO_Init(GPIOB, &GPIO_InitStruct);

    /* USER CODE BEGIN MX_GPIO_Init_2 */
    /* USER CODE END MX_GPIO_Init_2 */
}

/* USER CODE BEGIN 4 */
void HAL_ADC_ConvCpltCallback(ADC_HandleTypeDef* hadc)
{
    if(hadc->Instance == ADC1) i_sample_done_f = 1;
    if(hadc->Instance == ADC2) q_sample_done_f = 1;
}

void RotateMotor(uint16_t angle_increment) {

    static uint8_t motor_state = 1;

    for (uint16_t i = 0; i < 30.5*angle_increment; i++){
        switch (motor_state) {
            case(1):
                HAL_GPIO_WritePin(SMD1_GPIO_Port, SMD1_Pin, GPIO_PIN_SET);
                HAL_GPIO_WritePin(SMD2_GPIO_Port, SMD2_Pin, GPIO_PIN_RESET);
                HAL_GPIO_WritePin(SMD3_GPIO_Port, SMD3_Pin, GPIO_PIN_RESET);
                HAL_GPIO_WritePin(SMD4_GPIO_Port, SMD4_Pin, GPIO_PIN_SET);
                motor_state = 2;

```

```

        break;

    case(2):
        HAL_GPIO_WritePin(SMD1_GPIO_Port, SMD1_Pin, GPIO_PIN_SET);
        HAL_GPIO_WritePin(SMD2_GPIO_Port, SMD2_Pin, GPIO_PIN_SET);
        HAL_GPIO_WritePin(SMD3_GPIO_Port, SMD3_Pin, GPIO_PIN_RESET);
        HAL_GPIO_WritePin(SMD4_GPIO_Port, SMD4_Pin, GPIO_PIN_RESET);
        motor_state = 3;
        break;

    case(3):
        HAL_GPIO_WritePin(SMD1_GPIO_Port, SMD1_Pin, GPIO_PIN_RESET);
        HAL_GPIO_WritePin(SMD2_GPIO_Port, SMD2_Pin, GPIO_PIN_SET);
        HAL_GPIO_WritePin(SMD3_GPIO_Port, SMD3_Pin, GPIO_PIN_SET);
        HAL_GPIO_WritePin(SMD4_GPIO_Port, SMD4_Pin, GPIO_PIN_RESET);
        motor_state = 4;
        break;

    case(4):
        HAL_GPIO_WritePin(SMD1_GPIO_Port, SMD1_Pin, GPIO_PIN_RESET);
        HAL_GPIO_WritePin(SMD2_GPIO_Port, SMD2_Pin, GPIO_PIN_RESET);
        HAL_GPIO_WritePin(SMD3_GPIO_Port, SMD3_Pin, GPIO_PIN_SET);
        HAL_GPIO_WritePin(SMD4_GPIO_Port, SMD4_Pin, GPIO_PIN_SET);
        motor_state = 1;
        break;

    default:
        break;
}
HAL_Delay(1);
}

// Turn off all pins to prevent overheating of driver
HAL_GPIO_WritePin(SMD1_GPIO_Port, SMD1_Pin, GPIO_PIN_RESET);
HAL_GPIO_WritePin(SMD2_GPIO_Port, SMD2_Pin, GPIO_PIN_RESET);
HAL_GPIO_WritePin(SMD3_GPIO_Port, SMD3_Pin, GPIO_PIN_RESET);
HAL_GPIO_WritePin(SMD4_GPIO_Port, SMD4_Pin, GPIO_PIN_RESET);
}

/* USER CODE END 4 */

/**
 * @brief This function is executed in case of error occurrence.
 * @retval None
 */
void Error_Handler(void)
{
    /* USER CODE BEGIN Error_Handler_Debug */
    /* User can add his own implementation to report the HAL error return state */
    __disable_irq();
    while (1)
    {
    }
    /* USER CODE END Error_Handler_Debug */
}

#ifdef USE_FULL_ASSERT
/**
 * @brief Reports the name of the source file and the source line number
 * where the assert_param error has occurred.
 * @param file: pointer to the source file name
 * @param line: assert_param error line source number
 * @retval None
 */
void assert_failed(uint8_t *file, uint32_t line)
{
    /* USER CODE BEGIN 6 */
    /* User can add his own implementation to report the file name and line number,
    ex: printf("Wrong parameters value: file %s on line %d\r\n", file, line) */
    /* USER CODE END 6 */
}
#endif /* USE_FULL_ASSERT */

```